DATASET

This document describes how the dataset was constructed, revised, and discussed by three experts in logics for strategic reasoning in multi-agent systems. We provide this file to make our methodology fully transparent, and because we believe that the scientific community and the reviewers will help us improve it for future work along this research direction.

The dataset is provided in JSON format and, for each item, includes a natural-language specification, its translation into temporal logic (ATL/ATL*), and an estimated difficulty score for the translation, based on the well-known syntactic and semantic ambiguities in natural language.

Coalition member e.g. “User”

Strategic Operator <<.>>
Temporal Operator X U F G
Propositional Operator ! -> && ||

```
[
 {
  "id": "ex01",
  "input": "The user can guarantee that sooner or later the ticket will be printed.",
  "outputs": [
   {
    "formula": "<<User>>F ticket_printed"
   }
  ]
 },
 {
  "id": "ex02",
  "input": "The machine can guarantee that if the payment has been completed, then sooner or later the gate will open.",
  "outputs": [
   {
    "formula": "<<Machine>>(paid -> F gate_open)"
   }
  ]
 },
```

Reviewer 1 - “ex02”: Here, it can be observed that, since it does not appear in the natural language specification, the *Globally* operator could be presupposed in the interpretation (*Always* missing) and made explicit in the translation.

Reviewer 2: The presence of the conditional construction *if … then …* is interpreted as a universally quantified rule over time. Although the natural language sentence does not explicitly contain an adverb such as *always*, the use of the strategic verb *can guarantee* suggests a persistent capability, justifying the introduction of the G operator.

Reviewer 3: My opinion is that, since it is not explicitly stated in the natural-language property, both the version with G and the version without G are correct. Clearly, in the latter case we would be representing a dataset for ATL (I mean that, for the strategic operator, we use G; however, F appears without a strategic operator, which implicitly assumes an ATL* formula). If the dataset is intended for ATL, what we can assume for the LLM is the following: if there is no explicit temporal operator, assume G in order to make the formula well formed.
Reviewer 2: I agree with Reviewer 3, check the wider discussion at example 61.

```
{
  "id": "ex03",
  "input": "The user and the machine together can guarantee that the system will never enter an error state.",
  "outputs": [
    {
      "formula": "<<User,Machine>>G !error"
    }
  ]
},
```

Reviewer 1 - “ex03”: Here, we can find a reinforcement of the coalition concept conveyed by *together,* which supports the use of *and,* since *and* alone does not imply that the user and the machine are acting collaboratively. The use of *together* is clearly a piece of contextually derived information. Furthermore, it is non-trivial that the LLM is able to distinguish between the different syntactic roles of *and,* namely, being able to translate its occurrence as a separating comma within the coalition tuple, instead of *and* as a logical operator. It is worth noting that *never* simultaneously provides the Boolean negation of an atomic proposition and the *Globally* operator. A formally and semantically straightforward explanation is that the difference between negating a temporal operator and negating an atomic proposition dominated by a temporal operator is reflected in their respective syntactic trees. In the case presented here, *never* functions as a Verb Phrase (VP) level adjunct, with scope over the negation of the event of entering.
Reviewer 2: The adverb *together* explicitly enforces a cooperative interpretation, which is correctly reflected by the coalition operator <<A,B>>. Without this lexical cue, a non-cooperative or adversarial interpretation would also be plausible.
Reviewer 3: The word together reinforces the notion of a coalition, but even a simple and would have yielded the same semantic interpretation. Also in this case, G was automatically inserted, and my reasoning remains the same.

```
{
  "id": "ex04",
  "input": "The machine can guarantee that at the next step the ticket will be validated.",
  "outputs": [
    {
      "formula": "<<Machine>>X ticket_validated"
    }
  ]
},
```

Reviewer 1 - “ex04”: Added ticket_ for consistency (see “ex01”).

```
 {
   "id": "ex05",
   "input": "The user can guarantee that as long as they do not have the ticket, they will obtain it.",
   "outputs": [
     {
       "formula": "<<User>>(!has_ticket U has_ticket)"
     }
   ]
 },
```

Reviewer 2: Unlike conditional statements, the *until* construction naturally encodes a temporal progression starting from the current state. Therefore, the absence of a dominating G operator is consistent with both the linguistic structure and the standard use of the U operator in temporal logics.

```
 {
   "id": "ex06",
   "input": "If the user presses cancel, the machine can guarantee that sooner or later a refund will be issued.",
   "outputs": [
     {
       "formula": "user_cancel -> <<Machine>>F refund"
     }
   ]
 },
```

Original formula discussed: <<Machine>>(cancel -> F refund)
Reviewer 1 - “ex06”: In my opinion, it is a legitimate question to ask whether, in this case, the user is in coalition with the machine, as in *“ex03”,* or not. It should be noted that an interpretation of the user as an adversary may be favored by the absence of *and* and *together,* which are present in *“ex03”.* Nevertheless, in my view, the human and the machine are collaborating in this instance.
Reviewer 2: I do not agree with Reviewer 1. While a cooperative interpretation is possible at the pragmatic level, from a formal ATL perspective the absence of explicit coordination markers does not justify extending the coalition beyond the machine. Therefore, interpreting the user as part of the environment remains the most conservative and formally sound choice. The absence of and/together DOES NOT justify in any case to insert the user in the coalition.
Reviewer 1 - “ex06”: Alone a user cannot obtain a refund and alone a machine cannot give a refund. It is cooperative. If Reviewer 2 is right then the machine has a strategy that guarantees refunds after cancel, no matter what the user does, which is a contradiction given the fact that the action of the user, namely *cancel*, matters.
Reviewer 3: In my view, ‘If the user presses cancel’ represents an atomic proposition that should be defined a priori by the subsequent strategic operator. My formula would be: user_cancel -> <<Machine>> F refund.”

Reviewer 2: I agree with Reviewer 3. Case closed.


```
 {
   "id": "ex07",
   "input": "The machine can guarantee that the gate will never open without payment.",
   "outputs": [
     {
       "formula": "<<Machine>>G (!paid -> !gate_open)"
     }
   ]
 },
 {
   "id": "ex08",
   "input": "The user can guarantee that at the next step the system will not time out.",
   "outputs": [
     {
       "formula": "<<User>>X !timeout"
     }
   ]
 },
 {
   "id": "ex09",
   "input": "The machine can guarantee that if a card or cash is inserted, then sooner or later the payment will be completed.",
   "outputs": [
     {
       "formula": "<<Machine>>((card_inserted || cash_inserted) -> F paid)"
     }
   ]
 },
```


Reviewer 1 - “ex09”: Here, the human behavior is made explicit without an explicit performing subject. I still maintain that the machine and the user are not adversarial in this case, but rather cooperating, as in “*ex03*”. After a brief reality check, you can notice that || corresponds to an exclusive disjunction rather than an inclusive one, however, I am fairly certain that *or* is interpreted as an inclusive disjunction in ATL.
Reviewer 2: I agree with the reviewer 3.
Reviewer 3: The only thing I still notice is that we are in the ATL* language, and I don’t understand why a globally operator is always added even when it’s not necessary.


```
 {
   "id": "ex10",
   "input": "The user can guarantee that sooner or later the ticket will be printed and sooner or later the gate will open.",
   "outputs": [
     {
       "formula": "<<User>>F ticket_printed && <<User>>F gate_open"
     }
```

]
},

{
"id": "ex11",
"input": "The machine can guarantee that if the payment has been completed, then at the next step it will print the ticket.",
"outputs": [
{
"formula": "<<Machine>> (paid -> X ticket_printed)"
}
]
},

Reviewer 1 - “ex11”: Note that, in this case, the *Globally* operator is omitted from the natural language sentence.
Reviewer 2: see ex2
Reviewer 3: see exercise 2

{
"id": "ex12",
"input": "The user can guarantee that at the next step either a card or cash will be inserted.",
"outputs": [
{
"formula": "<<User>>X (card_inserted || cash_inserted)"
}
]
},

Reviewer 1 - “ex12”: Should we replace *“the user”* with *“the machine”*?
Reviewer 2: I agree
Reviewer 3: I agree that, from a logical point of view, it should be the machine and not the user, but here we are creating a translator from an informal language to a formal one, not an intelligent system that needs to improve the specifications in terms of logic.

{
"id": "ex13",
"input": "The machine can guarantee that the gate opens only if at the next step the ticket is printed.",
"outputs": [
{
"formula": "<<Machine>>(gate_open -> X ticket_printed)"
}
]
},

Reviewer 1 - “ex13”: See “ex11”.

Reviewer 2: see ex2
Reviewer 3: see exercise 2

```
 {
   "id": "ex14",
   "input": "The user can guarantee that if they do not have the ticket now, then at the next step they will never pass through the gate.",
   "outputs": [
     {
       "formula": "<<User>>G (!has_ticket -> X !pass_gate)"
     }
   ]
 },
```

Reviewer 1 - “ex14”: See “ex11”.
Reviewer 2: see ex2
Reviewer 3: see exercise 2

```
 {
   "id": "ex15",
   "input": "The machine can guarantee that if an error occurs at the next step, then sooner or later the system will be recovered.",
   "outputs": [
     {
       "formula": "<<Machine>>(X error -> F recovered)"
     }
   ]
 },
```

Reviewer 1 - “ex15”: See “ex11”.
Reviewer 2: see ex2
Reviewer 3: see exercise 2

```
 {
   "id": "ex16",
   "input": "The user can guarantee that after having obtained the ticket, they will sooner or later pass through the gate.",
   "outputs": [
     {
       "formula": "<<User>>(has_ticket && X F passed_through_gate)"
     }
   ]
 },
 {
   "id": "ex17",
   "input": "The machine can guarantee that if the payment does not occur, the ticket will never be printed.",
```

"outputs": [
{
"formula": "<<Machine>> (!paid -> G !ticket_printed)"

}
]
},

Reviewer 1 - “ex17”: Although the translation of the second *Globally* operator, the one dominating the negated consequent of the conditional, can be justified by the presence of the adverb *never*, it nevertheless seems problematic. The conditional itself is already under the scope of a *Globally* operator, so introducing a second one appears redundant. I therefore suggest eliminating the second *Globally* operator.
Reviewer 2: I do agree with Reviewer 1.
Reviewer 3: Our goal is to have a translation from natural language to formal language, keeping in mind that the user is capable of expressing clearly in their own language what they need.
Reviewer 1 - “ex17”: Solved; majority reached.
{
"id": "ex18",
"input": "The user can guarantee that at the next step there will be no error and that this condition will always remain true.",
"outputs": [
{
"formula": "<<User>>X G !error"
}
]
},
{
"id": "ex19",
"input": "The machine can guarantee that the system remains safe until cancellation is requested.",
"outputs": [
{
"formula": "<<Machine>>(safe U cancel)"
}
]
},
Reviewer 2:see ex05
{
"id": "ex20",
"input": "The user can guarantee that if they insert cash, then at the next step the payment will be registered.",
"outputs": [
{
"formula": "<<User>>(cash_inserted -> X paid)"
}
]

},
{
"id": "ex21",
"input": "The machine and the user together can guarantee that the ticket is printed before the system times out.",
"outputs": [
{
"formula": "<<Machine,User>>(timer_running U ticket_printed)"
}
]
},

Original formula discussed: <<User,Machine>>(!timeout U ticket_printed)
Reviewer 1 - “ex21”: Note that the negation is implicitly encoded in the semantics of *before* due to its negative entailment. If the ticket is printed *before* the system times out, then the timing out does not happen before the printing. Revision needed.
Reviewer 2: see ex03

{
"id": "ex22",
"input": "The machine can guarantee that every time the gate opens, it will close again at the next step.",
"outputs": [
{
"formula": "<<Machine>>G (gate_open -> X gate_closed)"
}
]
},
{
"id": "ex23",
"input": "The user can guarantee that they will continue attempting payment until it is completed.",
"outputs": [
{
"formula": "<<User>>(attempt_pay U paid)"
}
]
},
Reviewer 2:see ex05
{
"id": "ex24",
"input": "The machine can guarantee that if the ticket is printed, then sooner or later it will be validated.",
"outputs": [
{
"formula": "<<Machine>> (ticket_printed -> F validated)"
}
]

},
{
"id": "ex25",
"input": "The user can guarantee that at the next step they will perform a valid action or cancel the operation.",
"outputs": [
{
"formula": "<<User>>X (valid_action || cancel)"
}
]
},
{
"id": "ex26",
"input": "The drone can guarantee that sooner or later it will reach the destination waypoint.",
"outputs": [
{
"formula": "<<Drone>>F at_waypoint"
}
]
},
{
"id": "ex27",
"input": "The drone can guarantee that if the battery level is low, then sooner or later it will return to base.",
"outputs": [
{
"formula": "<<Drone>> (battery_low -> F at_base)"
}
]
},

Reviewer 1 - “ex27”: See “ex11”.
Reviewer 2: see ex2

{
"id": "ex28",
"input": "The drone can guarantee that it will never enter a no-fly zone.",
"outputs": [
{
"formula": "<<Drone>>G !in_no_fly_zone"
}
]
},
Reviewer 2: ok
{
"id": "ex29",

```
    "input": "The drone can guarantee that at the next step it will activate obstacle avoidance.",
    "outputs": [
      {
        "formula": "<<Drone>>X obstacle_avoidance_on"
      }
    ]
  },
```

Reviewer 2: ok

```
  {
    "id": "ex30",
    "input": "The drone can guarantee that it will continue searching until it detects the target.",
    "outputs": [
      {
        "formula": "<<Drone>>(searching U target_detected)"
      }
    ]
  },
```

Reviewer 2:see ex05

```
  {
    "id": "ex31",
    "input": "The drone can guarantee that if the target is detected, then at the next step it will take a photo.",
    "outputs": [
      {
        "formula": "<<Drone>> (target_detected -> X photo_taken)"
      }
    ]
  },
```

Reviewer 2: ok

```
  {
    "id": "ex32",
    "input": "The drone and the wind together can guarantee that sooner or later the drone will move away from the waypoint.",
    "outputs": [
      {
        "formula": "<<Drone,Wind>>F !at_waypoint"
      }
    ]
  },
```

Reviewer 2: see ex03

```
  {
    "id": "ex33",
    "input": "The drone can guarantee that if it loses GPS signal, then at the next step it will switch to inertial navigation.",
    "outputs": [
      {
```

```
      "formula": "<<Drone>> (gps_lost -> X inertial_nav_on)"
    }
  ]
},
```


Reviewer 1 - “ex33”: See “ex11”.
Reviewer 2: see ex2


```
{
  "id": "ex34",
  "input": "The drone can guarantee that if the battery is critical, then it will never start a delivery mission.",
  "outputs": [
    {
      "formula": "<<Drone>> (battery_critical -> G !delivery_started)"
    }
  ]
},
```


Reviewer 1 - “ex34”: See “ex17”
Reviewer 2: See why I agree in ex17


```
{
  "id": "ex35",
  "input": "The drone can guarantee that sooner or later it will land safely.",
  "outputs": [
    {
      "formula": "<<Drone>>F safe_landed"
    }
  ]
},

{
  "id": "ex36",
  "input": "The mobile robot can guarantee that sooner or later it will reach the charging station.",
  "outputs": [
    {
      "formula": "<<Robot>>F at_charging_station"
    }
  ]
},
{
  "id": "ex37",
  "input": "The mobile robot can guarantee that it will never collide with an obstacle.",
  "outputs": [
    {
      "formula": "<<Robot>>G !collision"
```

```
    }
  ]
},
{
  "id": "ex38",
  "input": "The mobile robot can always guarantee that if the path is blocked, then sooner or later it will replan a new route.",
  "outputs": [
    {
      "formula": "<<Robot>>G (path_blocked -> F replanned)"
    }
  ]
},
{
  "id": "ex39",
  "input": "The mobile robot can always guarantee that at the next step it will slow down if it detects a pedestrian.",
  "outputs": [
    {
      "formula": "<<Robot>>G (pedestrian_detected -> X speed_low)"
    }
  ]
},
```

Reviewer 1 - “ex39”: See “ex11”.

```
{
  "id": "ex40",
  "input": "The mobile robot can guarantee that it will continue following the line until it reaches the loading area.",
  "outputs": [
    {
      "formula": "<<Robot>>(following_line U at_loading_area)"
    }
  ]
},
```

Reviewer 2:see ex05

```
{
  "id": "ex41",
  "input": "The mobile robot can always guarantee that if it is in emergency mode, then it will always remain stopped.",
  "outputs": [
    {
      "formula": "<<Robot>>G (emergency_mode -> G stopped)"
    }
  ]
},
```

Reviewer 1 - “ex41”: See “ex17”.
Reviewer 2: See why I do agree in ex17

```
{
  "id": "ex42",
  "input": "The robot and the operator together can guarantee that sooner or later the robot will resume the mission after a pause.",
  "outputs": [
    {
      "formula": "<<Robot,Operator>>(paused -> XF mission_resumed)"
    }
  ]
},
```

Reviewer 1 - “ex42”: In my opinion, as it stands, the translation is incomplete with respect to the natural language meaning. I think the formula should resemble that the coalition has a strategy to eventually reach a state where *mission resumed* holds. I propose the following translation: <<Robot,Operator>> (paused -> F mission_resumed)
Reviewer 2: NL contains contextual guards that must be lifted to the logical level, so I agree with Francesco’s opinion about this. Sentence “after a pause” requests contextual conditions, so this one: <<Robot,Operator>> (paused -> F mission_resumed) is more correct.
Reviewer 3: I almost fully agree with your interpretation. Indeed, in natural language the order is reversed in this case, but logically we are talking about an implication. However, the temporal aspect is quite strange here. The information we have is “sooner or later” and "after". Given this information, it seems incorrect to write "paused -> F mission_resumed" since the property would also be satisfied in a state where both paused and mission_resumed are true, without respecting the "after." Therefore, I would say that the correct specification is: paused -> XF mission_resumed.
Reviwer 1: I agree with Reviewer 3.
Reviewer 2: Me too.

```
{
  "id": "ex43",
  "input": "The mobile robot can always guarantee that if the map is invalid, then at the next step it will enter exploration mode.",
  "outputs": [
    {
      "formula": "<<Robot>>G (map_invalid -> X exploration_mode)"
    }
  ]
},

{
  "id": "ex44",
  "input": "The mobile robot can guarantee that it will never enter a restricted zone.",
```

```
    "outputs": [
      {
        "formula": "<<Robot>>G !in_restricted_zone"
      }
    ]
  },
```

Reviewer 1 - “ex44”: See “ex03”
Reviewer 2: see ex03

```
  {
    "id": "ex45",
    "input": "The mobile robot can guarantee that sooner or later it will reach a safe spot if a
fault is detected.",
    "outputs": [
      {
        "formula": "<<Robot>>(fault_detected -> F at_safe_spot)"
      }
    ]
  },
```

Reviewer 1 - “ex45”: See “ex11”.
Reviewer 2: see ex2

```
  {
    "id": "ex46",
    "input": "The Mars rover can guarantee that sooner or later it will reach the sampling site.",
    "outputs": [
      {
        "formula": "<<Rover>>F at_sampling_site"
      }
    ]
  },
  {
    "id": "ex47",
    "input": "The rover can always guarantee that if a wheel slips, then at the next step it will
reduce speed.",
    "outputs": [
      {
        "formula": "<<Rover>>G (wheel_slip -> X speed_low)"
      }
    ]
  },
```

Reviewer 1 - “ex47”: See “ex11”.
Reviewer 2: see ex2

```
 {
   "id": "ex48",
   "input": "The rover can guarantee that it will never enter a high-risk area.",
   "outputs": [
     {
       "formula": "<<Rover>>G !in_high_risk_area"
     }
   ]
 },
```

Reviewer 1 - “ex48”: See “ex03”

```
 {
   "id": "ex49",
   "input": "The rover can guarantee that it will keep scanning the terrain until it finds a safe path.",
   "outputs": [
     {
       "formula": "<<Rover>>(scanning_terrain U safe_path_found)"
     }
   ]
 },
```

Reviewer 2:see ex05

```
 {
   "id": "ex50",
   "input": "The rover can guarantee that if communication with the base is lost, then sooner or later it will activate the backup antenna.",
   "outputs": [
     {
       "formula": "<<Rover>> (comm_lost -> F backup_antenna_on)"
     }
   ]
 },
```

Reviewer 1 - “ex50”: See “ex03”

```
 {
   "id": "ex51",
   "input": "The rover can guarantee that when it is ready to sample, then at the next step it will start sampling.",
   "outputs": [
     {
       "formula": "<<Rover>> (ready_to_sample -> X sampling_started)"
     }
   ]
 },
```

Reviewer 1 - “ex51”: Note that *when* is used as a substitute for *if* in the conditional.

```
 {
   "id": "ex52",
   "input": "The rover can always guarantee that if the battery is low, then it will never start drilling.",
   "outputs": [
     {
       "formula": "<<Rover>>G (battery_low -> G !drilling_started)"
     }
   ]
 },
```

Reviewer 1 - “ex52”: See “ex17”

Reviewer 2: See why I do agree in ex17

```
 {
   "id": "ex53",
   "input": "The rover and the base together can guarantee that sooner or later the sample will be transmitted.",
   "outputs": [
     {
       "formula": "<<Rover,Base>>F sample_transmitted"
     }
   ]
 },
 {
   "id": "ex54",
   "input": "The rover can guarantee that if it detects a sandstorm, then at the next step it will enter shelter mode.",
   "outputs": [
     {
       "formula": "<<Rover>> (sandstorm_detected -> X shelter_mode)"
     }
   ]
 },
 {
   "id": "ex55",
   "input": "The rover can guarantee that sooner or later it will reach an area with good visibility.",
   "outputs": [
     {
       "formula": "<<Rover>>F good_visibility_area"
     }
   ]
 },
 {
```

```
    "id": "ex56",
    "input": "The robotic arm can guarantee that sooner or later it will grasp the object.",
    "outputs": [
      {
        "formula": "<<Arm>>F object_grasped"
      }
    ]
  },
  {
    "id": "ex57",
    "input": "The robotic arm can always guarantee that if the object is fragile, then it will never exceed a maximum force.",
    "outputs": [
      {
        "formula": "<<Arm>>G (fragile_object -> G !force_over_limit)"
      }
    ]
  },
```

Reviewer 1 - “ex57”: See “ex17”

```
  {
    "id": "ex58",
    "input": "The robotic arm can always guarantee that if the object is aligned, then at the next step it will close the gripper.",
    "outputs": [
      {
        "formula": "<<Arm>>G (object_aligned -> X gripper_closed)"
      }
    ]
  },
```

Reviewer 1 - “ex58”: see “ex11”
Reviewer 2: see ex2

```
  {
    "id": "ex59",
    "input": "The robotic arm can guarantee that it will keep aligning until the object is aligned.",
    "outputs": [
      {
        "formula": "<<Arm>>(aligning U object_aligned)"
      }
    ]
  },
  {
    "id": "ex60",
```

```
    "input": "The robotic arm can always guarantee that if it detects a collision, then at the next step it will stop.",
    "outputs": [
      {
        "formula": "<<Arm>>G (collision_detected -> X stopped)"
      }
    ]
  },
```

Reviewer 1 - “ex60”: see “ex11”
Reviewer 2: see ex2

```
  {
    "id": "ex61",
    "input": "The robotic arm can guarantee that if the gripper is closed, then sooner or later the object will be lifted.",
    "outputs": [
      {
        "formula": "<<Arm>>(gripper_closed -> F object_lifted)"
      }
    ]
  },
```

Reviewer 1 - “ex61”: see “ex11”. Perhaps, at this stage of the analysis, it is worth expanding the discussion of the *Globally* operator inferred in the logical form, despite its absence in the natural language sentence. It is not obvious whether *can guarantee that* already embeds a global quantification or whether *Globally* is an additional modeling choice. In my opinion *Globally* could also be an interpretative assumption regarding what kind of *guarantee* is meant. Therefore, to me, the sentence has two readings: (i) at the current state, if the gripper is closed now, the arm has a strategy to eventually lift the object = <<Arm>> (gripper_closed -> F object_lifted); (ii) No matter when the gripper becomes closed, the arm is always able to ensure that the object is eventually lifted = <<Arm>>G (gripper_closed -> F object_lifted).
It also appears that the presence of the *Globally* operator is systematically triggered by conditional constructions. A possible explanation could be that in formal methods and specification languages conditionals are rarely meant as one-off. The issue could be addressed by introducing adverbs or other functional words that explicitly signal the presence of the *Globally* operator, like *always*.
Reviewer 2: I don’t agree. I don’t think it has to be seen as an overgeneralization. This example highlights a genuine semantic ambiguity rather than an error. The natural language sentence admits both a local and a global interpretation, which motivates the need for interactive clarification or multiple candidate specifications. As far as I concern, both formulas capture local capacity and persistent capacity respectively.
Reviewer 3: My opinion in this regard is tied to the expressiveness of the logic. If we are in ATL, it is necessary to have a temporal operator, and consequently, "globally" represents the best choice among all temporal operators. If we are in ATL*, in my view, the best interpretation is to have no temporal operator unless it is explicitly stated in the specification.
Reviewer 2: I align myself with reviewer 3. Case closed.

```
{
  "id": "ex62",
  "input": "The robotic arm can guarantee that it will never enter a singular configuration.",
  "outputs": [
    {
      "formula": "<<Arm>>G !in_singularity"
    }
  ]
},
```

```
{
  "id": "ex63",
  "input": "The arm and the controller together can guarantee that sooner or later the object will be placed into the container.",
  "outputs": [
    {
      "formula": "<<Arm,Controller>>F object_placed"
    }
  ]
},
```

Reviewer 1 - “ex63”: see “ex03”

Reviewer 2: see ex03

```
{
  "id": "ex64",
  "input": "The robotic arm can guarantee that if the target position changes, then at the next step it will start replanning.",
  "outputs": [
    {
      "formula": "<<Arm>>(target_changed -> X replanning)"
    }
  ]
},
```

Reviewer 1 - “ex64”: see “ex60”

```
{
  "id": "ex65",
  "input": "The robotic arm can guarantee that sooner or later it will return to the home position.",
  "outputs": [
    {
      "formula": "<<Arm>>F at_home"
    }
  ]
},
```

```
 {
   "id": "ex66",
   "input": "The collaborative robot can always guarantee that if a human enters the area, then at the next step it will reduce speed.",
   "outputs": [
     {
       "formula": "<<Cobot>>G (human_in_area -> X speed_low)"
     }
   ]
 },
```


Reviewer 1 - “ex66”: see “ex60”


```
 {
   "id": "ex67",
   "input": "The collaborative robot can guarantee that it will never cause a collision with a human.",
   "outputs": [
     {
       "formula": "<<Cobot>>G !human_collision"
     }
   ]
 },
```


Reviewer 1 - "ex67": see “ex03”
Reviewer 2: see ex03


```
 {
   "id": "ex68",
   "input": "The collaborative robot can always guarantee that if the safety barrier is on, then it will always remain stopped.",
   "outputs": [
     {
       "formula": "<<Cobot>>G (safety_barrier_on -> G stopped)"
     }
   ]
 },
 {
   "id": "ex69",
   "input": "The collaborative robot can guarantee that it will keep running the cycle until a stop is requested.",
   "outputs": [
     {
       "formula": "<<Cobot>>(cycle_running U stop_requested)"
     }
   ]
```

```
},

{
  "id": "ex70",
  "input": "The collaborative robot can always guarantee that if a stop is requested, then at the next step it will stop moving.",
  "outputs": [
    {
      "formula": "<<Cobot>>G (stop_requested -> X stopped)"
    }
  ]
},
```

Reviewer 1 - “ex70”: see “ex60”

```
{
  "id": "ex71",
  "input": "The collaborative robot can guarantee that sooner or later it will complete the assigned task.",
  "outputs": [
    {
      "formula": "<<Cobot>>F task_done"
    }
  ]
},
{
  "id": "ex72",
  "input": "The collaborative robot can guarantee that it will never start picking if the grip is not stable.",
  "outputs": [
    {
      "formula": "<<Cobot>>(!grip_stable -> G !pick_started)"
    }
  ]
},
```

Reviewer 1 - “ex72”: Note that, in the natural language specification, the antecedent and the consequent are positionally inverted.

```
{
  "id": "ex73",
  "input": "The robot and the human together can guarantee that sooner or later the part will be assembled.",
  "outputs": [
    {
      "formula": "<<Cobot,Human>>F assembled"
    }
```

],

Reviewer 1 - “ex72”: see “ex03”
Reviewer 2: see ex03

{
"id": "ex74",
"input": "The collaborative robot can always guarantee that if a sensor fails, then at the next step it will enter safe mode.",
"outputs": [
{
"formula": "<<Cobot>>G (sensor_fault -> X safe_mode)"
}
]
},

Reviewer 1 - “ex74”: see “ex60”

{
"id": "ex75",
"input": "The collaborative robot can guarantee that sooner or later it will resume work after a pause.",
"outputs": [
{
"formula": "<<Cobot>>(pause && XF work_resumed)"
}
]
},

Reviewer 1 - “ex75”: *after a pause* missing. Do you have any translation proposals? I see a clear issue, but I don’t want to influence your suggestions. Solved.
Reviewer 2: I propose this translation "<<Cobot>>pause && XF work_resumed"
Reviewer 3: I agree with Reviewer 2.

{
"id": "ex76",
"input": "The surveillance system can always guarantee that if an intruder enters the area, then sooner or later it will detect the intruder.",
"outputs": [
{
"formula": "<<System>>G (intruder_entered -> F intruder_detected)"
}
]
},

Reviewer 1 - “ex76”: see “ex60”

```
{
  "id": "ex77",
  "input": "The system can always  guarantee that if an intruder is detected, then at the next step it will send an alarm.",
  "outputs": [
    {
      "formula": "<<System>>G (intruder_detected -> X alarm_sent)"
    }
  ]
},
```


Reviewer 1 - “ex77”: see “ex60”


```
{
  "id": "ex78",
  "input": "The system can guarantee that there will never be false positives if calibration is valid.",
  "outputs": [
    {
      "formula": "<<System>> (calibration_valid -> G !false_positive)"
    }
  ]
},
```


Reviewer 1 - “ex78”: see “ex60” and “ex75”.


```
{
  "id": "ex79",
  "input": "The system can guarantee that it will keep monitoring until the observation window ends.",
  "outputs": [
    {
      "formula": "<<System>>(monitoring U window_closed)"
    }
  ]
},
{
  "id": "ex80",
  "input": "The system and the operator together can guarantee that sooner or later the incident will be classified correctly.",
  "outputs": [
    {
      "formula": "<<System,Operator>>F incident_classified"
    }
  ]
},
```


Reviewer 1 - “ex80”: see “ex03”

Reviewer 2: see ex03

```
{
  "id": "ex81",
  "input": "The drone can always guarantee that if it enters return mode, then it will never land outside the base.",
  "outputs": [
    {
      "formula": "<<Drone>>G (return_mode -> G !landed_outside_base)"
    }
  ]
},
```

Reviewer 1 - “ex81”: see “ex17”

```
{
  "id": "ex82",
  "input": "The rover can always guarantee that if it finds an interesting sample, then at the next step it will tag the sample.",
  "outputs": [
    {
      "formula": "<<Rover>>G (interesting_sample -> X sample_tagged)"
    }
  ]
},
```

Reviewer 1 - “ex82”: see “ex60”.

```
{
  "id": "ex83",
  "input": "The robotic arm can always guarantee that if the object is grasped, then at the next step it will start transport.",
  "outputs": [
    {
      "formula": "<<Arm>>G (object_grasped -> X transport_started)"
    }
  ]
},
```

Reviewer 1 - “ex83”: see “ex60”.

```
{
  "id": "ex84",
  "input": "The mobile robot can guarantee that if it receives a new goal, then sooner or later it will reach that goal.",
  "outputs": [
    {
      "formula": "<<Robot>> (new_goal -> F at_goal)"
```

```
    }
  ]
},
```


Reviewer 1 - “ex85”: see “ex60”.


```
{
  "id": "ex85",
  "input": "The drone can guarantee that it will keep holding altitude until it receives a descend command.",
  "outputs": [
    {
      "formula": "<<Drone>>(holding_altitude U descend_command)"
    }
  ]
},

{
  "id": "ex86",
  "input": "The rover can guarantee that it will never remain stuck forever in a dune.",
  "outputs": [
    {
      "formula": "<<Rover>>G !stuck_forever"
    }
  ]
},
{
  "id": "ex87",
  "input": "The collaborative robot can guarantee that if the grip fails, then sooner or later it will retry.",
  "outputs": [
    {
      "formula": "<<Cobot>> (grip_failed -> F retry_grip)"
    }
  ]
},
```


Reviewer 1 - “ex87”: see “ex60”


```
{
  "id": "ex88",
  "input": "The robotic arm can guarantee that it will never exceed the torque limit.",
  "outputs": [
    {
      "formula": "<<Arm>> G !torque_over_limit"
    }
```

```
    ]
  },
  {
    "id": "ex89",
    "input": "The mobile robot can always guarantee that when it is near the base, at the next step it will enable docking mode.",
    "outputs": [
      {
        "formula": "<<Robot>>G (near_base -> X docking_mode)"
      }
    ]
  },
```

Reviewer 1 - “ex89”: see “ex60” and “ex75”

```
  {
    "id": "ex90",
    "input": "The drone and the base together can guarantee that sooner or later the mission will be completed.",
    "outputs": [
      {
        "formula": "<<Drone,Base>>F mission_completed"
      }
    ]
  },
```

Reviewer 1 - “ex90”: see “ex03”

Reviewer 2: see ex03

```
  {
    "id": "ex91",
    "input": "The drone can always guarantee that if wind speed is high, then at the next step it will reduce altitude.",
    "outputs": [
      {
        "formula": "<<Drone>>G (wind_high -> X altitude_low)"
      }
    ],
```

Reviewer 1 - “ex91”: see “ex60”

```
  },
  {
    "id": "ex92",
    "input": "The rover can always guarantee that if the temperature is too high, then at the next step it will activate cooling.",
    "outputs": [
```

```
      {
        "formula": "<<Rover>>G (temp_high -> X cooling_on)"
      }
    ]
  },
```

Reviewer 1 - “ex92”: see “ex60”

```
  {
    "id": "ex93",
    "input": "The robotic arm can guarantee that it will keep moving toward the target until it reaches it.",
    "outputs": [
      {
        "formula": "<<Arm>>(moving_to_target U at_target)"
      }
    ]
  },
```

```
  {
    "id": "ex94",
    "input": "The mobile robot can guarantee that if the battery is low, then at the next step it will reduce speed.",
    "outputs": [
      {
        "formula": "<<Robot>>(battery_low -> X reduce_speed)"
      }
    ]
  },
```

Reviewer 1 - “ex94”: see “ex60”

```
  {
    "id": "ex95",
    "input": "The collaborative robot can guarantee that if the human confirms, then at the next step it will start assembly.",
    "outputs": [
      {
        "formula": "human_confirmed -> <<Cobot>>X assembly_started"
      }
    ]
  },
```

Reviewer 1 - “ex95”: see “ex60” and note that the formal translation must not contain an easily inferrable cooperation in the coalition between a human and the collaborative robot.
Reviewer 2: Approved.

```
{
  "id": "ex96",
  "input": "The surveillance system can guarantee that sooner or later it will archive the logs after the shift ends.",
  "outputs": [
    {
      "formula": "<<System>> (shift_ended && XF logs_archived)"
    }
  ]
},
```

Reviewer 1 - “ex96”: *after the shift ends* missing, see “ex75”. Solved.

```
{
  "id": "ex97",
  "input": "The drone can guarantee that if the target is lost, then sooner or later it will return to searching.",
  "outputs": [
    {
      "formula": "<<Drone>> (target_lost -> F searching)"
    }
  ]
},
```

Reviewer 1 - “ex97”: see “ex60”.

```
{
  "id": "ex98",
  "input": "The rover can guarantee that if the slope is too steep, then it will never move forward in that direction.",
  "outputs": [
    {
      "formula": "<<Rover>> (slope_too_high -> G !moving_forward)"
    }
  ]
},
```

Reviewer 1 - “ex98”: see “ex17”.

```
{
  "id": "ex99",
  "input": "The robotic arm can always guarantee that if the force exceeds the limit, then at the next step it will release the object.",
  "outputs": [
    {
      "formula": "<<Arm>>G (force_over_limit -> X object_released)"
    }
  ]
```

```
},
```

Reviewer 1 - “ex99”: see “ex60”

```
{
  "id": "ex100",
  "input": "The mobile robot can guarantee that sooner or later it will deliver the package to
the drop-off point.",
  "outputs": [
    {
      "formula": "<<Robot>>F delivered"
    }
  ]
},


{
  "id": "ex101",
  "input": "The autonomous vehicle can guarantee that sooner or later it will reach the
destination.",
  "outputs": [
    {
      "formula": "<<Vehicle>>F at_destination"
    }
  ]
},
{
  "id": "ex102",
  "input": "The autonomous vehicle can guarantee that it will never exceed the speed limit.",
  "outputs": [
    {
      "formula": "<<Vehicle>>G !speed_over_limit"
    }
  ]
},
{
  "id": "ex103",
  "input": "The autonomous vehicle can guarantee that if a pedestrian is crossing, then at
the next step it will brake.",
  "outputs": [
    {
      "formula": "<<Vehicle>> (pedestrian_crossing -> X braking)"
    }
  ]
},
```

Reviewer 1 - “ex103”: see “ex60”

```
 {
   "id": "ex104",
   "input": "The vehicle can guarantee that it will keep searching for parking until it finds a free spot.",
   "outputs": [
     {
       "formula": "<<Vehicle>>(searching_parking U parking_found)"
     }
   ]
 },
```

Reviewer 1 - “ex104”: see “ex79”

```
 {
   "id": "ex105",
   "input": "The vehicle can always guarantee that if a sensor fails, then sooner or later it will enter safe mode.",
   "outputs": [
     {
       "formula": "<<Vehicle>>G (sensor_fault -> F safe_mode)"
     }
   ]
 },
```

Reviewer 1 - “ex105”: see “ex60”

```
 {
   "id": "ex106",
   "input": "The medical system can guarantee that sooner or later the patient will receive the treatment.",
   "outputs": [
     {
       "formula": "<<MedicalSystem>>F treatment_given"
     }
   ]
 },
 {
   "id": "ex107",
   "input": "The medical system can guarantee that it will never administer the wrong medication.",
   "outputs": [
     {
       "formula": "<<MedicalSystem>>G !wrong_medication"
     }
   ]
 },
 {
   "id": "ex108",
```

```
   "input": "The medical system can guarantee that if the heart rate is abnormal, then at the next step it will send an alarm.",
   "outputs": [
    {
     "formula": "<<MedicalSystem>> (heart_rate_abnormal -> X alarm_sent)"
    }
   ]
  },
```

Reviewer 1 - “ex108”: see “ex60”

```
  {
   "id": "ex109",
   "input": "The medical system can guarantee that it will keep monitoring the patient until the patient is discharged.",
   "outputs": [
    {
     "formula": "<<MedicalSystem>>(monitoring U discharged)"
    }
   ]
  },
```

Reviewer 1 - “ex109”: see “ex79”

```
  {
   "id": "ex110",
   "input": "The doctor and the system together can guarantee that sooner or later a correct diagnosis will be made.",
   "outputs": [
    {
     "formula": "<<Doctor,MedicalSystem>>F diagnosis_done"
    }
   ]
  },
```

Reviewer 1 - “ex110”: see “ex03”
Reviewer 2: see ex03

```
  {
   "id": "ex111",
   "input": "The smart grid can guarantee that sooner or later the load will be balanced.",
   "outputs": [
    {
     "formula": "<<Grid>>F load_balanced"
    }
   ]
  },
  {
```

"id": "ex112",
"input": "The smart grid can guarantee that overload will never occur.",
"outputs": [
{
"formula": "<<Grid>>G !overload"
}
]
},
{
"id": "ex113",
"input": "The smart grid can always guarantee that if demand increases, then at the next step it will increase production.",
"outputs": [
{
"formula": "<<Grid>>G (demand_high -> X production_increased)"
}
]
},

Reviewer 1 - “ex113”: see “ex60”

{
"id": "ex114",
"input": "The smart grid can guarantee that it will keep compensating until the frequency becomes stable.",
"outputs": [
{
"formula": "<<Grid>>(compensating U frequency_stable)"
}
]
},
{
"id": "ex115",
"input": "The grid and the controller together can guarantee that sooner or later power supply will be restored.",
"outputs": [
{
"formula": "<<Grid,Controller>>F power_restored"
}
]
},

Reviewer 1 - “ex115”: see “ex03”
Reviewer 2: see ex03

{
"id": "ex116",

```
    "input": "The satellite can guarantee that sooner or later it will establish a link with the ground station.",
    "outputs": [
      {
        "formula": "<<Satellite>>F link_established"
      }
    ]
  },
  {
    "id": "ex117",
    "input": "The satellite can guarantee that it will never collide with debris.",
    "outputs": [
      {
        "formula": "<<Satellite>>G !collision"
      }
    ]
  },
  {
    "id": "ex118",
    "input": "The satellite can guarantee that if the battery is low, then at the next step it will enable power-saving mode.",
    "outputs": [
      {
        "formula": "<<Satellite>>(battery_low -> X power_save_on)"
      }
    ]
  },
```


Reviewer 1 - “ex118”: see “ex60”


```
  {
    "id": "ex119",
    "input": "The satellite can guarantee that it will keep adjusting attitude until the antenna is aligned.",
    "outputs": [
      {
        "formula": "<<Satellite>>(adjusting_attitude U antenna_aligned)"
      }
    ]
  },
```


Reviewer 1 - “ex119”: see “ex79”


```
  {
    "id": "ex120",
    "input": "The satellite and the ground station together can guarantee that sooner or later the data will be downloaded.",
    "outputs": [
```

{
"formula": "<<Satellite,GroundStation>>F data_downloaded"
}
]
},

Reviewer 1 - “ex120”: see “ex03”
Reviewer 2: see ex03

{
"id": "ex121",
"input": "The financial system can guarantee that sooner or later the transaction will be completed.",
"outputs": [
{
"formula": "<<FinanceSystem>>F transaction_completed"
}
]
},
{
"id": "ex122",
"input": "The financial system can guarantee that an unauthorized transaction will never be executed.",
"outputs": [
{
"formula": "<<FinanceSystem>>G !unauthorized_transaction"
}
]
},
{
"id": "ex123",
"input": "The financial system can always guarantee that if fraud is detected, then at the next step it will block the account.",
"outputs": [
{
"formula": "<<FinanceSystem>>G (fraud_detected -> X account_blocked)"
}
]
},

Reviewer 1 - “ex123”: see “ex60”.

{
"id": "ex124",
"input": "The financial system can guarantee that it will keep verifying until the payment is valid.",
"outputs": [
{

```
      "formula": "<<FinanceSystem>>(verifying U payment_valid)"
    }
  ]
},
```

Reviewer 1 - “ex124”: see “ex79”

```
{
  "id": "ex125",
  "input": "The system and the user together can guarantee that sooner or later the refund will be credited.",
  "outputs": [
    {
      "formula": "<<System,User>>F refund_credited"
    }
  ]
},
```

Reviewer 1 - “ex125”: see “ex03”

```
{
  "id": "ex126",
  "input": "The cybersecurity system can guarantee that sooner or later the attack will be blocked.",
  "outputs": [
    {
      "formula": "<<SecuritySystem>>F attack_blocked"
    }
  ]
},
{
  "id": "ex127",
  "input": "The security system can guarantee that unauthorized access will never occur.",
  "outputs": [
    {
      "formula": "<<SecuritySystem>>G !unauthorized_access"
    }
  ]
},
{
  "id": "ex128",
  "input": "The security system can guarantee that if an intrusion is detected, then at the next step it will isolate the node.",
  "outputs": [
```

```
    {
      "formula": "<<SecuritySystem>>(intrusion_detected -> X node_isolated)"
    }
  ]
},
```

Reviewer 1 - “ex128”: see “ex60”.

```
{
  "id": "ex129",
  "input": "The security system can guarantee that it will keep monitoring until the threat is removed.",
  "outputs": [
    {
      "formula": "<<SecuritySystem>>(monitoring U threat_removed)"
    }
  ]
},
```

Reviewer 1 - “ex129”: see “ex79”

```
{
  "id": "ex130",
  "input": "The system and the administrator together can guarantee that sooner or later security will be restored.",
  "outputs": [
    {
      "formula": "<<SecuritySystem,Admin>>F security_restored"
    }
  ]
},
```

Reviewer 1 - “ex130”: see “ex03”
Reviewer 2: see ex03

```
{
  "id": "ex131",
  "input": "The automated warehouse can guarantee that sooner or later the order will be prepared.",
  "outputs": [
    {
      "formula": "<<Warehouse>>F order_prepared"
    }
  ]
},
{
  "id": "ex132",
  "input": "The automated warehouse can guarantee that it will never pick the wrong item.",
```

"outputs": [
{
"formula": "<<Warehouse>>G !wrong_item_picked"
}
]
},
{
"id": "ex133",
"input": "The warehouse can always guarantee that if a shelf is empty, then at the next step it will request restocking.",
"outputs": [
{
"formula": "<<Warehouse>>G (shelf_empty -> X restock_requested)"
}
]
},

Reviewer 1 - “ex133”: see “ex60”.

{
"id": "ex134",
"input": "The warehouse can guarantee that it will keep sorting until the exit area is clear.",
"outputs": [
{
"formula": "<<Warehouse>>(sorting U exit_clear)"
}
]
},

Reviewer 1 - “ex134”: see “ex79”.

{
"id": "ex135",
"input": "The warehouse and the courier together can guarantee that sooner or later the package will be shipped.",
"outputs": [
{
"formula": "<<Warehouse,Courier>>F package_shipped"
}
]
},

Reviewer 1 - “ex135”: see “ex03”
Reviewer 2: see ex03

{
"id": "ex136",

```
    "input": "The smart city system can guarantee that sooner or later traffic will flow smoothly.",
    "outputs": [
      {
        "formula": "<<CitySystem>>F traffic_flowing"
      }
    ]
  },
  {
    "id": "ex137",
    "input": "The smart city system can guarantee that critical congestion will never occur.",
    "outputs": [
      {
        "formula": "<<CitySystem>>G !critical_congestion"
      }
    ]
  },
  {
    "id": "ex138",
    "input": "The smart city system can guarantee that if an intersection is congested, then at the next step it will retime the traffic lights.",
    "outputs": [
      {
        "formula": "<<CitySystem>>(junction_congested -> X lights_retimed)"
      }
    ]
  },
```

Reviewer 1 - “ex138”: see “ex60”.

```
  {
    "id": "ex139",
    "input": "The smart city system can guarantee that it will keep optimizing until the flow is stable.",
    "outputs": [
      {
        "formula": "<<CitySystem>>(optimizing U flow_stable)"
      }
    ]
  },
```

Reviewer 1 - “ex139”: see “ex79”.

```
  {
    "id": "ex140",
    "input": "The system and the vehicles together can guarantee that sooner or later pollution will be reduced.",
```

```
    "outputs": [
      {
        "formula": "<<CitySystem,Vehicle>>F pollution_reduced"
      }
    ]
  },
```

Reviewer 1 - “ex140”: see “ex03”.
Reviewer 2: see ex03

```
  {
    "id": "ex141",
    "input": "The game system can guarantee that sooner or later the level will be completed.",
    "outputs": [
      {
        "formula": "<<GameSystem>>F level_completed"
      }
    ]
  },

  {
    "id": "ex142",
    "input": "The game system can, but the score monitor cannot, guarantee that a critical bug
will never occur.",
    "outputs": [
      {
        "formula": "<<GameSystem>>G !critical_bug && !<<ScoreMonitor>>G !critical_bug"
      }
    ]
  },
  {
    "id": "ex143",
    "input": "The game system can, but the spectator interface cannot, guarantee that if the
boss spawns, then at the next step the fight will start."
    "outputs": [
      {
        "formula": "<<GameSystem>>(boss_spawned -> X fight_started) &&
!<<SpectatorInterface>>(boss_spawned -> X fight_started)"
      }
    ]
  },
  {
    "id": "ex144",
    "input": "The game system can, but the session logger cannot, guarantee that enemy
spawning will continue until the player becomes inactive.",
    "outputs": [
      {
```

```
      "formula": "<<GameSystem>>(spawning_enemies U player_inactive) &&
!<<SessionLogger>>(spawning_enemies U player_inactive)"
    }
  ]
 },
 {
  "id": "ex145",
  "input": "The game system and the player together can, but the NPC alone cannot,
guarantee that sooner or later an achievement will be unlocked.",
  "outputs": [
    {
      "formula": "<<GameSystem,Player>>F achievement_unlocked && !<<NPC>>F
achievement_unlocked"
    }
  ]
 },
```

Reviewer 1 - “ex145”: see “ex03”

Reviewer 2: see ex03

```
 {
  "id": "ex146",
  "input": "The drone swarm can, but the field operator cannot, guarantee that sooner or
later the area will be fully explored.",
  "outputs": [
    {
      "formula": "<<Swarm>>F area_explored && !<<FieldOperator>>F area_explored"
    }
  ]
 },
```

Reviewer 1 - Right Node Raising.

```
 {
  "id": "ex147",
  "input": "The drone swarm can, but the backup controller cannot, guarantee that global
connectivity will never be lost.",
  "outputs": [
    {
      "formula": "<<Swarm>>G !network_disconnected && !<<BackupController>>G
!network_disconnected"
    }
  ]
 },
 {
  "id": "ex148",
  "input": "The swarm can, but the monitoring station cannot, guarantee that if a drone fails,
then at the next step the formation will be reconfigured.",
  "outputs": [
    {
```

```
        "formula": "<<Swarm>>(drone_failed -> X formation_reconfigured) &&
!<<MonitoringStation>>(drone_failed -> X formation_reconfigured)"
      }
    ]
  },
  {
    "id": "ex149",
    "input": "The swarm can, but the remote operator cannot, guarantee that coordination will
continue until the mission is completed.",
    "outputs": [
      {
        "formula": "<<Swarm>>(coordinating U mission_completed) &&
!<<RemoteOperator>>(coordinating U mission_completed)"
      }
    ]
  },
  {
    "id": "ex150",
    "input": "The swarm and the controller together can, but the operator cannot, guarantee
that sooner or later the required data will be collected.",
    "outputs": [
      {
        "formula": "<<Swarm,Controller>>F data_collected && !<<Operator>>F data_collected"
      }
    ]
  },

  {
    "id": "ex151",
    "input": "The industrial system can, but the floor terminal cannot, guarantee that sooner or
later production will be completed.",
    "outputs": [
      {
        "formula": "<<IndustrySystem>>F production_done && !<<FloorTerminal>>F
production_done"
      }
    ]
  },
  {
    "id": "ex152",
    "input": "The industrial system can, but the external dashboard cannot, guarantee that a
critical failure will never occur.",
    "outputs": [
      {
        "formula": "<<IndustrySystem>>G !critical_failure && !<<ExternalDashboard>>G
!critical_failure"
      }
    ]
```

```
 },
 {
  "id": "ex153",
  "input": "The industrial system can, but the maintenance display cannot, always guarantee
that if a machine overheats, then at the next step it will shut down.",
  "outputs": [
   {
    "formula": "<<IndustrySystem>>G (overheating -> X shutdown) &&
!<<MaintenanceDisplay>>G (overheating -> X shutdown)"
   }
  ]
 },
 {
  "id": "ex154",
  "input": "The industrial system can, but the autonomous warehouse cannot, guarantee
that calibration will continue until quality is acceptable.",
  "outputs": [
   {
    "formula": "<<IndustrySystem>>(calibrating U quality_ok) &&
!<<Warehouse>>(calibrating U quality_ok)"
   }
  ]
 },
 {
  "id": "ex155",
  "input": "The industrial system and the operator together can, but the collaborative robot
alone cannot, guarantee that sooner or later production will be restored.",
  "outputs": [
   {
    "formula": "<<IndustrySystem,Operator>>F production_restored &&
!<<CollaborativeRobot>>F production_restored"
   }
  ]
 },

 {
  "id": "ex156",
  "input": "The climate control system can, but the wall sensor cannot, guarantee that
sooner or later the temperature will return to a stable state.",
  "outputs": [
   {
    "formula": "<<ClimateSystem>>F temperature_stable && !<<WallSensor>>F
temperature_stable"
   }
  ]
 },
 {
  "id": "ex157",
```

```
    "input": "The climate control system can, but the user panel cannot, guarantee that it will never exceed a critical threshold.",
    "outputs": [
      {
        "formula": "<<ClimateSystem>>G !critical_temperature && !<<UserPanel>>G !critical_temperature"
      }
    ]
  },
  {
    "id": "ex158",
    "input": "The climate control system can, but the humidity display cannot, always guarantee that if humidity is high, then at the next step ventilation will be enabled.",
    "outputs": [
      {
        "formula": "<<ClimateSystem>>G (humidity_high -> X ventilation_on) && !<<HumidityDisplay>>G (humidity_high -> X ventilation_on)"
      }
    ]
  },
  {
    "id": "ex159",
    "input": "The climate control system can, but the temperature logger cannot, guarantee that adjustment will continue until comfort is reached.",
    "outputs": [
      {
        "formula": "<<ClimateSystem>>(adjusting U comfort_reached) && !<<TemperatureLogger>>(adjusting U comfort_reached)"
      }
    ]
  },
  {
    "id": "ex160",
    "input": "The climate control system and the controller together can, but the collaborative robot alone cannot, guarantee that sooner or later consumption will be reduced.",
    "outputs": [
      {
        "formula": "<<ClimateSystem,Controller>>F consumption_reduced && !<<CollaborativeRobot>>F consumption_reduced"
      }
    ]
  },

  {
    "id": "ex161",
    "input": "The network system can, but the traffic monitor cannot, guarantee that sooner or later the packet will be delivered.",
    "outputs": [
```

{
"formula": "<<Network>>F packet_delivered && !<<TrafficMonitor>>F packet_delivered"
}
]
},
{
"id": "ex162",
"input": "The network system can, but the diagnostic probe cannot, guarantee that critical packet loss will never occur.",
"outputs": [
{
"formula": "<<Network>>G !critical_packet_loss && !<<DiagnosticProbe>>G !critical_packet_loss"
}
]
},
{
"id": "ex163",
"input": "The network system can, but the logging service cannot, guarantee that if a link fails, then at the next step rerouting will occur.",
"outputs": [
{
"formula": "<<Network>>(link_failed -> X rerouting) && !<<LoggingService>>(link_failed -> X rerouting)"
}
]
},
{
"id": "ex164",
"input": "The network system can, but the packet sniffer cannot, guarantee that transmission will continue until the connection is closed.",
"outputs": [
{
"formula": "<<Network>>(transmitting U connection_closed) && !<<PacketSniffer>>(transmitting U connection_closed)"
}
]
},
{
"id": "ex165",
"input": "The network system and the operator together can, but the collaborative robot alone cannot, guarantee that sooner or later the connection will be restored.",
"outputs": [
{
"formula": "<<Network,Operator>>F connection_restored && !<<CollaborativeRobot>>F connection_restored"
}
]

},

{
"id": "ex166",
"input": "The logistics system can, but the depot display cannot, guarantee that sooner or later the delivery will be completed.",
"outputs": [
{
"formula": "<<LogisticsSystem>>F delivery_done && !<<DepotDisplay>>F delivery_done"
}
]
},
{
"id": "ex167",
"input": "The logistics system can, but the tracking screen cannot, guarantee that it will never lose a package.",
"outputs": [
{
"formula": "<<LogisticsSystem>>G !package_lost && !<<TrackingScreen>>G !package_lost"
}
]
},
{
"id": "ex168",
"input": "The logistics system can, but the route board cannot, always guarantee that if a delay is detected, then at the next step the customer will be notified.",
"outputs": [
{
"formula": "<<LogisticsSystem>>G (delay_detected -> X customer_notified) && !<<RouteBoard>>G (delay_detected -> X customer_notified)"
}
]
},
{
"id": "ex169",
"input": "The logistics system can, but the courier terminal cannot, guarantee that planning will continue until the route is optimal.",
"outputs": [
{
"formula": "<<LogisticsSystem>>(planning U route_optimal) && !<<CourierTerminal>>(planning U route_optimal)"
}
]
},
{
"id": "ex170",

```
    "input": "The logistics system and the courier together can, but the collaborative robot
alone cannot, guarantee that sooner or later the package will be delivered.",
    "outputs": [
      {
        "formula": "<<LogisticsSystem,Courier>>F package_delivered &&
!<<CollaborativeRobot>>F package_delivered"
      }
    ]
  },

  {
    "id": "ex171",
    "input": "The education system can, but the attendance register cannot, guarantee that
sooner or later the student will complete the course.",
    "outputs": [
      {
        "formula": "<<EduSystem>>F course_completed && !<<AttendanceRegister>>F
course_completed"
      }
    ]
  },
  {
    "id": "ex172",
    "input": "The education system can, but the grading dashboard cannot, guarantee that it
will never issue an invalid certificate.",
    "outputs": [
      {
        "formula": "<<EduSystem>>G !invalid_certificate && !<<GradingDashboard>>G
!invalid_certificate"
      }
    ]
  },
  {
    "id": "ex173",
    "input": "The education system can, but the student portal cannot, guarantee that if a
student fails a test, then at the next step remediation will be proposed.",
    "outputs": [
      {
        "formula": "<<EduSystem>>(test_failed -> X recovery_proposed) &&
!<<StudentPortal>>(test_failed -> X recovery_proposed)"
      }
    ]
  },
  {
    "id": "ex174",
    "input": "The education system can, but the exam archive cannot, guarantee that
evaluation will continue until competence is reached.",
    "outputs": [
```

{
"formula": "<<EduSystem>>(evaluating U competence_reached) &&
!<<ExamArchive>>(evaluating U competence_reached)"
}
]
},
{
"id": "ex175",
"input": "The education system and the teacher together can, but the AI tutor alone
cannot, guarantee that sooner or later the student will be certified.",
"outputs": [
{
"formula": "<<EduSystem,Teacher>>F certified && !<<AITutor>>F certified"
}
]
},

{
"id": "ex176",
"input": "The agricultural system can, but the moisture display cannot, guarantee that
sooner or later irrigation will be completed.",
"outputs": [
{
"formula": "<<AgriSystem>>F irrigation_done && !<<MoistureDisplay>>F
irrigation_done"
}
]
},
{
"id": "ex177",
"input": "The agricultural system can, but the field sensor cannot, guarantee that it will
never waste water.",
"outputs": [
{
"formula": "<<AgriSystem>>G !water_waste && !<<FieldSensor>>G !water_waste"
}
]
},
{
"id": "ex178",
"input": "The agricultural system can, but the weather station cannot, guarantee that if the
soil is dry, then at the next step irrigation will be enabled.",
"outputs": [
{
"formula": "<<AgriSystem>>(soil_dry -> X irrigation_on) &&
!<<WeatherStation>>(soil_dry -> X irrigation_on)"
}
]

},
{
"id": "ex179",
"input": "The agricultural system can, but the irrigation display cannot, guarantee that monitoring will continue until humidity is acceptable.",
"outputs": [
{
"formula": "<<AgriSystem>>(monitoring U humidity_ok) && !<<IrrigationDisplay>>(monitoring U humidity_ok)"
}
]
},
{
"id": "ex180",
"input": "The agricultural system and the farmer together can, but the agricultural robot alone cannot, guarantee that sooner or later the harvest will be saved.",
"outputs": [
{
"formula": "<<AgriSystem,Farmer>>F harvest_saved && !<<AgriRobot>>F harvest_saved"
}
]
},

{
"id": "ex181",
"input": "The emergency system can, but the public dashboard cannot, guarantee that sooner or later rescue teams will arrive.",
"outputs": [
{
"formula": "<<EmergencySystem>>F rescue_arrived && !<<PublicDashboard>>F rescue_arrived"
}
]
},
{
"id": "ex182",
"input": "The emergency system can, but the call log cannot, guarantee that it will never ignore a request for help.",
"outputs": [
{
"formula": "<<EmergencySystem>>G !help_ignored && !<<CallLog>>G !help_ignored"
}
]
},
{
"id": "ex183",

```
  "input": "The emergency system can, but the incident board cannot, guarantee that if an
alarm is active, then at the next step a team will be dispatched.",
  "outputs": [
   {
    "formula": "<<EmergencySystem>>(alarm_active -> X team_dispatched) &&
!<<IncidentBoard>>(alarm_active -> X team_dispatched)"
   }
  ]
 },
 {
  "id": "ex184",
  "input": "The emergency system can, but the radio display cannot, guarantee that
coordination will continue until the incident is resolved.",
  "outputs": [
   {
    "formula": "<<EmergencySystem>>(coordinating U incident_resolved) &&
!<<RadioDisplay>>(coordinating U incident_resolved)"
   }
  ]
 },
 {
  "id": "ex185",
  "input": "The emergency system and the operators together can, but the emergency robot
alone cannot, guarantee that sooner or later the area will be secured.",
  "outputs": [
   {
    "formula": "<<EmergencySystem,Operators>>F area_secured &&
!<<EmergencyRobot>>F area_secured"
   }
  ]
 },

 {
  "id": "ex186",
  "input": "The control system can, but the observation panel cannot, guarantee that sooner
or later the process will reach a steady state.",
  "outputs": [
   {
    "formula": "<<ControlSystem>>F steady_state && !<<ObservationPanel>>F
steady_state"
   }
  ]
 },
 {
  "id": "ex187",
  "input": "The control system can, but the monitoring display cannot, guarantee that it will
never cause instability.",
  "outputs": [
```

```
    {
      "formula": "<<ControlSystem>>G !unstable && !<<MonitoringDisplay>>G !unstable"
    }
  ]
},
{
  "id": "ex188",
  "input": "The control system can, but the diagnostic screen cannot, guarantee that if the error increases, then at the next step compensation will be applied.",
  "outputs": [
    {
      "formula": "<<ControlSystem>>(error_high -> X compensation_applied) && !<<DiagnosticScreen>>(error_high -> X compensation_applied)"
    }
  ]
},
{
  "id": "ex189",
  "input": "The control system can, but the status logger cannot, guarantee that regulation will continue until the error becomes zero.",
  "outputs": [
    {
      "formula": "<<ControlSystem>>(regulating U error_zero) && !<<StatusLogger>>(regulating U error_zero)"
    }
  ]
},
{
  "id": "ex190",
  "input": "The control system and the supervisor together can, but the clerk alone cannot, guarantee that sooner or later the system will be stabilized.",
  "outputs": [
    {
      "formula": "<<ControlSystem,Supervisor>>F stabilized && !<<Clerk>>F stabilized"
    }
  ]
},
{
  "id": "ex191",
  "input": "The recommender system can, but the preference viewer cannot, guarantee that sooner or later it will suggest relevant content.",
  "outputs": [
    {
      "formula": "<<Recommender>>F relevant_content && !<<PreferenceViewer>>F relevant_content"
    }
  ]
},
```

```
  {
    "id": "ex192",
    "input": "The recommender system can, but the catalogue browser cannot, guarantee that it will never suggest forbidden content.",
    "outputs": [
      {
        "formula": "<<Recommender>>G !forbidden_content && !<<CatalogueBrowser>>G !forbidden_content"
      }
    ]
  },
  {
    "id": "ex193",
    "input": "The recommender system can, but the user profile page cannot, always guarantee that if preferences change, then at the next step the model will be updated.",
    "outputs": [
      {
        "formula": "<<Recommender>>G (preferences_changed -> X model_updated) && !<<UserProfilePage>>G (preferences_changed -> X model_updated)"
      }
    ]
  },
  {
    "id": "ex194",
    "input": "The recommender system can, but the click logger cannot, guarantee that learning will continue until accuracy improves.",
    "outputs": [
      {
        "formula": "<<Recommender>>(learning U accuracy_improved) && !<<ClickLogger>>(learning U accuracy_improved)"
      }
    ]
  },
  {
    "id": "ex195",
    "input": "The recommender system and the user together can, but the preference viewer alone cannot, guarantee that sooner or later a good recommendation will be found.",
    "outputs": [
      {
        "formula": "<<Recommender,User>>F good_recommendation && !<<PreferenceViewer>>F good_recommendation"
      }
    ]
  },
  {
    "id": "ex196",
    "input": "The management system can, but the request dashboard cannot, guarantee that sooner or later the request will be fulfilled.",
```

"outputs": [
{
"formula": "<<ManagementSystem>>F request_fulfilled && !<<RequestDashboard>>F request_fulfilled"
}
]
},
{
"id": "ex197",
"input": "The management system can, but the intake form cannot, guarantee that it will never reject a valid request.",
"outputs": [
{
"formula": "<<ManagementSystem>>G !valid_request_rejected && !<<IntakeForm>>G !valid_request_rejected"
}
]
},
{
"id": "ex198",
"input": "The management system can, but the planning board cannot, guarantee that if a priority changes, then at the next step the schedule will be updated.",
"outputs": [
{
"formula": "<<ManagementSystem>>(priority_changed -> X schedule_updated) && !<<PlanningBoard>>(priority_changed -> X schedule_updated)"
}
]
},
{
"id": "ex199",
"input": "The management system can, but the resource monitor cannot, guarantee that planning will continue until resources are exhausted.",
"outputs": [
{
"formula": "<<ManagementSystem>>(planning U resources_exhausted) && !<<ResourceMonitor>>(planning U resources_exhausted)"
}
]
},
{
"id": "ex200",
"input": "The management system and the manager together can, but the collaborative robot alone cannot, guarantee that sooner or later the goal will be reached.",
"outputs": [
{
"formula": "<<ManagementSystem,Manager>>F goal_reached && !<<CollaborativeRobot>>F goal_reached"

}
]
}

The Art of War by Sun Tzu (MIT Classics Archive) - **Chapter VI, Weak Points and Strong, verse 11:** : https://classics.mit.edu/Tzu/artwar.html

{
"id": "ex201",
"input": "If we wish to fight, the enemy can be forced to an engagement even though he be sheltered behind a high rampart and a deep ditch. All we need do is attack some other place that he will be obliged to relieve.",
"outputs": [
{
"formula": "<<We>>(wish_to_fight -> F (attack_other_place && F engagement))"
}
]
},
Reviewer 3: Modified and used Until: <<We>> sheltered_high_rampart && sheltered_deep_ditch U attack_other_place
Reviewer 2: I don't agree, I think this is the correct translation: "(wish_to_fight && enemy_sheltered_behind_rampart && enemy_sheltered_behind_ditch) -> <<We>>F engagement"
Reviewer 1: I agree with Reviewer 2. Maybe "wish_to_fight", triggered by "if" (conditional), should be in the coalition scope?
Reviewer 2: Ok, what about this? "<<We>>(wish_to_fight -> F (attack_other_place && F engagement))"
Reviewer 1: Approved

The Art of War by Sun Tzu (MIT Classics Archive) - **Chapter VI, Weak Points and Strong, verse 12:** : https://classics.mit.edu/Tzu/artwar.html

{
"id": "ex202",
"input": "If we do not wish to fight, we can prevent the enemy from engaging us even though the lines of our encampment be merely traced out on the ground. All we need do is to throw something odd and unaccountable in his way.",
"outputs": [
{
"formula": "<<We>>(!wish_to_fight -> F (throw_something_odd_in_his_way && G !enemy_engages_us))"
}
]
},
Original formula discussed: "weak_defense -> <<We>> G !engaged"
Reviewer 3: For me the "weak_defense -> <<We>> G !engaged" it's ok.

Reviewer 2: I don't agree. This is the version I propose "<<We>>(!wish_to_fight -> F (throw_something_odd_in_his_way && G !enemy_engages_us))"

The Art of War by Sun Tzu (MIT Classics Archive) - **Chapter III, Attack by Stratagem, verses 17**: https://classics.mit.edu/Tzu/artwar.html

```
{
   "id": "ex203",
   "input": "He will win whose army is animated by the same spirit throughout all its ranks.",
   "outputs": [
     {
       "formula": "same_spirit_in_ranks  ->  <<Army>>F achieve_victory"
     }
   ]
 },
```

Reviewer 3: If "same_spirit" means same strategy, then we need Strategy Logic. Having just ATL, the formula is not precise, but fine for me.

The Prince by Niccolò Machiavelli (Project Gutenberg) - **Chapter III (Mixed Principalities)**: https://www.gutenberg.org/files/1232/1232-h/1232-h.htm

```
{
   "id": "ex204",
   "input": "The Romans, in the countries which they annexed, observed closely these measures; they sent colonies and maintained friendly relations with the minor powers, without increasing their strength; they kept down the greater, and did not allow any strong foreign powers to gain authority.",
   "outputs": [
     {
       "formula": "<<Romans>>G(send_colony && friendly_with_minor_powers && keep_down_the_greater )"
     }
   ]
 },
```

Reviewer 3: Ok. In case we want to stick with ATL, than it is fine.
Reviewer 1: Excellent.
Reviewer 2: I agree.

The Prince by Niccolò Machiavelli (Project Gutenberg) - **Chapter XXV (What Fortune Can Effect):** https://www.gutenberg.org/files/1232/1232-h/1232-h.htm

```
{
   "id": "ex205",
```

"input": "I conclude therefore that, fortune being changeful and mankind steadfast in their ways, so long as the two are in agreement men are successful, but unsuccessful when they fall out. For my part I consider that it is better to be adventurous than cautious, because fortune is a woman, and if you wish to keep her under it is necessary to beat and ill-use her; and it is seen that she allows herself to be mastered by the adventurous rather than by those who go to work more coldly.",
"outputs": [
{
"formula": "<<Men>>(G ((aligned_with_fortune) -> success) && G ((out_of_agreement) -> unsuccessful) && G (wish_to_master_fortune -> F adventurous) && G (adventurous -> F master_fortune))"
}
],
G ((aligned_with_fortune) -> success) &&
G ((out_of_agreement) -> unsuccessful) &&
G (wish_to_master_fortune -> F adventurous) &&
G (adventurous -> F master_fortune)
)”
},

Reviewer 1 - “ex205”: I recognize that this paragraph is misogynistic. If it poses a potential issue for anyone, we can eliminate it from the dataset.
Reviewer 2 - You cannot give an entire period wishing to translate just a part out of it? How does the llm understands this preference?
Reviewer 1 - “ex205”: It is interesting to see if it can infer what needs to be translated. Non expert users could add additional information not related to the translation due to ignorance or lack of confidence.
Reviewer 3: It is ok but can be improved. Fortune/fate is usually adversarial. To win, then it is required that fate is always good for men (the match).

Reviewer2: I would just write: <<Men>>(
G ((aligned_with_fortune) -> success) &&
G ((out_of_agreement) -> unsuccessful) &&
G (wish_to_master_fortune -> F adventurous) &&
G (adventurous -> F master_fortune)
)
“
Reviewer 1 Ok

U.S. Constitution (National Archives) - **Article II, Section 3**: The Constitution of the United States: A Transcription | National Archives

{
"id": "ex206",
"input": "He [the President] shall from time to time give to the Congress Information of the State of the Union, and recommend to their Consideration such Measures as he shall judge necessary and expedient.",
"outputs": [

{
"formula": "<<President>>F(give_info && reccommend)"
}
]
},
Reviewer 1: Perfect.
Reviewer2:Ok

Moby Dick: or The Whale by Herman Melville (Project Gutenberg) - **Chapter 36. The Quarter Deck**: Moby Dick; or The Whale | Project Gutenberg

{
"id": "ex207",
"input": "Aye, aye! and I'll chase him round Good Hope, and round the Horn, and round the Norway Maelstrom, and round perdition's flames before I give him up. And this is what ye have shipped for, men! to chase that white whale on both sides of land, and over all sides of earth, till he spouts black blood and rolls fin out.",
"outputs": [
{
"formula": "<<I>>(chase_round_Good_Hope && chase_round_the_Horn && chase_round_the_Norway_Maelstrom && chase_round_perditions_flame && chase_both_sides_of_land && chase_over_all_sides_of_earth U enemy_spouts_black_blood && enemy_rolls_fin_out)"
}
]
},
Reviewer 3: It is ok.

The North Atlantic Treaty (1949), **Article 5** : Collective defence and Article 5 | NATO Topic

{
"id": "ex208",
"input": "The Parties agree that an armed attack against one or more of them in Europe or North America shall be considered an attack against them all and consequently they agree that, if such an armed attack occurs, each of them, in exercise of the right of individual or collective self-defence recognised by Article 51 of the Charter of the United Nations, will assist the Party or Parties so attacked by taking forthwith, individually and in concert with the other Parties, such action as it deems necessary, including the use of armed force, to restore and maintain the security of the North Atlantic area.",
"outputs": [
{
"formula": "<<Parties>>(armed_attack_against_one_or_more_in_europe_or_north_america -> F(attack_against_all && F (each_party_assists_attacked_parties && F security_restored && G security_maintained)))"
}
],

&& F (each_party_assists_attacked_parties && F security_restored && G security_maintained)))"

},
Reviewer 3: It is ok.
Reviewer 2: For me it is something like: <<Parties>>(armed_attack_against_one_or_more_in_europe_or_north_america -> F(attack_against_all && F (each_party_assists_attacked_parties && F security_restored && G security_maintained)))

The Art of War by Sun Tzu (MIT Classics Archive) - **Chapter VII: Maneuvering, verses 65 - 68:** : https://classics.mit.edu/Tzu/artwar.html

```
{
  "id": "ex209",
  "input": "If the enemy leaves a door open, you must rush in. Forestall your opponent by seizing what he holds dear, and subtly contrive to time his arrival on the ground. Walk in the path defined by rule, and accommodate yourself to the enemy until you can fight a decisive battle.",
  "outputs": [
    {
      "formula": "<<You>> (enemy_leaves_door_open -> G rush_in && !enemy_leaves_door_open -> idle U decisive_battle)"
    }
  ]
},
```

Reviewer 3: It is ok.
Reviewer 1: Again, “must” seems far from being associated with “G” to me, I agree that is inferrable but I don’t believe that it is a direct translation, the fact that I must do something doesn’t follow the fact that that thing has to be done globally, it would be more natural to use F or FX...

Runaround by Isaac Asimov: [Runaround]() modified from the original “A robot may not injure a human being or, through inaction, allow a human being to come to harm”.

```
{
  "id": "ex210",
  "input": "A robot always not injure a human being or, through inaction, allow a human being to come to harm.",
  "outputs": [
    {
      "formula": "<<Robot>> G (!injure && !allow_harm)"
    }
  ]
},
```

Reviewer 3: It is ok.
]
The Merchant of Venice by William Shakespear (Act 1, Scene 3): [The Merchant of Venice - Act 1, scene 3 | Folger Shakespeare Library]()

{
"id": "ex211",
"input": "Shylock : Go with me to a notary, seal me there your single bond; and, in a merry sport, if you repay me not on such a day, in such a place, such sum or sums as are expressed in the condition, let the forfeit be nominated for an equal pound of your fair flesh, to be cut off and taken in what part of your body pleaseth me.",
"outputs": [
{
"formula": "<<Shylock>>(bond_sealed && (!repayment -> F pound_of_flesh))"
}
]
},

Original Output: <<Shylock>>F(!repayment -> pound_of_flesh)
Reviewer 3: it is ok, but more precise would have been: <<Shylock>>F((such_a_day && !repayment) -> pound_of_flesh). Another possibility:<<Shylock>>(F repayment) || (G !repayment -> F pound_of_flesh)
Reviewer 1 - “ex211”: I agree with Reviewer 3.
Reviewer 2 - I think it misses the bond sealing aspect, actually this would be better: “<<Shylock>>(bond_sealed && (!repayment -> F pound_of_flesh) && (pound_of_flesh -> F flesh_cut_off))“
Reviewer 1 - “ex211”: Maybe without the final conjunct “[...] && (pound_of_flesh -> F flesh_cut_off))”

Brave New World by Aldous Huxley: Brave New World by Aldous Huxley, from Project Gutenberg Canada

{
"id": "ex212",
"input": "The world’s stable now. People are happy; they get what they want, and they never want what they can’t get. They’re well off; they’re safe; they’re never ill; they’re not afraid of death; they’re blissfully ignorant of passion and old age; they’re plagued with no mothers or fathers; they’ve got no wives, or children, or lovers to feel strongly about; they’re so conditioned that they practically can’t help behaving as they ought to behave.",
"outputs": [
{
"formula": "<<World>>(G stable && G (happy && get_what_they_want && !want_what_they_cannot_get && well_off && safe && !ill && !afraid_of_death && ignorant_of_passion && ignorant_of_old_age && !plagued_by_mothers_or_fathers && !have_wives && !have_children && !have_lovers_to_feel_strongly_about && conditioned_to_behave_as_they_ought))"
}
]
},
Original Output: <<World>>G stable
Reviewer 3: it is ok.

Reviewer 2: It lacks of all details after the stability clause, i.e.: “<<World>>G stable && G (happy && get_what_they_want && !want_what_they_cannot_get && well_off && safe && !ill && !afraid_of_death && ignorant_of_passion && ignorant_of_old_age && !plagued_by_mothers_or_fathers && !have_wives && !have_children && !have_lovers_to_feel_strongly_about && conditioned_to_behave_as_they_ought)”
Reviewer 1: Perfect. Note that “stable now” triggers “Globally”.

Antigone by Sophocles: [Antigone : Sophocles : Free Download, Borrow, and Streaming : Internet Archive]

{
  "id": "ex213",
  "input": "I will bury him myself. And even if I die in the act, that death will be a glory. I will rest with the one I love, having broken the law for a holy cause. I must please the dead far longer than the living, for with them I shall dwell forever.",
  "outputs": [
    {
      "formula": "<<I>>(F bury_him && break_law_for_holy_cause && G (die_in_act -> glory) && F rest_with_loved_one && G please_dead_longer_than_living && G dwell_with_dead)"
    }
  ],
"
 },
Original Output: "<<I>>(F bury && G(death -> glory))"
Reviewer 2: It misses a lot here, take a look: <<I>>(F bury && G (die_in_act -> glory) && F rest_with_loved_one && F break_law_for_holy_cause && G please_dead_longer_than_living && G dwell_with_dead_forever)
Reviewer 1: “having broken the law for a holy cause” with an F? I think that it should be added as a conjunct to “bury”: <<I>>(F bury-him && break_law_for_holy_cause && G (die_in_act -> glory) && F rest_with_loved_one && G please_dead_longer_than_living && G dwell_with_dead)
Reviewer 2: Ok good.

The Trial by Franz Kafka: [The Project Gutenberg eBook of The Trial, by Franz Kafka—A Project Gutenberg eBook]

{
  "id": "ex214",
  "input": “Nobody else could have got in this way, as this entrance was meant only for you [K.]. Now I'll go and close it.”,
  "outputs": [
    {
      "formula": "<<K>> (F enter && X entrance_closed)"
    }
  ]
 },
Original Output: "<<K>>F enter”

Reviewer 2: Again, it's not completed, it should be "<<K>>(F enter && G !others_enter_this_way && F entrance_closed)"

Reviewer 1: I do not agree due to the fact that "nobody else" is literally a translation of "no matter the others do". I agree with the insert of "&& F entrance_closed" but with a X: && X entrance_closed)

Reviewer 2: ok

The Odyssey by Homer - BOOK XII, THE SIRENS, SCYLLA AND CHARYBDIS, THE CATTLE OF THE SUN: [The Project Gutenberg eBook of The Odyssey, by Homer]

{
  "id": "ex215",
  "input": "Therefore, take me and bind me to the crosspiece half way up the mast; bind me as I stand upright, with a bond so fast that I cannot possibly break away, and lash the rope's ends to the mast itself. If I beg and pray you to set me free, then bind me more tightly still.",
  "outputs": [
    {
      "formula": "<<I>> (G(binded && !break_away && F(praying -> bind_tightly)))"
    }
  ]
},

The Odyssey by Homer - BOOK I: [The Project Gutenberg eBook of The Odyssey, by Homer]

{
  "id": "ex216",
  "input": "he [Odysseus] is a man of many resources, and even though he were in chains of iron he would find some way of getting home again.",
  "outputs": [
    {
      "formula": "chains_of_iron -> <<Odysseus>>F(reach_home)"
    }
  ]
},

The Iliad by Homer - BOOK IX: [The Project Gutenberg eBook of The Iliad, by Homer]

{
  "id": "ex217",
  "input": "My mother Thetis tells me that there are two ways in which I [Achilles] may meet my end. If I stay here and fight, I shall not return alive but my name will live for ever:",
  "outputs": [
    {

"formula": "<<Achilles>>F(is_here && fight -> die && G name_lives)"
}
]
},
Reviewer 2: Ok

Inspired by **The Iliad by Homer - BOOK IX**: The Project Gutenberg eBook of The Iliad, by Homer

{
"id": "ex218",
"input": "If I choose to go home, I can always guarantee that my name will die and that death will come to me only much later.",
"outputs": [
{
"formula": "<<I>>(go_home -> G (name_dies && delayed_death))"
}
]
},
Reviewer 2: Ok

{
"id": "ex219",
"input": "My mother Thetis tells me that there are two ways in which I [Achilles] may meet my end. If I stay here and fight, I shall not return alive but my name will live for ever: whereas if I go home my name will die, but it will be long ere death shall take me.",
"outputs": [
{
"formula": "<<Achilles>>G(stay_here && fight -> die && name_lives) || <<Achilles>>(go_home -> name_dies && live_longer)"
}
]
}
Reviewer 2: Ok

The Bible - Psalm 138:8: Psalm 138 ESV

{
"id": "ex220",
"input": "The Lord will fulfill his purpose for me.",
"outputs": [
{
"formula": "<<Lord>> F fulfilled"
}
]
},

{
"id": "ex221",
"input": "The robot number 1 has a strategy to ensure that eventually reach position 1.",
"outputs": [
{
"formula": "<<Robot1>> F pos1"
}
]
}

{
"id": "ex222",
"input": "The robot number 2 has a strategy to ensure that globally reach position 2.",
"outputs": [
{
"formula": "<<Robot2>> G pos2"
}
]
},

{
"id": "ex223",
"input": "The robot number 3 has a strategy to guarantee that they will reach position 2 in the next step.",
"outputs": [
{
"formula": "<<Robot3>> X pos2"
}
]
}

{
"id": "ex224",
"input": "The robots number 1 and number 2 do not have a strategy to ensure that they will reach position 2 in the next step.",
"outputs": [
{
"formula": "!<<Robot1, Robot2>> X pos2"
}
]
},

{
"id": "ex225",
"input": "The robot number 1 has a strategy to guarantee that eventually it will reach position 1 and not position 2.",
"outputs": [

```
      {
        "formula": "<<Robot1>> F (pos1 && !pos2)"
      }
    ]
  }

{
    "id": "ex226",
    "input": "The robot number 2 has a strategy to Globally reach position 1 and not position 2
and not position 3.",
    "outputs": [
      {
        "formula": "<<Robot2>> G (pos1 && pos2 && !pos3)"
      }
    ]
},

{
    "id": "ex227",
    "input": "Eventually, The robot number 3 has a strategy to guarantee that it will reach
position 2 if it reaches position 1",
    "outputs": [
      {
        "formula": "<<Robot3>> F (pos1 -> pos2)"
      }
    ]
  },
```

Reviewer 1 - “ex227”: Note the right dislocation of ‘Eventually’.

```
{
"id": "ex228",
"input": "The robot number 1 has a strategy to guarantee that it will eventually reach position
1 or position 2.",
    "outputs": [
      {
        "formula": "<<Robot1>> F (pos1 || pos2)"
      }
    ]
},

{
"id": "ex229",
"input": "Robots number 1 and number 2 have a joint strategy to always avoid position 3.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot2>> G !pos3"
      }
```

```
    ]
},

{
"id": "ex230",
"input": "No coalition of robots 2 and 3 has a strategy to guarantee that they will eventually
reach position 1.",
    "outputs": [
      {
        "formula": "!<<Robot2, Robot3>> F pos1"
      }
    ]
},

{
"id": "ex231",
"input": "The robot number 1 has a strategy to ensure garbage collection until garbage
retirement is commanded.",
    "outputs": [
      {
        "formula": "<<Robot1>> (garbage_collection U garbage_retirement)"
      }
    ]
},

{
"id": "ex232",
"input": "The robot number 2 has a strategy to always eventually reach position 3.",
    "outputs": [
      {
        "formula": "<<Robot2>> G F pos3"
      }
    ]
},
{
"id": "ex233",
"input": "Robots number 1, number 2 and number 3 have a strategy to ensure that they will
reach either position 1 or position 2 in the next step.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot2, Robot3>> X (pos1 || pos2)"
      }
    ]
},
{
"id": "ex234",
"input": "The robot number 3 does not have a strategy to guarantee position 1 holds
globally.",
```

"outputs": [
{
"formula": "!<<Robot3>> G pos1"
}
]
},
{
"id": "ex235",
"input": "The robot number 1 has a strategy to always ensure that if it is in position 2 then it eventually reaches position 3.",
"outputs": [
{
"formula": "<<Robot1>> G (pos2 -> F pos3)"
}
]
},

Reviewer 1 - “ex235”: Check this, please.
Reviewer 3: It seems perfectly specified to me.

{
"id": "ex236",
"input": "Robots number 2 and number 3 have a strategy to guarantee that they will eventually reach position 1 and eventually reach position 2.",
"outputs": [
{
"formula": "<<Robot2, Robot3>> (F pos1 && F pos2)"
}
]
},
{
"id": "ex237",
"input": "The robot number 1 has a strategy to guarantee that it will keep position 1 until move is true and while not attack.",
"outputs": [
{
"formula": "<<Robot1>> (pos1 U (move && !attack))"
}
]
},
{
"id": "ex238",
"input": "Robots number 1 and number 3 do not have a strategy to ensure that always position 1 or position 2 holds.",
"outputs": [
{
"formula": "!<<Robot1, Robot3>> G (pos1 || pos2)"
}

]
},
{
"id": "ex239",
"input": "The robot number 2 has a strategy to ensure that in the next step it does not move in position 3.",
"outputs": [
{
"formula": "<<Robot2>> X !pos3"
}
]
},
{
"id": "ex240",
"input": "Eventually, robots number 1 and number 2 have a strategy to ensure the reaching of position 3.",
"outputs": [
{
"formula": "<<Robot1, Robot2>> F pos3"
}
]
},

Reviewer 1 - “ex240”: see “ex227”.

{
"id": "ex241",
"input": "The robot number 3 has a strategy to guarantee that either it always keeps position 1 or eventually it reaches position 2.",
"outputs": [
{
"formula": "<<Robot3>> (G pos1 || F pos2)"
}
]
},

{
"id": "ex242",
"input": "Robots number 1 and number 2 have a strategy to guarantee that either in the next step attack holds or eventually defend holds.",
"outputs": [
{
"formula": "<<Robot1, Robot2>> (X attack || F defend)"
}
]
},
{
"id": "ex243",

```
"input": "None of the robots number 1, 2, or 3 has a strategy to ensure that position 1 is globally avoided.",
   "outputs": [
    {
     "formula": "!<<Robot1>> (G !pos1 && !<<Robot2>> G !pos1 && !<<Robot3>> G !pos1)"
    }
   ]
},
```

Reviewer 1 - “ex243”: ‘Avoided’ is semantically interpreted relative to the negation at position 1, with the initial ‘none’ serving as the coalition's negating operator. Please, check if the difficulty is right.

```
{
"id": "ex244",
"input": "The robots number 1 and number 2 have a strategy to ensure that eventually position 1 holds and always position 2 holds thereafter.",
   "outputs": [
    {
     "formula": "<<Robot1, Robot2>> F (pos1 && X G pos2)"
    }
   ]
},
```

Reviewer 1 - “ex244”: The formula does not efficiently capture thereafter. Something like that could: <<Robot1, Robot2>> F (pos1 && X G pos2). Should this problem be addressed as a hard one?

```
{
"id": "ex245",
"input": "The robot number 1 has a strategy to guarantee that eventually it reaches position 2 without ever reaching position 3.",
   "outputs": [
    {
     "formula": "<<Robot1>> F (pos2 && G !pos3)"
    }
   ]
},

{
"id": "ex246",
"input": "The robot number 2 has a strategy to ensure that if position 1 eventually holds then position 3 will eventually hold too.",
   "outputs": [
    {
     "formula": "<<Robot2>> (F pos1 -> F pos3)"
    }
```

```
    ]
},

{
"id": "ex247",
"input": "Robots number 1 and number 3 have a strategy to ensure that defend holds until attack holds and then always attack holds.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot3>> ((defend U attack) && G attack)"
      }
    ]
},
{
"id": "ex248",
"input": "The robot number 3 does not have a strategy to ensure that in the next state either position 1 or position 2 are valid.",
    "outputs": [
      {
        "formula": "!<<Robot3>> X (pos1 || pos2)"
      }
    ]
},
{
"id": "ex249",
"input": "The robot number 1 has a strategy to always avoid position 2 or eventually reach position 3.",
    "outputs": [
      {
        "formula": "<<Robot1>> (G !pos2 || F pos3)"
      }
    ]
},
{
"id": "ex250",
"input": "Robots number 2 and number 3 have no strategy to guarantee that position 1 holds until position 4.",
    "outputs": [
      {
        "formula": "!<<Robot2, Robot3>> (pos1 U pos4)"
      }
    ]
},
{
"id": "ex251",
"input": "The robot number 1 has a strategy to ensure that eventually position 2 and not position 3.",
    "outputs": [
```

```
      {
        "formula": "<<Robot1>> F (pos2 && !pos3)"
      }
    ]
},

{
"id": "ex252",
"input": "The coalition of robots 1 and 2 has a strategy to ensure globally that if position 1 then next position 2.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot2>> G (pos1 -> X pos2)"
      }
    ]
},

{
"id": "ex253",
"input": "It is not the case that robots 1 and 3 have a strategy to always eventually reach position 2.",
    "outputs": [
      {
        "formula": "!<<Robot1, Robot3>> G F pos2"
      }
    ]
},
{
"id": "ex254",
"input": "The robot number 2 has a strategy to ensure that in the next step it attacks and does not defend.",
    "outputs": [
      {
        "formula": "<<Robot2>> X (attack && !defend)"
      }
    ]
},
{
"id": "ex255",
"input": "Robots 1, 2 and 3 have a strategy to ensure eventually position 4 and then always avoid position 5.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot2, Robot3>> F (pos4 && X G !pos5) "
      }
    ]
},
```

Reviewer 1 - “ex255”: In my opinion, the natural language "eventually position 4 **and then** always avoid position 5" suggests a temporal/causal sequence where, first, position 4 is reached and then, namely after that, position 5 is always avoided. To me, the current formula doesn't capture this sequencing. An alternative could be something like <<Robot1, Robot2, Robot3>> F (pos4 && X G !pos5).
Reviewer 3: I agree on the issue with *then*. In this case, I would also use *next*. However, I have some doubts about the *eventually* that contains the whole implication. Isn’t *eventually* only related to the left-hand side of the implication?
Reviewer 1 - “ex255”: I agree.
Reviewer 2: I agree

{
"id": "ex256",
"input": "The robot number 1 does not have a strategy to ensure that position 3 will eventually hold when position 2 holds now.",
  "outputs": [
   {
    "formula": "!<<Robot1>> (pos2 -> F pos3)"
   }
  ]
},

Reviewer 1 - “ex256”: I find this translation problematic because *now* carries a temporal connotation incompatible with conditional semantics. The phrase "pos2 holds *now*" is not well-formalized. When the antecedent is false and the consequent is true, the conditional remains true, which obscures the temporal relationship the original statement intended to express.
Reviewer 3: I also find it difficult to understand the logical structure of the sentence in English. However, with respect to the property, the specification seems to be the best possible. Again, as I mentioned before, in my view we have to assume that the user writes correctly. Question: could we envision an interaction with an LLM? For instance, the NLP component detects a warning (an unclear sentence) and the user tries to ‘clean up’ the property by interacting with an LLM. I think this would be more in the realm of future work, but it is something worth keeping in mind.
Reviewer 2: case closed, I agree with reviewer 1.

{
"id": "ex257",
"input": "Robots number 2 and number 3 have a strategy to ensure that either always position 1 holds or eventually position 2 holds.",
  "outputs": [
   {
    "formula": "<<Robot2, Robot3>> (G pos1 || F pos2)"
   }
  ]
},
{
"id": "ex258",

```
"input": "The robot number 3 has a strategy to ensure that position 1 holds until position 2
holds or position 4 holds.",
   "outputs": [
    {
     "formula": "<<Robot3>> (pos1 U (pos2 || pos4))"
    }
   ]
},
{
"id": "ex259",
"input": "Coalition of robots 1, 2 and 3 do not have a strategy to ensure next not position 1.",
   "outputs": [
    {
     "formula": "!<<Robot1, Robot2, Robot3>> X !pos1"
    }
   ]
},
{
"id": "ex260",
"input": "The robot number 1 has a strategy to guarantee globally that position 2 implies
eventually position 3.",
   "outputs": [
    {
     "formula": "<<Robot1>> G (pos2 -> F pos3)"
    }
   ]
},
{
"id": "ex261",
"input": "Robots number 1 and number 2 have a strategy to ensure eventually position 1 and
always position 3 afterwards.",
   "outputs": [
    {
     "formula": "<<Robot1, Robot2>> F (pos1 && X G pos3)"
    }
   ]
},
```


Reviewer 1 - “ex261”: see “ex244”; "<<Robot1, Robot2>> F (pos1 && X G pos3)" Should this problem be addressed as a hard one?
Reviewer 2:no
Reviewer 1 - ok


```
{
"id": "ex262",
"input": "The robot number 2 does not have a strategy to ensure that position 3 will hold until
position 1 holds.",
   "outputs": [
```

```
    {
      "formula": "!<<Robot2>> (pos3 U pos1)"
    }
  ]
},
{
"id": "ex263",
"input": "The coalition of robots 1 and 3 has a strategy to ensure that in the next step either position 2 or position 4 holds.",
  "outputs": [
    {
      "formula": "<<Robot1, Robot3>> X (pos2 || pos4)"
    }
  ]
},
{
"id": "ex264",
"input": "The robot number 1 has a strategy to ensure eventually get to position 2 and then eventually to position 3.",
  "outputs": [
    {
      "formula": "<<Robot1>> (F pos2 && F pos3)"
    }
  ]
},
```

Reviewer 1 - “ex264”: Unfortunately, I have been fulgurated on the road to Damascus just a few meters from the city... I don't think "has a strategy" alone captures the semantics of the strategic operator in ATL/ATL*, which takes into account all the strategic behaviors of the opponents. I believe "has a strategy" works better when it immediately precedes verbs like "ensure", "guarantee", etc. Edit: solved.
Reviewer 3: “I don’t see any particular problems with the mention of ‘has a strategy’; it seems fairly explicit to me. I am rather puzzled by the ‘then’, which, in my view, is not being taken into account.
Reviewer 2: I agree with reviewer 1.

```
{
"id": "ex265",
"input": "Robots number 2 and number 3 have a strategy to ensure that always not position 4.",
  "outputs": [
    {
      "formula": "<<Robot2, Robot3>> G !pos4"
    }
  ]
},
{
"id": "ex266",
```

```
"input": "It is not the case that robot number 3 has a strategy to guarantee eventually reach
position 1 or position 2.",
    "outputs": [
      {
        "formula": "!<<Robot3>> F (pos1 || pos2)"
      }
    ]
},
```

Reviewer 1 - “ex266”: see “ex264”.

```
{
"id": "ex267",
"input": "The robot number 1 has a strategy to ensure that if position 1 ever occurs then
position 2 will hold in the next step.",
    "outputs": [
      {
        "formula": "<<Robot1>> (F pos1 -> X pos2)"
      }
    ]
},
{
"id": "ex268",
"input": "Robots number 1 and number 2 have a strategy to ensure that position 3 holds until
position 4 holds and not position 5.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot2>> ((pos3 U pos4) && !pos5)"
      }
    ]
},
{
"id": "ex269",
"input": "The robot number 2 has a strategy to ensure always either position 1 or position 3
holds.",
    "outputs": [
      {
        "formula": "<<Robot2>> G (pos1 || pos3)"
      }
    ]
},
{
"id": "ex270",
"input": "Robots number 1 and number 3 together do not have a strategy to ensure
eventually not position 2.",
    "outputs": [
      {
        "formula": "!<<Robot1, Robot3>> F !pos2"
```

    }
  ]
},

Reviewer 1 - “ex270”: The absence of *together* / *coalition* is problematic.
Reviewer 2: I agree w/ r1
Reviewer 3: Shouldn’t the initial negation be external? I mean, something like ![rest of the formula]? The rest seems consistent to me.
Reviewer 2: as far as I may concern also for my implementation background in this parsing semantics, if we extend the negation explicitly to the whole formula, it will also negate the goal (indeed the second negation) not reflecting the meaning of spec.
Reviewer 1 - “ex270”: I agree with Reviewer 2.

{
"id": "ex271",
"input": "The robot number 3 has a strategy to force that eventually position 2 holds and never position 4 afterwards.",
  "outputs": [
    {
     "formula": "<<Robot3>> F (pos2 && XG !pos4)"
    }
  ]
},

Reviewer 1 - “ex271”: see “ex244”. Should this problem be addressed as a hard one?

{
"id": "ex272",
"input": "The coalition of robots 1 and 2 has a strategy to ensure that whenever position 1 holds, position 2 will hold in the next step.",
  "outputs": [
    {
     "formula": "<<Robot1, Robot2>> G (pos1 -> X pos2)"
    }
  ]
},

{
"id": "ex273",
"input": "None of the robots number 1, 2, or 3 has a strategy to ensure that attack holds globally and attack never holds.",
  "outputs": [
    {

"formula": "!<<Robot1>> (G attack && G !attack) && !<<Robot2>> (G attack && G !attack) && !<<Robot3>> (G attack && G !attack)"
}
]
},
Reviewer 1 - “ex273”: see “ex243”. Please, check if the difficulty is right.

{
"id": "ex274",
"input": "The robot number 1 has a strategy to ensure that in the next step dance and not stay hold.",
"outputs": [
{
"formula": "<<Robot1>> X (dance && !stay)"
}
]
},
{
"id": "ex275",
"input": "Robots number 2 and number 3 together have a strategy to always guarantee that if position 4 then eventually position 1.",
"outputs": [
{
"formula": "<<Robot2, Robot3>> G (pos4 -> F pos1)"
}
]
},

Reviewer 1 - “ex275”: see “ex270”.

{
"id": "ex276",
"input": "The robot number 3 does not have a strategy to ensure that position 2 holds until position 5 holds.",
"outputs": [
{
"formula": "!<<Robot3>> (pos2 U pos5)"
}
]
},
{
"id": "ex277",
"input": "The robot number 1 has a strategy to ensure eventually reach position 4 or to always stay in position 1.",
"outputs": [
{
"formula": "<<Robot1>> (F pos4 || G pos1)"
}

]
},

Reviewer 1 - “ex277”: see “ex264”.

{
"id": "ex278",
"input": "Robots number 1 and number 2 together have a strategy to ensure next that if defend then retreat.",
"outputs": [
{
"formula": "<<Robot1, Robot2>> X (defend -> retreat)"
}
]

},

Reviewer 1 - “ex278”: see “ex270”.

{
"id": "ex279",
"input": "The coalition of robots 2 and 3 has a strategy to ensure that eventually either attack or defend holds and not retreat.",
"outputs": [
{
"formula": "<<Robot2, Robot3>> F ((attack || defend) && !retreat)"
}
]
},

{
"id": "ex280",
"input": "The robot number 1 has a strategy to ensure that always eventually position 2 or always eventually position 3.",
"outputs": [
{
"formula": "<<Robot1>> (G F pos2 || G F pos3)"
}
]
},

{
"id": "ex281",
"input": "It is not the case that robots 1, 2 and 3 have a strategy to enforce globally position 1.",
"outputs": [
{
"formula": "!<<Robot1, Robot2, Robot3>> G pos1"
}

```
    ]
},

{
"id": "ex282",
"input": "The robot number 2 has a strategy to ensure that position 1 holds until position 2 holds or position 3 holds.",
    "outputs": [
      {
        "formula": "<<Robot2>> (pos1 U (pos2 || pos3))"
      }
    ]
},

{
"id": "ex283",
"input": "Robots number 1 and number 3 together have a strategy to ensure that eventually position 1 happens and afterwards globally position 2 holds.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot3>> F (pos1 && X G pos2)"
      }
    ]
},
```


Reviewer 1 - “ex283”: see “ex270” and “ex244”.


```
{
"id": "ex284",
"input": "The robot number 3 does not have a strategy to ensure next that position 2 and position 3 both hold.",
    "outputs": [
      {
        "formula": "!<<Robot3>> X (pos2 && pos3)"
      }
    ]
},
{
"id": "ex285",
"input": "The robot number 1 has a strategy to guarantee that if position 1 holds now then position 4 will eventually hold.",
    "outputs": [
      {
        "formula": "<<Robot1>> (pos1 -> F pos4)"
      }
    ]
},
```

```
{
"id": "ex286",
"input": "Robots number 2 and number 3 together have a strategy to ensure that always position 2 implies next position 3.",
  "outputs": [
   {
    "formula": "<<Robot2, Robot3>> G (pos2 -> X pos3)"
   }
  ]
},
```

Reviewer 1 - “ex286”: see “ex270”.

```
{
"id": "ex287",
"input": "The robot number 2 has a strategy to ensure that eventually position 1 holds and not position 2 at that moment.",
  "outputs": [
   {
    "formula": "<<Robot2>> F (pos1 && !pos2)"
   }
  ]
},

{
"id": "ex288",
"input": "The coalition of robots 1 and 2 does not have a strategy to ensure that position 3 will hold globally.",
  "outputs": [
   {
    "formula": "!<<Robot1, Robot2>> G pos3"
   }
  ]
},
{
"id": "ex289",
"input": "The robot number 3 has a strategy to ensure that position 1 holds until it eventually reaches position 6.",
  "outputs": [
   {
    "formula": "<<Robot3>> (pos1 U pos6)"
   }
  ]
},
{
"id": "ex290",
"input": "Robots number 1 and number 2 together have a strategy to ensure in the next step not position 5 and not position 4.",
  "outputs": [
```

```
      {
        "formula": "<<Robot1, Robot2>> X (!pos5 && !pos4)"
      }
    ]
},
```

Reviewer 1 - “ex290”: see “ex270”.

```
{
"id": "ex291",
"input": "The robot number 1 has a strategy to eventually ensure position 2 or else always ensure position 3.",
    "outputs": [
      {
        "formula": "<<Robot1>> (F pos2 || G pos3)"
      }
    ]
},
{
"id": "ex292",
"input": "Given three robots, none of them has a strategy to ensure that whenever position 2 holds, position 1 will hold in the next step.",
    "outputs": [
      {
        "formula": "!<<Robot1>> (G (pos2 -> X pos1) && !<<Robot2>> G (pos2 -> X pos1)) &&
!<<Robot3>> G (pos2 -> X pos1)"
      }
    ]
},

{
"id": "ex293",
"input": "The coalition of robots 2 and 3 has a strategy to ensure eventually position 7 and then always position 7.",
    "outputs": [
      {
        "formula": "<<Robot2, Robot3>> F (pos7 && X G pos7)"
      }
    ]
},
```

Reviewer 1 - “ex293”: see “ex244”.

```
{
"id": "ex294",
```

"input": "The robot number 1 does not have a strategy to ensure either eventually position 3 or eventually position 4.",
"outputs": [
{
"formula": "!<<Robot1>> (F pos3 || F pos4)"
}
]
},

{
"id": "ex295",
"input": "Robots number 1, 2 and 3 together have a strategy to ensure globally that if position 5 then eventually position 6.",
"outputs": [
{
"formula": "<<Robot1, Robot2, Robot3>> G (pos5 -> F pos6)"
}
]
},

Reviewer 1 - “ex295”: see “ex270”.

{
"id": "ex296",
"input": "The robot number 2 has a strategy to ensure that position 3 will never hold.",
"outputs": [
{
"formula": "<<Robot2>> G !pos3"
}
]
},
{
"id": "ex297",
"input": "The robot number 3 has a strategy to ensure that eventually position 2 holds and then in the next step position 1 holds.",
"outputs": [
{
"formula": "<<Robot3>> F (pos2 && X pos1)"
}
]
},
{
"id": "ex298",
"input": "Robots number 1 and number 2 have a strategy to ensure that position 2 holds until either position 3 or position 4 holds.",
"outputs": [
{
"formula": "<<Robot1, Robot2>> (pos2 U (pos3 || pos4))"

```
      }
    ]
},
{
"id": "ex299",
"input": "The robot number 1 does not have a strategy to guarantee that always eventually position 5 holds.",
    "outputs": [
      {
        "formula": "!<<Robot1>> G F pos5"
      }
    ]
},
{
"id": "ex300",
"input": "The coalition of robots 2 and 3 has a strategy to ensure that either next position 1 or globally position 2 holds.",
    "outputs": [
      {
        "formula": "<<Robot2, Robot3>> (X pos1 || G pos2)"
      }
    ]
},
{
"id": "ex301",
"input": "The robot number 2 has a strategy to ensure that if position 3 ever holds then position 4 will hold until position 1 holds.",
    "outputs": [
      {
        "formula": "<<Robot2>> (F pos3 -> (pos4 U pos1))"
      }
    ]
},

{
"id": "ex302",
"input": "Robots number 1 and number 3 together have a strategy to make sure that eventually position 2 holds and position 3 never holds afterwards.",
    "outputs": [
      {
        "formula": "<<Robot1, Robot3>> F (pos2 && X G !pos3)"
      }
    ]
}
```

Reviewer 1 - “ex302”: see “ex270” and “ex244”.

```
{
```

```
"id": "ex303",
"input": "They can eventually ensure the game ends, Bob and Richard.",
  "outputs": [
   {
    "formula": "<<Bob, Richard>> F game_ends"
   }
  ]
}
```


Reviewer 1 - Right dislocation. The translation is correct.
Reviewer 2 - I am ok with that.


```
{
"id": "ex304",
"input": "The laundry machine, Mary can ensure it will work in the next turn of work",
  "outputs": [
   {
    "formula": "<<Mary>>X working_laundry_machine"
   }
  ]
}
```


Reviewer 1 - Left dislocation. The translation seems fine and pretty straightforward to me.
Reviewer 2 - I am ok with that.


```
{
"id": "ex305",
"input": "Amanda has a strategy to surely sell all the toys and Leo does too",
  "outputs": [
   {
    "formula": "<<Amanda>> G sell_all_toys && <<Leo>> G sell_all_toys"
   }
  ]
}
```


Reviewer 1 - VP ellipsis. The translation is fine.
Reviewer 2 - I am ok with that.


```
{
"id": "ex306",
"input": "Every agent can eventually prevent a breach",
  "outputs": [
   {
    "formula": "<<agent_1>>F prevent_breach_1 && <<agent_2>>F prevent_breach_2"
   },
   {
```

```
        "formula": "<<agent_1,agent_2>>F prevent_breach"
      }
    ]
}
```

Reviewer 1 - Quantifier scope ambiguity. Being able to reason, the LLM needs to assume the presence of n > 1 and n = 1 breaches in order to correctly suggest both the interpretations. Same reasoning for all quantifier scope ambiguity examples. Obviously, this happens if and only if the LLM does not enough informations.

Reviewer 2 - I am ok with that, moreover, it is a good practice to me to vary the number of agents inside the coalition since and “every” statement is aleatory due to the interpretation of the LLM circa the complete coalition.

```
  {
    "id": "ex307",
    "input": "They can guarantee that sooner or later the gate will open, the user and the machine.",
    "outputs": [
      {
        "formula": "<<User,Machine>>F gate_open"
      }
    ]
  },
```

Reviewer 1 - Right dislocation see ex. 303.

Reviewer 2 - see ex. 303

```
  {
    "id": "ex308",
    "input": "It can guarantee that it will never enter a restricted zone, the mobile robot.",
    "outputs": [
      {
        "formula": "<<Robot>>G !in_restricted_zone"
      }
    ]
  },
```

Reviewer 1 - Right dislocation see ex. 303.

Reviewer 2 - Okay

```
  {
    "id": "ex309",
    "input": "They can guarantee that at the next step the alarm will be sent, the surveillance system and the operator.",
    "outputs": [
      {
        "formula": "<<System,Operator>>X alarm_sent"
      }
    ]
  },
```

Reviewer 1 - Right dislocation see ex. 303.

{
"id": "ex310",
"input": "It can always guarantee that if a sensor fails, then sooner or later safe mode will be entered, the vehicle.",
"outputs": [
{
"formula": "<<Vehicle>>G (sensor_fault -> F safe_mode)"
}
]
},
Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - Note: This output is an ATL* formula example of right dislocation.

{
"id": "ex311",
"input": "The gate, the machine can guarantee that it will open at the next step.",
"outputs": [
{
"formula": "<<Machine>>X gate_open"
}
]
},
Reviewer 1 - Left dislocation. See ex. 304. Correct.
Reviewer 2 - It's good to me
{
"id": "ex312",
"input": "The charging station, the mobile robot can guarantee that it will eventually reach it.",
"outputs": [
{
"formula": "<<Robot>>F at_charging_station"
}
]
},
Reviewer 1 - See ex. 304. Fine.
Reviewer 2 - It's good to me

{
"id": "ex313",
"input": "The ticket, the user can guarantee that it will eventually be printed.",
"outputs": [
{
"formula": "<<User>>F ticket_printed"
}
]
},
Reviewer 2 - It's good to me

```
{
  "id": "ex314",
  "input": "The backup antenna, the rover can guarantee that it will activate it at the next step if communication with the base is lost.",
  "outputs": [
    {
      "formula": "<<Rover>> (comm_lost -> X backup_antenna_on)"
    }
  ]
},
```

Reviewer 2 - It’s good to me, basically we translate “guarantee” with G as the only way to effectively and formally guarantee a property is to check all the states.
Reviewer 1 - Exactly. Furthermore, note the consequent and antecedent inversion.

```
{
  "id": "ex315",
  "input": "Robot number 1 has a strategy to ensure that eventually position 3 holds, and robot number 2 does too.",
  "outputs": [
    {
      "formula": "<<Robot1>>F pos3 && <<Robot2>>F pos3"
    }
  ]
},
```

Reviewer 2 - It’s okay to me.
Reviewer 1 - VP ellipsis. see ex. 305, seems fine.

```
{
  "id": "ex316",
  "input": "The jet can guarantee that it will land safely sooner or later, and the drone can too.",
  "outputs": [
    {
      "formula": "<<Jet>>F safe_landed && <<Drone>>F safe_landed"
    }
  ]
},
```

Reviewer 2 - It’s good to me
Reviewer 1 - VP ellipsis. see ex. 305, okay.

```
{
  "id": "ex317",
  "input": "The robotic arm can guarantee that it will never exceed the torque limit, and the collaborative robot can too.",
  "outputs": [
    {
      "formula": "<<Arm>>G !torque_over_limit && <<Cobot>>G !torque_over_limit"
    }
  ]
```

```
 },
```

Reviewer 2 - It’s good to me

Reviewer 1 - VP ellipsis. see ex. 305. correct

```
 {
   "id": "ex318",
   "input": "The system can guarantee that if an intruder is detected, then at the next step it will send an alarm, and the operator can too.",
   "outputs": [
     {
       "formula": "<<System>> (intruder_detected -> X alarm_sent) && <<Operator>> (intruder_detected -> X alarm_sent)"
     }
   ]
 },
```

Reviewer 1 - VP ellipsis. see ex. 305. Fine.

Reviewer 2 - It’s good to me

```
 {
"id": "ex319",
"input": "Every robot can guarantee that it will eventually reach a safe spot.",
   "outputs": [
     {
       "formula": "<<Robot1>>F at_safe_spot_1 && <<Robot2>>F at_safe_spot_2 && <<Robot3>>F at_safe_spot_3"
     },
     {
       "formula": "<<Robot1,Robot2,Robot3>>F at_safe_spot"
     }
   ]
},
```

Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

Reviewer 2 - I am ok with that.

```
{
"id": "ex320",
"input": "Every operator can eventually prevent an alarm failure.",
   "outputs": [
     {
       "formula": "<<Operator1>>F !alarm_failure && <<Operator2>>F !alarm_failure && <<Operator3>>F !alarm_failure"
     },
     {
       "formula": "<<Operator1,Operator2,Operator3>>F !alarm_failure"
     }
   ]
},
```

Reviewer 2 - It’s good to me

Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

```
{
"id": "ex321",
"input": "Every drone can guarantee that it will eventually reach a waypoint.",
   "outputs": [
     {
       "formula": "<<Drone1>>F at_waypoint && <<Drone2>>F at_waypoint && <<Drone3>>F at_waypoint"
     },
     {
       "formula": "<<Drone1,Drone2,Drone3>>F at_waypoint"
     }
   ]
},
```

Reviewer 2 - It’s good to me

Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

```
{
"id": "ex322",
"input": "Every rover can guarantee that it will never enter a high-risk area.",
   "outputs": [
     {
       "formula": "<<Rover1>>G !in_high_risk_area_1 && <<Rover2>>G !in_high_risk_area_2 && <<Rover3>>G !in_high_risk_area_3"
     },
     {
       "formula": "<<Rover1,Rover2,Rover3>>G !in_high_risk_area"
     }
   ]
} ,
```

Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

Reviewer 2 - It’s good to me

```
  {
   "id": "ex323",
   "input": "They can guarantee that globally the sample will never be lost, the rover and the base.",
   "outputs": [
     {
       "formula": "<<Rover,Base>>G !sample_lost"
     }
   ]
  },
```

Reviewer 1 - Right dislocation see ex. 303.

Reviewer 2 - It’s good to me

```
  {
   "id": "ex324",
```

```
    "input": "It can guarantee that at the next step docking mode will be enabled, the mobile robot.",
    "outputs": [
      {
        "formula": "<<Robot>>X docking_mode"
      }
    ]
  },
```

Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - It's good to me

```
  {
    "id": "ex325",
    "input": "They can guarantee that sooner or later the package will be delivered, the robot and the operator.",
    "outputs": [
      {
        "formula": "<<Robot,Operator>>F delivered"
      }
    ]
  },
```

Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - It's good to me

```
  {
    "id": "ex326",
    "input": "It can guarantee that if the battery is low, then at the next step it will reduce speed, the autonomous vehicle.",
    "outputs": [
      {
        "formula": "<<Vehicle>> (battery_low -> X speed_low)"
      }
    ]
  },
```

Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - It's good to me

```
  {
    "id": "ex327",
    "input": "The mission, James Bond can guarantee that he will eventually complete it.",
    "outputs": [
      {
        "formula": "<<jamesbond>>F mission_completed"
      }
    ]
  },
```

Reviewer 1 - Left dislocation. See ex. 304. Correct.
Reviewer 2 - It's good to me

{
"id": "ex328",
"input": "The restricted zone, the rover can guarantee that it will never enter it.",
"outputs": [
{
"formula": "<<Rover>>G !in_restricted_zone"
}
]
},
Reviewer 1 - Left dislocation. See ex. 304. Correct.
Reviewer 2 - It’s good to me

{
"id": "ex329",
"input": "The charging station, the robot can guarantee that it will reach it sooner or later.",
"outputs": [
{
"formula": "<<Robot>>F at_charging_station"
}
]
},
Reviewer 1 - Left dislocation. See ex. 304. Correct.
Reviewer 2 - It’s good to me

{
"id": "ex330",
"input": "The alarm, the system can always guarantee that it will send it at the next step if an intruder is detected.",
"outputs": [
{
"formula": "<<System>>G (intruder_detected -> X alarm_sent)"
}
]
},
Reviewer 1 - Left dislocation. See ex. 304. Correct.
Reviewer 2 - It’s good to me

{
"id": "ex331",
"input": "The drone can guarantee that it will return to base sooner or later, and the rover can too.",
"outputs": [
{
"formula": "<<Drone>>F at_base && <<Rover>>F at_base"
}
]
},

Reviewer 1 - VP ellipsis. see ex. 305.
Reviewer 2 - It’s good to me

{
"id": "ex332",
"input": "Robot number 1 has a strategy to ensure that position 4 will eventually hold, and robot number 3 does too.",
"outputs": [
{
"formula": "<<Robot1>>F pos4 && <<Robot3>>F pos4"
}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
Reviewer 2 - It’s good to me

{
"id": "ex333",
"input": "The collaborative robot can guarantee that it will never enter standby mode, and the surveillance system can too.",
"outputs": [
{
"formula": "<<Cobot>>G !standby_mode && <<System>>G !standby_mode"
}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
Reviewer 2 - It’s good to me

{
"id": "ex334",
"input": "The robotic arm can always guarantee that if a collision is detected, then at the next step it will stop, and the mobile robot can too.",
"outputs": [
{
“formula”: “<<Arm>> G(collision_detected -> X stopped) && <<Robot>> G(collision_detected -> X stopped)”

}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
Reviewer 2 - It’s good to me

{
"id": "ex335",
"input": "Every drone can guarantee that it will eventually reach a safe landing zone.",

```
    "outputs": [
      {
        "formula": "<<Drone1>>F safe_landing_zone_1 && <<Drone2>>F
safe_landing_zone_2 && <<Drone3>>F safe_landing_zone_3 && <<Drone4>>F
safe_landing_zone_4"
      },
      {
        "formula": "<<Drone1,Drone2,Drone3>>F safe_landing_zone"
      }
    ]
  },
```

Reviewer 2 - It's good to me

Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

```
  {
    "id": "ex336",
    "input": "Every plane can prevent a collision.",
    "outputs": [
      {
        "formula": "<<plane1>>G !collision && <<plane2>>G !collision && <<plane3>>G
!collision"
      },
      {
        "formula": "<<plane1,plane2,plane3>>G !collision"
      }
    ]
  },
```

Reviewer 2 - It's good to me

Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

```
  {
    "id": "ex337",
    "input": "Every operator can guarantee that an alarm will eventually be acknowledged.",
    "outputs": [
      {
        "formula": "<<Operator1>>F alarm_acknowledged && <<Operator2>>F
alarm_acknowledged && <<Operator3>>F alarm_acknowledged"
      },
      {
        "formula": "<<Operator1,Operator2,Operator3>>F alarm_acknowledged"
      }
    ]
  },
```

Reviewer 2 - It's good to me

Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

```
  {
    "id": "ex338",
    "input": "Every rover can guarantee that it will never enter a hazardous area.",
    "outputs": [
      {
```

```
      "formula": "<<Rover1>>G !in_hazardous_area_1 && <<Rover2>>G
!in_hazardous_area_2 && <<Rover3>>G !in_hazardous_area_3"
    },
    {
      "formula": "<<Rover1,Rover2,Rover3>>G !in_hazardous_area"
    }
  ]
}
```

Reviewer 2 - It’s good to me
Reviewer 1 - Quantifier scope ambiguity. See ex. 306.

```
{
  "id": "ex339",
  "input": "They can guarantee that the gate will remain closed until the ticket is validated, the user and the machine.",
  "outputs": [
    {
      "formula": "<<User,Machine>>(gate_closed U ticket_validated)"
    }
  ]
},
```

Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - It’s a good one

```
{
  "id": "ex340",
  "input": "It can guarantee that the alarm will stay silent until smoke is detected, the control unit.",
  "outputs": [
    {
      "formula": "<<ControlUnit>>(alarm_off U smoke_detected)"
    }
  ]
},
```

Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - It’s good to me

```
{
  "id": "ex341",
  "input": "They can guarantee that the room will remain locked until the code is entered, the resident and the smart lock.",
  "outputs": [
    {
      "formula": "<<Resident,SmartLock>>(door_locked U code_entered)"
    }
  ]
},
```

Reviewer 1 - Right dislocation see ex. 303.

Reviewer 2 - Approved

```
 {
   "id": "ex342",
   "input": "It can guarantee that the report will stay pending until the doctor approves it, the hospital system.",
   "outputs": [
    {
     "formula": "<<HospitalSystem>>(report_pending U report_approved)"
    }
   ]
 },
```

Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - Approved

```
 {
   "id": "ex343",
   "input": "The payment, the banking system can guarantee that it will remain suspended until the identity is verified.",
   "outputs": [
    {
     "formula": "<<BankingSystem>>(payment_suspended U identity_verified)"
    }
   ]
 },
```

Reviewer 1 - Left dislocation. See ex. 304.
Reviewer 2 - Approved

```
 {
   "id": "ex344",
   "input": "The crossing light, the traffic controller can guarantee that it will stay red until the road is clear.",
   "outputs": [
    {
     "formula": "<<TrafficController>>(red_light U road_clear)"
    }
   ]
 },
```

Reviewer 1 - Left dislocation. See ex. 304
Reviewer 2 - Approved

```
 {
   "id": "ex345",
   "input": "The heater, the smart home controller can guarantee that it will stay off until the temperature drops.",
   "outputs": [
    {
     "formula": "<<HomeController>>(heater_off U temp_low)"
```

```
      }
    ]
  },
```
Reviewer 1 - Left dislocation. See ex. 304.
Reviewer 2 - Approved

```
  {
    "id": "ex346",
    "input": "The vault, the security system can guarantee that it will remain closed until the badge is recognized.",
    "outputs": [
      {
        "formula": "<<SecuritySystem>>(vault_closed U badge_recognized)"
      }
    ]
  },
```
Reviewer 1 - Left dislocation. See ex. 304
Reviewer 2 - Approved

```
  {
    "id": "ex347",
    "input": "The cashier can guarantee that the receipt will remain unavailable until the payment is confirmed, and the kiosk can too.",
    "outputs": [
      {
        "formula": "<<Cashier>>( receipt_unavailable U payment_confirmed) && <<Kiosk>>( receipt_unavaialble U payment_confirmed)"
      }
    ]
  },
```
Reviewer 1 - VP ellipsis. see ex. 305.
Reviewer 2 - Approved

```
  {
    "id": "ex348",
    "input": "The platform manager can guarantee that boarding will remain closed until the train arrives, and the station system can too.",
    "outputs": [
      {
        "formula": "<<PlatformManager>>( boarding_closed U train_arrived) && <<StationSystem>>( boarding_closed U train_arrived)"
      }
    ]
  },
```
Reviewer 1 - VP ellipsis. see ex. 305
Reviewer 2 - Approved

```
  {
```

"id": "ex349",
"input": "The teacher can guarantee that the exam will remain inaccessible until the key is released, and the exam server can too.",
"outputs": [
{
"formula": "<<Teacher>>(exam_inaccessible U key_released) && <<ExamServer>>(exam_inaccessible U key_released)"
}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
Reviewer 2 - Approved

{
"id": "ex350",
"input": "The dispatcher can guarantee that the shipment will remain on hold until customs clearance is completed, and the logistics platform can too.",
"outputs": [
{
"formula": "(<<Dispatcher>>(shipment_on_hold U customs_cleared) && <<LogisticsPlatform>>(shipment_on_hold U customs_cleared))"
}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
Reviewer 2 - Approved

{
"id": "ex351",
"input": "Every teller can guarantee that the transaction will remain blocked until a supervisor authorizes it.",
"outputs": [
{
"formula": "<<Teller1>>(transaction_blocked U supervisor_auth) && <<Teller2>>(transaction_blocked U supervisor_auth)"
},
{
"formula": "<<Teller1,Teller2>>(transaction_blocked U supervisor_auth)"
}
]
},
Reviewer 2 - Approved
Reviewer 1 - Quantifier scope ambiguity. See ex. 306
{
"id": "ex352",
"input": "Every nurse can guarantee that the medication will not be administered until the doctor signs.",

"outputs": [
{
"formula": "<<Nurse1>>(!med_administered U doctor_signed) && <<Nurse2>>(!med_administered U doctor_signed) && <<Nurse3>>(!med_administered U doctor_signed)"
},
{
"formula": "<<Nurse1,Nurse2,Nurse3>>(!med_administered U doctor_signed)"
}
]
},
Reviewer 1 - Quantifier scope ambiguity. See ex. 306.
Reviewer 2 - Approved

{
"id": "ex353",
"input": "Every clerk can guarantee that the file will remain closed until the client arrives.",
"outputs": [
{
"formula": "<<Clerk1>>(file_closed U client_arrived) && <<Clerk2>>(file_closed U client_arrived) && <<Clerk3>>(file_closed U client_arrived) && <<Clerk4>>(file_closed U client_arrived)"
},
{
"formula": "<<Clerk1,Clerk2,Clerk3,Clerk4>>(file_closed U client_arrived)"
}
]
},
Reviewer 1 - Quantifier scope ambiguity. See ex. 306.
Reviewer 2 - Approved

{
"id": "ex354",
"input": "Every controller can guarantee that the circuit will stay inactive until maintenance is completed.",
"outputs": [
{
"formula": "<<Controller1>>(circuit_inactive U maintenance_done) && <<Controller2>>(circuit_inactive U maintenance_done) && <<Controller3>>(circuit_inactive U maintenance_done)"
},
{
"formula": "<<Controller1,Controller2,Controller3>>(circuit_inactive U maintenance_done)"
}
]
},
Reviewer 1 - Quantifier scope ambiguity. See ex. 306

Reviewer 2 - Approved

{
  "id": "ex355",
  "input": "They can guarantee that the hearing will remain suspended until the judge arrives, the court clerk and the court system.",
  "outputs": [
    {
      "formula": "<<CourtClerk,CourtSystem>>(hearing_suspended U judge_arrived)"
    }
  ]
},

Reviewer 2 - Approved

Reviewer 1 - Right dislocation see ex. 303.

{
  "id": "ex356",
  "input": "It can guarantee that the boarding gate will stay closed until the final security check is completed, the airport system.",
  "outputs": [
    {
      "formula": "<<AirportSystem>>(gate_closed U security_check_completed)"
    }
  ]
},

Reviewer 2 - Approved

Reviewer 1 - Right dislocation see ex. 303.

{
  "id": "ex357",
  "input": "They can guarantee that the lecture recording will remain unavailable until consent is collected, the lecturer and the platform.",
  "outputs": [
    {
      "formula": "<<Lecturer,Platform>>(recording_unavailable U consent_collected)"
    }
  ]
},

Reviewer 1 - Right dislocation see ex. 303.

Reviewer 2 - Approved

{
  "id": "ex358",
  "input": "It can guarantee that the package will remain blocked until the customs fee is paid, the delivery service.",
  "outputs": [
    {
      "formula": "<<DeliveryService>>(package_blocked U customs_fee_paid)"
    }

]
},
Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - Approved

{
"id": "ex359",
"input": "The invoice, the accounting system can guarantee that it will remain pending until the manager approves it.",
"outputs": [
{
"formula": "<<AccountingSystem>>(invoice_pending U manager_approved)"
}
]
},
Reviewer 2 - Approved
Reviewer 1 - Left dislocation. See ex. 304
{
"id": "ex360",
"input": "The exam portal, the university platform can guarantee that it will stay inaccessible until enrollment is confirmed.",
"outputs": [
{
"formula": "<<UniversityPlatform>>(portal_inaccessible U enrollment_confirmed)"
}
]
},
Reviewer 2 - Approved
Reviewer 1 - Left dislocation. See ex. 304
{
"id": "ex361",
"input": "The prescription, the pharmacy system can guarantee that it will remain inactive until the physician signs it.",
"outputs": [
{
"formula": "<<PharmacySystem>>(prescription_inactive U physician_signed)"
}
]
},
Reviewer 2 - Approved
Reviewer 1 - Left dislocation. See ex. 304
{
"id": "ex362",
"input": "The refund, the customer service system can guarantee that it will remain on hold until the complaint is reviewed.",
"outputs": [
{
"formula": "<<CustomerService>>(refund_on_hold U complaint_reviewed)"

}
]
},
Reviewer 2 - Approved
Reviewer 1 - Left dislocation. See ex. 304

{
"id": "ex363",
"input": "The registrar can guarantee that the application will remain unprocessed until the fee is paid, and the admissions office can too.",
"outputs": [
{
"formula": "<<Registrar>>(application_unprocessed U fee_paid) && <<AdmissionsOffice>>(application_unprocessed U fee_paid)"
}
]
},
Reviewer 2 - Approved
Reviewer 1 - VP ellipsis. see ex. 305.
{
"id": "ex364",
"input": "The hotel manager can guarantee that the room will not be available until cleaning is completed, and the booking platform can too.",
"outputs": [
{
"formula": "<<HotelManager>>(!room_available U cleaning_completed) && <<BookingPlatform>>(!room_available U cleaning_completed)"
}
]
},
Reviewer 2 - Approved
Reviewer 1 - VP ellipsis. see ex. 305.
{
"id": "ex365",
"input": "The librarian can guarantee that the archive will remain closed until authorization is granted, and the access system can too.",
"outputs": [
{
"formula": "<<Librarian>>(archive_closed U authorization_granted) && <<AccessSystem>>(archive_closed U authorization_granted)"
}
]
},
Reviewer 2 - Approved
Reviewer 1 - VP ellipsis. see ex. 305.
{
"id": "ex366",

"input": "The warehouse supervisor can guarantee that the shipment will remain unsent until the label is printed, and the inventory system can too.",
"outputs": [
{
"formula": "<<WarehouseSupervisor>>(shipment_unsent U label_printed) && <<InventorySystem>>(shipment_unsent U label_printed)"
}
]
},

Reviewer 2 - Approved

Reviewer 1 - VP ellipsis. see ex. 305

{
"id": "ex367",
"input": "Every cashier can guarantee that the register will remain locked until identification is verified.",
"outputs": [
{
"formula": "<<Cashier1>>(register_locked U id_verified) && <<Cashier2>>(register_locked U id_verified) && <<Cashier3>>(register_locked U id_verified) && <<Cashier4>>(register_locked U id_verified) && <<Cashier5>>(register_locked U id_verified)"
},
{
"formula": "<<Cashier1,Cashier2,Cashier3,Cashier4,Cashier5>>(register_locked U id_verified)"
}
]
},

Reviewer 2 - Approved

Reviewer 1 - Quantifier scope ambiguity. See ex. 306

{
"id": "ex368",
"input": "Every examiner can guarantee that the oral test will not begin until the candidate is identified.",
"outputs": [
{
"formula": "<<Examiner1>>(!oral_test_started U candidate_identified) && <<Examiner2>>(!oral_test_started U candidate_identified) && <<Examiner3>>(!oral_test_started U candidate_identified) && <<Examiner4>>(!oral_test_started U candidate_identified)"
},
{
"formula": "<<Examiner1,Examiner2,Examiner3,Examiner4>>(!oral_test_started U candidate_identified)"
}
]
},

Reviewer 1 - Quantifier scope ambiguity. See ex. 306
Reviewer 2 - Approved

{
"id": "ex369",
"input": "Every dispatcher can guarantee that the route will remain inactive until the vehicle is assigned.",
"outputs": [
{
"formula": "<<Dispatcher1>>(route_inactive U vehicle_assigned) && <<Dispatcher2>>(route_inactive U vehicle_assigned) && <<Dispatcher3>>(route_inactive U vehicle_assigned)"
},
{
"formula": "<<Dispatcher1,Dispatcher2,Dispatcher3>>(route_inactive U vehicle_assigned)"
}
]
},
Reviewer 1 - Quantifier scope ambiguity. See ex. 306
Reviewer 2 - Approved

{
"id": "ex370",
"input": "Every administrator can guarantee that the server will remain offline until the update is validated.",
"outputs": [
{
"formula": "<<Admin1>>(server_offline U update_validated) && <<Admin2>>(server_offline U update_validated) "
},
{
"formula": "<<Admin1,Admin2>>(server_offline U update_validated)"
}
]
}
Reviewer 1 - Quantifier scope ambiguity. See ex. 306
Reviewer 2 - Approved

{
"id": "ex371",
"input": "They can guarantee that at the next step the goal will be scored, the striker and the midfielder.",
"outputs": [
{
"formula": "<<Striker,Midfielder>>X goal_scored"
}
]

},
Reviewer 2 - Ok
Reviewer 1 - Right dislocation see ex. 303.
{
"id": "ex372",
"input": "It can guarantee that at the next step the penalty will be saved, the goalkeeper.",
"outputs": [
{
"formula": "<<Goalkeeper>>X penalty_saved"
}
]
},
Reviewer 1 - Right dislocation see ex. 303.
Reviewer 2 - Ok

{
"id": "ex373",
"input": "They can guarantee that at the next step the relay baton will be passed correctly, the first runner and the second runner.",
"outputs": [
{
"formula": "<<Runner1,Runner2>>X baton_passed"
}
]
},
Reviewer 2 - Ok
Reviewer 1 - Right dislocation see ex. 303.
{
"id": "ex374",
"input": "She can guarantee that at the next service the ball will land in, the tennis player.",
"outputs": [
{
"formula": "<<TennisPlayer>>X serve_in"
}
]
},
Reviewer 2 - Ok
Reviewer 1 - Right dislocation see ex. 303.
{
"id": "ex375",
"input": "The ball, the point guard can guarantee that at the next step he will pass it.",
"outputs": [
{
"formula": "<<PointGuard>>X ball_passed"
}
]
},
Reviewer 2 - Ok

Reviewer 1 - Left dislocation. See ex. 304.

```
 {
   "id": "ex376",
   "input": "The end zone, the wide receiver can guarantee that at the next step he will catch the ball in it.",
   "outputs": [
     {
       "formula": "<<WideReceiver>>X finish_crossed"
     }
   ]
 },
```

Reviewer 1 - Left dislocation. See ex. 304
Reviewer 2 - Ok

```
 {
   "id": "ex377",
   "input": "The corner kick, the defender and the goalkeeper can guarantee that at the next step they will clear it.",
   "outputs": [
     {
       "formula": "<<Defender,Goalkeeper>>X corner_cleared"
     }
   ]
 },
```

Reviewer 2 - Ok
Reviewer 1 - Left dislocation. See ex. 304

```
 {
   "id": "ex378",
   "input": "The pick-and-pop, the power forward can guarantee that at the next step he will set a screen for the point guard and then pop out and then take the shot.",
   "outputs": [
     {
       "formula": "<<PowerForward>>X (screen_set && X (pop_out && (X shot_taken)))"
     }
   ]
 },
```

Reviewer 1 - Left dislocation. See ex. 304

```
 {
   "id": "ex379",
   "input": "The striker can guarantee that at the next step the net will be hit, and the winger can too.",
   "outputs": [
     {
       "formula": "<<Striker>>X net_hit && <<Winger>>X net_hit"
     }
   ]
 },
```

Reviewer 1 - VP ellipsis. see ex. 305.

{
"id": "ex380",
"input": "The server can guarantee that at the next step the ball will cross the line, and the doubles partner can too.",
"outputs": [
{
"formula": "<<Server>>X ball_in_court && <<Partner>>X ball_in_court"
}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
{
"id": "ex381",
"input": "The captain can guarantee that at the next step possession will be recovered, and the midfielder can too.",
"outputs": [
{
"formula": "<<Captain>>X possession_recovered && <<Midfielder>>X possession_recovered"
}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
{
"id": "ex382",
"input": "The goalkeeper can guarantee that at the next step the shot will be blocked, and the defender can too.",
"outputs": [
{
"formula": "<<Goalkeeper>>X shot_blocked && <<Defender>>X shot_blocked"
}
]
},
Reviewer 1 - VP ellipsis. see ex. 305.
{
"id": "ex383",
"input": "Every striker can guarantee that at the next step the ball will be shotted.",
"outputs": [
{
"formula": "<<Striker1>>X shot_1 && <<Striker2>>X shot_2 && <<Striker3>>X shot_3"
},
{
"formula": "<<Striker1,Striker2,Striker3>>X shot"
}
]
},
Reviewer 1 - Quantifier scope ambiguity. See ex. 306
{

```
  "id": "ex384",
  "input": "Every defender can guarantee that at the next step the shot will be blocked.",
  "outputs": [
    {
      "formula": "<<defender1>>X shot_block_1 && <<defender2>>X shot_block_2 &&
<<defender3>>X shot_block_3"
    },
    {
      "formula": "<<defender1,defender2,defender3>>X shot_block"
    }
  ]
},
```

Reviewer 1 - Quantifier scope ambiguity. See ex. 306

```
{
  "id": "ex385",
  "input": "Every runner can guarantee that at the next step the baton will be received.",
  "outputs": [
    {
      "formula": "<<Runner1>>X baton_received_1 && <<Runner2>>X baton_received_2 &&
<<Runner3>>X baton_received_3"
    },
    {
      "formula": "<<Runner1,Runner2,Runner3>>X baton_received"
    }
  ]
},
```

Reviewer 1 - Quantifier scope ambiguity. See ex. 306

```
{
  "id": "ex386",
  "input": "Every player can guarantee that at the next step the pass will be completed.",
  "outputs": [
    {
      "formula": "<<Player1>>X pass_completed_1 && <<Player2>>X pass_completed_2 &&
<<Player3>>X pass_completed_3"
    },
    {
      "formula": "<<Player1,Player2,Player3>>X pass_completed"
    }
  ]
}
```

Reviewer 1 - Quantifier scope ambiguity. See ex. 306

```
{
  "id": "ex387",
  "input": "They can guarantee that the request will be retried until the server accepts it, the
client and the transport layer.",
  "outputs": [
    {
```

```
        "formula": "<<Client,TransportLayer>>(request_retried U request_accepted)"
      }
    ]
  },
  {
    "id": "ex388",
    "input": "It can guarantee that the recovery controller will keep compensating until the plant becomes stable, the supervisory module.",
    "outputs": [
      {
        "formula": "<<Supervisor>>(compensating U plant_stable)"
      }
    ]
  },
  {
    "id": "ex389",
    "input": "They can guarantee that the health-check protocol will keep collecting metrics until the control plane persists them, the node agent and the cloud manager.",
    "outputs": [
      {
        "formula": "<<NodeAgent,CloudManager>>(collecting_metrics U metrics_persisted)"
      }
    ]
  },
  {
    "id": "ex390",
    "input": "It can guarantee that the proposal will remain under validation until a quorum acknowledges it, the consensus replica.",
    "outputs": [
      {
        "formula": "<<Replica>>(proposal_validating U quorum_acknowledged)"
      }
    ]
  },
  {
    "id": "ex391",
    "input": "The connection attempt, the client can guarantee that it will be repeated until an error response is received.",
    "outputs": [
      {
        "formula": "<<Client>>(connection_attempt_repeated U error_response_received)"
      }
    ]
  },
  {
    "id": "ex392",
    "input": "The telemetry stream, the monitoring system can guarantee that it will remain active until the anomaly score becomes conclusive.",
```

```
    "outputs": [
      {
        "formula": "<<MonitoringSystem>>(telemetry_active U anomaly_score_conclusive)"
      }
    ]
  },
  {
    "id": "ex393",
    "input": "The balancing action, the grid controller can guarantee that it will continue until
the frequency becomes stable.",
    "outputs": [
      {
        "formula": "<<GridController>>(balancing_action U frequency_stable)"
      }
    ]
  },
  {
    "id": "ex394",
    "input": "The pending transaction, the replicated ledger can guarantee that it will remain
unresolved until the block is finalized.",
    "outputs": [
      {
        "formula": "<<Ledger>>(transaction_pending U block_finalized)"
      }
    ]
  },

  {
    "id": "ex395",
    "input": "The proposer can guarantee that the slot will remain open until the value is
decided, and the acceptor can too.",
    "outputs": [
      {
        "formula": "<<Proposer>>(slot_open U value_decided) && <<Acceptor>>(slot_open U
value_decided)"
      }
    ]
  },
  {
    "id": "ex396",
    "input": "The gateway can guarantee that the packet will remain buffered until the route is
confirmed, and the edge router can too.",
    "outputs": [
      {
        "formula": "<<Gateway>>(packet_buffered U route_confirmed) &&
<<EdgeRouter>>(packet_buffered U route_confirmed)"
      }
    ]
```

```
 },
 {
   "id": "ex397",
   "input": "The runtime monitor can guarantee that the trace will keep being observed until
the specification is decided, and the verifier can too.",
   "outputs": [
     {
       "formula": "<<RuntimeMonitor>>(trace_observed U specification_decided) &&
<<Verifier>>(trace_observed U specification_decided)"
     }
   ]
 },
 {
   "id": "ex398",
   "input": "The scheduler can guarantee that the task will remain deferred until resources
are released, and the orchestrator can too.",
   "outputs": [
     {
       "formula": "<<Scheduler>>(task_deferred U resources_released) &&
<<Orchestrator>>(task_deferred U resources_released)"
     }
   ]
 },
 {
   "id": "ex399",
   "input": "Every replica can guarantee that the request will remain pending until a quorum
acknowledges it.",
   "outputs": [
     {
       "formula": "<<Replica1>>(request_pending_1 U quorum_acknowledged_1) &&
<<Replica2>>(request_pending_2 U quorum_acknowledged_2) &&
<<Replica3>>(request_pending_3 U quorum_acknowledged_3)"
     },
     {
       "formula": "<<Replica1,Replica2,Replica3>>(request_pending U
quorum_acknowledged)"
     }
   ]
 },
 {
   "id": "ex400",
   "input": "Every controller can guarantee that compensation will continue until the plant
reaches a stability.",
   "outputs": [
     {
       "formula": "<<Controller1>>(compensating_1 U stability_1) &&
<<Controller2>>(compensating_2 U stability_2) && <<Controller3>>(compensating_3 U
stability_3)"
```

```
    },
    {
      "formula": "<<Controller1,Controller2,Controller3>>(compensating U stability)"
    }
  ]
},
{
  "id": "ex401",
  "input": "Every monitor can guarantee that telemetry will keep being collected until the
alert is validated.",
  "outputs": [
    {
      "formula": "<<Monitor1>>(telemetry_collected_1 U alert_validated_1) &&
<<Monitor2>>(telemetry_collected_2 U alert_validated_2) &&
<<Monitor3>>(telemetry_collected_3 U alert_validated_3)"
    },
    {
      "formula": "<<Monitor1,Monitor2,Monitor3>>(telemetry_collected U alert_validated)"
    }
  ]
},
{
  "id": "ex402",
  "input": "Every participant can guarantee that the proposal remains active until a
consensus is reached.",
  "outputs": [
    {
      "formula": "<<Participant1>>(proposal_active_1 U consensus_reached_1) &&
<<Participant2>>(proposal_active_2 U consensus_reached_2) &&
<<Participant3>>(proposal_active_3 U consensus_reached_3)"
    },
    {
      "formula": "<<Participant1,Participant2,Participant3>>(proposal_active U
consensus_reached)"
    }
  ]
},
{
  "id": "ex403",
  "input": "They can guarantee that the checkpoint will keep being retried until the storage
backend confirms it, the database engine and the replication manager.",
  "outputs": [
    {
      "formula": "<<DatabaseEngine,ReplicationManager>>(checkpoint_retried U
checkpoint_confirmed)"
    }
  ]
},
```

```
{
  "id": "ex404",
  "input": "It can guarantee that the scheduler will keep reallocating resources until the latency bound is satisfied, the orchestration layer.",
  "outputs": [
    {
      "formula": "<<Orchestrator>>(reallocating_resources U latency_bound_satisfied)"
    }
  ]
},
{
  "id": "ex405",
  "input": "They can guarantee that the encrypted session will remain renegotiated until the certificate chain is validated, the client and the security gateway.",
  "outputs": [
    {
      "formula": "<<Client,SecurityGateway>>(session_renegotiated U certificate_chain_validated)"
    }
  ]
},
{
  "id": "ex406",
  "input": "It can guarantee that the control loop will keep updating the actuator until the tracking error becomes negligible, the controller module.",
  "outputs": [
    {
      "formula": "<<ControllerModule>>(actuator_updated U tracking_error_negligible)"
    }
  ]
},
{
  "id": "ex407",
  "input": "The migration process, the hypervisor can guarantee that it will remain active until the memory image is synchronized.",
  "outputs": [
    {
      "formula": "<<Hypervisor>>(migration_active U memory_image_synchronized)"
    }
  ]
},
{
  "id": "ex408",
  "input": "The anomaly score, the diagnosis engine can guarantee that it will remain provisional until the final evidence is aggregated.",
  "outputs": [
    {
```

"formula": "<<DiagnosisEngine>>(anomaly_score_provisional U final_evidence_aggregated)"
}
]
},
{
"id": "ex409",
"input": "The routing update, the control plane can guarantee that it will remain pending until every dependency is resolved.",
"outputs": [
{
"formula": "<<ControlPlane>>(routing_update_pending U dependencies_resolved)"
}
]
},
{
"id": "ex410",
"input": "The execution trace, the verifier can guarantee that it will keep being extended until the safety obligation is discharged.",
"outputs": [
{
"formula": "<<Verifier>>(trace_extended U safety_obligation_discharged)"
}
]
},
{
"id": "ex411",
"input": "The broker can guarantee that the message will remain buffered until the consumer acknowledges it, and the relay can too.",
"outputs": [
{
"formula": "<<Broker>>(message_buffered U consumer_acknowledged) && <<Relay>>(message_buffered U consumer_acknowledged)"
}
]
},
{
"id": "ex412",
"input": "The coordinator can guarantee that the transaction will remain open until the commit decision is propagated, and the participant can too.",
"outputs": [
{
"formula": "<<Coordinator>>(transaction_open U commit_decision_propagated) && <<Participant>>(transaction_open U commit_decision_propagated)"
}
]
},
{

```
    "id": "ex413",
    "input": "The stream processor can guarantee that the window will remain active until the watermark arrives, and the ingestion layer can too.",
    "outputs": [
      {
        "formula": "<<StreamProcessor>>(window_active U watermark_arrived) && <<IngestionLayer>>(window_active U watermark_arrived)"
      }
    ]
  },
  {
    "id": "ex414",
    "input": "The planner can guarantee that the task will remain queued until the dependency graph is closed, and the runtime can too.",
    "outputs": [
      {
        "formula": "<<Planner>>(task_queued U dependency_graph_closed) && <<Runtime>>(task_queued U dependency_graph_closed)"
      }
    ]
  },

  {
    "id": "ex415",
    "input": "Every replica can guarantee that the log entry remains tentative until the term is committed.",
    "outputs": [
      {
        "formula": "<<Replica1>>(log_entry_tentative_1 U term_committed_1) && <<Replica2>>(log_entry_tentative_2 U term_committed_2) && <<Replica3>>(log_entry_tentative_3 U term_committed_3)"
      },
      {
        "formula": "<<Replica1,Replica2,Replica3>>(log_entry_tentative U term_committed)"
      }
    ]
  },
  {
    "id": "ex416",
    "input": "Every validator can guarantee that the block remains unverifiable until the proof is attached.",
    "outputs": [
      {
        "formula": "<<Validator1>>(block_unverifiable_1 U proof_attached_1) && <<Validator2>>(block_unverifiable_2 U proof_attached_2)"
      },
      {
        "formula": "<<Validator1,Validator2>>(block_unverifiable U proof_attached)"
```

}
]
},
{
"id": "ex417",
"input": "Every monitor can guarantee that the alert remains suppressed until the confidence score exceeds the threshold.",
"outputs": [
{
"formula": "<<Monitor1>>(alert_suppressed_1 U confidence_above_threshold_1) && <<Monitor2>>(alert_suppressed_2 U confidence_above_threshold_2) && <<Monitor3>>(alert_suppressed_3 U confidence_above_threshold_3)"
},
{
"formula": "<<Monitor1,Monitor2,Monitor3>>(alert_suppressed U confidence_above_threshold)"
}
]
},
{
"id": "ex418",
"input": "Every node can guarantee that the synchronization phase remains active until the global snapshot is complete.",
"outputs": [
{
"formula": "<<Node1>>(sync_phase_active_1 U global_snapshot_complete_1) && <<Node2>>(sync_phase_active_2 U global_snapshot_complete_2) && <<Node3>>(sync_phase_active_3 U global_snapshot_complete_3)"
},
{
"formula": "<<Node1,Node2,Node3>>(sync_phase_active U global_snapshot_complete)"
}
]
},
{
"id": "ex419",
"input": "They can guarantee that the leader election phase will remain active until a unique coordinator is recognized, the replicas and the failure detector.",
"outputs": [
{
"formula": "<<Replicas,FailureDetector>>(leader_election_active U unique_coordinator_recognized)"
}
]
},
{
"id": "ex420",

```
    "input": "It can guarantee that the symbolic exploration will keep expanding until the counterexample is discharged, the model checker.",
    "outputs": [
      {
        "formula": "<<ModelChecker>>(symbolic_exploration_expanding U counterexample_discharged)"
      }
    ]
  },
  {
    "id": "ex421",
    "input": "They can guarantee that the reconfiguration protocol will remain enabled until the new membership view is installed, the controller and the replicas.",
    "outputs": [
      {
        "formula": "<<Controller,Replicas>>(reconfiguration_enabled U membership_view_installed)"
      }
    ]
  },
  {
    "id": "ex422",
    "input": "It can guarantee that the refinement loop will continue until the abstraction error becomes acceptable, the synthesis engine.",
    "outputs": [
      {
        "formula": "<<SynthesisEngine>>(refinement_loop_active U abstraction_error_acceptable)"
      }
    ]
  },
  {
    "id": "ex423",
    "input": "The commit certificate, the consensus module can guarantee that it will remain incomplete until the final vote is collected.",
    "outputs": [
      {
        "formula": "<<ConsensusModule>>(commit_certificate_incomplete U final_vote_collected)"
      }
    ]
  },
  {
    "id": "ex424",
    "input": "The repair action, the recovery manager can guarantee that it will remain provisional until the fault hypothesis is confirmed.",
    "outputs": [
      {
```

"formula": "<<RecoveryManager>>(repair_action_provisional U fault_hypothesis_confirmed)"
}
]
},
{
"id": "ex425",
"input": "The trace abstraction, the verifier can guarantee that it will remain coarse until the witness is reconstructed.",
"outputs": [
{
"formula": "<<Verifier>>(trace_abstraction_coarse U witness_reconstructed)"
}
]
},
{
"id": "ex426",
"input": "The pending checkpoint, the storage controller can guarantee that it will remain unsealed until the durable write is observed.",
"outputs": [
{
"formula": "<<StorageController>>(checkpoint_unsealed U durable_write_observed)"
}
],
"difficulty": "hard"
},
{
"id": "ex427",
"input": "The proposer can guarantee that the decision remains uncommitted until the quorum certificate is assembled, and the coordinator can too.",
"outputs": [
{
"formula": "<<Proposer>>(decision_uncommitted U quorum_certificate_assembled) && <<Coordinator>>(decision_uncommitted U quorum_certificate_assembled)"
}
]
},
{
"id": "ex428",
"input": "The monitor can guarantee that the violation report remains suppressed until the confidence interval stabilizes, and the analyzer can too.",
"outputs": [
{
"formula": "<<Monitor>>(violation_report_suppressed U confidence_interval_stabilized) && <<Analyzer>>(violation_report_suppressed U confidence_interval_stabilized)"
}
]
},

{
"id": "ex429",
"input": "The scheduler can guarantee that the job remains migratable until the destination node is warmed up, and the runtime can too.",
"outputs": [
{
"formula": "<<Scheduler>>(job_migratable U destination_node_warmed_up) && <<Runtime>>(job_migratable U destination_node_warmed_up)"
}
]
},
{
"id": "ex430",
"input": "The observer can guarantee that the execution remains partially ordered until the causal relation is saturated, and the logger can too.",
"outputs": [
{
"formula": "<<Observer>>(execution_partially_ordered U causal_relation_saturated) && <<Logger>>(execution_partially_ordered U causal_relation_saturated)"
}
]
},
{
"id": "ex431",
"input": "Every replica can guarantee that the ballot remains undecided until the leader proof is delivered.",
"outputs": [
{
"formula": "<<Replica1>>(ballot_undecided_1 U leader_proof_delivered_1) && <<Replica2>>(ballot_undecided_2 U leader_proof_delivered_2) && <<Replica3>>(ballot_undecided_3 U leader_proof_delivered_3)"
},
{
"formula": "<<Replica1,Replica2,Replica3>>(ballot_undecided U leader_proof_delivered)"
}
]
},
{
"id": "ex432",
"input": "Every verifier can guarantee that the obligation remains open until the ranking argument is found.",
"outputs": [
{
"formula": "<<Verifier1>>(obligation_open_1 U ranking_argument_found_1) && <<Verifier2>>(obligation_open_2 U ranking_argument_found_2)"
},
{

```
        "formula": "<<Verifier1,Verifier2>>(obligation_open U ranking_argument_found)"
      }
    ]
  },
  {
    "id": "ex433",
    "input": "Every node can guarantee that the snapshot remains partial until the channel
state is recorded.",
    "outputs": [
      {
        "formula": "<<Node1>>(snapshot_partial_1 U channel_state_recorded_1) &&
<<Node2>>(snapshot_partial_2 U channel_state_recorded_2) &&
<<Node3>>(snapshot_partial_3 U channel_state_recorded_3)"
      },
      {
        "formula": "<<Node1,Node2,Node3>>(snapshot_partial U channel_state_recorded)"
      }
    ]
  },
  {
    "id": "ex434",
    "input": "Every planner can guarantee that the repair plan remains incomplete until the
conflict set is minimized.",
    "outputs": [
      {
        "formula": "<<Planner1>>(repair_plan_incomplete_1 U conflict_set_minimized_1) &&
<<Planner2>>(repair_plan_incomplete_2 U conflict_set_minimized_2) &&
<<Planner3>>(repair_plan_incomplete_3 U conflict_set_minimized_3) &&
<<Planner4>>(pair_plan_incomplete_4 U conflict_set_minimized_4)"
      },
      {
        "formula": "<<Planner1,Planner2,Planner3,Planner4>>(repair_plan_incomplete U
conflict_set_minimized)"
      }
    ]
  },
  {
    "id": "ex435",
    "input": "They can guarantee that the authentication exchange will remain pending until
the challenge response is verified, the client and the identity provider.",
    "outputs": [
      {
        "formula": "<<Client,IdentityProvider>>(authentication_pending U
challenge_response_verified)"
      }
    ]
  },
  {
```

"id": "ex436",
"input": "It can guarantee that the containment procedure will remain active until the malicious process is isolated, the endpoint agent.",
"outputs": [
{
"formula": "<<EndpointAgent>>(containment_active U malicious_process_isolated)"
}
]
},
{
"id": "ex437",
"input": "They can guarantee that the access request will remain quarantined until the policy engine evaluates it, the gateway and the authorization server.",
"outputs": [
{
"formula": "<<Gateway,AuthorizationServer>>(access_request_quarantined U policy_evaluated)"
}
]
},
{
"id": "ex438",
"input": "It can guarantee that the forensic collection will continue until the evidence image is sealed, the incident response module.",
"outputs": [
{
"formula": "<<IncidentResponseModule>>(forensic_collection_active U evidence_image_sealed)"
}
]
},
{
"id": "ex439",
"input": "The session token, the authentication service can guarantee that it will remain invalid until the second factor is confirmed.",
"outputs": [
{
"formula": "<<AuthenticationService>>(session_token_invalid U second_factor_confirmed)"
}
]
},
{
"id": "ex440",
"input": "The suspicious flow, the intrusion detector can guarantee that it will remain under inspection until the signature is matched.",
"outputs": [
{

"formula": "<<IntrusionDetector>>(suspicious_flow_inspected U signature_matched)"
}
]
},
{
"id": "ex441",
"input": "The privilege escalation alert, the security monitor can guarantee that it will remain unresolved until the audit trail is reconstructed.",
"outputs": [
{
"formula": "<<SecurityMonitor>>(privilege_escalation_alert_unresolved U audit_trail_reconstructed)"
}
]
},
{
"id": "ex442",
"input": "The revoked certificate, the trust manager can guarantee that it will remain unusable until the revocation list is synchronized.",
"outputs": [
{
"formula": "<<TrustManager>>(revoked_certificate_unusable U revocation_list_synchronized)"
}
]
},
{
"id": "ex443",
"input": "The firewall can guarantee that the packet remains blocked until the rule set is refreshed, and the proxy can too.",
"outputs": [
{
"formula": "<<Firewall>>(packet_blocked U rule_set_refreshed) && <<Proxy>>(packet_blocked U rule_set_refreshed)"
}
]
},
{
"id": "ex444",
"input": "The verifier can guarantee that the credential remains untrusted until the attestation is validated, and the policy engine can too.",
"outputs": [
{
"formula": "<<Verifier>>(credential_untrusted U attestation_validated) && <<PolicyEngine>>(credential_untrusted U attestation_validated)"
}
]
},

{
"id": "ex445",
"input": "The sandbox can guarantee that the executable remains constrained until the behavior profile is classified, and the analyzer can too.",
"outputs": [
{
"formula": "<<Sandbox>>(executable_constrained U behavior_profile_classified) && <<Analyzer>>(executable_constrained U behavior_profile_classified)"
}
]
},
{
"id": "ex446",
"input": "The key manager can guarantee that the decryption request remains deferred until the enclave measurement is accepted, and the enclave can too.",
"outputs": [
{
"formula": "<<KeyManager>>(decryption_request_deferred U enclave_measurement_accepted) && <<Enclave>>(decryption_request_deferred U enclave_measurement_accepted)"
}
]
},
{
"id": "ex447",
"input": "Every monitor can guarantee that the anomaly report remains hidden until the confidence score exceeds the release threshold.",
"outputs": [
{
"formula": "<<Monitor1>>(anomaly_report_hidden_1 U confidence_above_release_threshold_1) && <<Monitor2>>(anomaly_report_hidden_2 U confidence_above_release_threshold_2)"
},
{
"formula": "<<Monitor1,Monitor2>>(anomaly_report_hidden U confidence_above_release_threshold)"
}
]
},
{
"id": "ex448",
"input": "Every validator can guarantee that the access token remains unusable until the issuer proof is confirmed.",
"outputs": [
{
"formula": "<<Validator1>>(access_token_unusable_1 U issuer_proof_confirmed_1) && <<Validator2>>(access_token_unusable_2 U issuer_proof_confirmed_2) &&

```
<<Validator3>>(access_token_unusable_3 U issuer_proof_confirmed_3) &&
<<Validator4>>(access_token_unusable_4 U issuer_proof_confirmed_4)"
    },
    {
     "formula": "<<Validator1,Validator2,Validator3,Validator4>>(access_token_unusable U
issuer_proof_confirmed)"
    }
   ]
  },
  {
   "id": "ex449",
   "input": "Every responder can guarantee that the host remains quarantined until the
integrity scan is completed.",
   "outputs": [
    {
     "formula": "<<Responder1>>(host_quarantined_1 U integrity_scan_completed_1) &&
<<Responder2>>(host_quarantined_2 U integrity_scan_completed_2) &&
<<Responder3>>(host_quarantined_3 U integrity_scan_completed_3)"
    },
    {
     "formula": "<<Responder1,Responder2,Responder3>>(host_quarantined U
integrity_scan_completed)"
    }
   ]
  },
  {
   "id": "ex450",
   "input": "Every gateway can guarantee that the request remains unforwarded until the
threat score is recomputed.",
   "outputs": [
    {
     "formula": "<<Gateway1>>(request_unforwarded_1 U threat_score_recomputed_1) &&
<<Gateway2>>(request_unforwarded_2 U threat_score_recomputed_2) &&
<<Gateway3>>(request_unforwarded_3 U threat_score_recomputed_3)"
    },
    {
     "formula": "<<Gateway1,Gateway2,Gateway3>>(request_unforwarded U
threat_score_recomputed)"
    }
   ]
  },
  {
   "id": "ex451",
   "input": "They can guarantee that the triage assessment will remain open until the
attending physician confirms the severity level, the nurse and the clinical support system.",
   "outputs": [
    {
```

```
        "formula": "<<Nurse,ClinicalSupportSystem>>(triage_assessment_open U
severity_level_confirmed)"
      }
    ]
  },
  {
    "id": "ex452",
    "input": "It can guarantee that the medication order will remain suspended until the allergy
profile is validated, the prescribing module.",
    "outputs": [
      {
        "formula": "<<PrescribingModule>>(medication_order_suspended U
allergy_profile_validated)"
      }
    ]
  },
  {
    "id": "ex453",
    "input": "They can guarantee that the monitoring protocol will remain active until the vital
signs trend becomes stable, the bedside monitor and the clinician.",
    "outputs": [
      {
        "formula": "<<BedsideMonitor,Clinician>>(monitoring_protocol_active U
vitals_trend_stable)"
      }
    ]
  },
  {
    "id": "ex454",
    "input": "It can guarantee that the discharge workflow will continue until the follow-up plan
is acknowledged, the hospital information system.",
    "outputs": [
      {
        "formula": "<<HospitalInformationSystem>>(discharge_workflow_active U
followup_plan_acknowledged)"
      }
    ]
  },
  {
    "id": "ex455",
    "input": "The laboratory request, the diagnostic platform can guarantee that it will remain
pending until the specimen integrity is verified.",
    "outputs": [
      {
        "formula": "<<DiagnosticPlatform>>(laboratory_request_pending U
specimen_integrity_verified)"
      }
    ]
```

```
},
{
  "id": "ex456",
  "input": "The infusion plan, the therapy controller can guarantee that it will remain
provisional until the dosage constraint is approved.",
  "outputs": [
    {
      "formula": "<<TherapyController>>(infusion_plan_provisional U
dosage_constraint_approved)"
    }
  ]
},
{
  "id": "ex457",
  "input": "The patient record, the hospital system can guarantee that it will remain locked
until the attending team completes reconciliation.",
  "outputs": [
    {
      "formula": "<<HospitalSystem>>(patient_record_locked U reconciliation_completed)"
    }
  ]
},
{
  "id": "ex458",
  "input": "The alert threshold, the clinical decision engine can guarantee that it will remain
adaptive until the risk score is finalized.",
  "outputs": [
    {
      "formula": "<<ClinicalDecisionEngine>>(alert_threshold_adaptive U
risk_score_finalized)"
    }
  ]
},
{
  "id": "ex459",
  "input": "The physician can guarantee that the treatment plan remains tentative until the
imaging report is signed, and the radiology system can too.",
  "outputs": [
    {
      "formula": "<<Physician>>(treatment_plan_tentative U imaging_report_signed) &&
<<RadiologySystem>>(treatment_plan_tentative U imaging_report_signed)"
    }
  ]
},
{
  "id": "ex460",
  "input": "The monitor can guarantee that the alarm remains silenced until the clinician
acknowledges it, and the nursing station can too.",
```

```
    "outputs": [
      {
        "formula": "<<Monitor>>(alarm_silenced U clinician_acknowledged) &&
<<NursingStation>>(alarm_silenced U clinician_acknowledged)"
      }
    ]
  },
  {
    "id": "ex461",
    "input": "The scheduler can guarantee that the operating room assignment remains
unconfirmed until the preoperative checklist is completed, and the surgical coordinator can
too.",
    "outputs": [
      {
        "formula": "<<Scheduler>>(or_assignment_unconfirmed U preop_checklist_completed)
&& <<SurgicalCoordinator>>(or_assignment_unconfirmed U preop_checklist_completed)"
      }
    ]
  },
  {
    "id": "ex462",
    "input": "The pharmacy interface can guarantee that the dispense request remains
deferred until the interaction screen is cleared, and the medication safety service can too.",
    "outputs": [
      {
        "formula": "<<PharmacyInterface>>(dispense_request_deferred U
interaction_screen_cleared) && <<MedicationSafetyService>>(dispense_request_deferred
U interaction_screen_cleared)"
      }
    ]
  },
  {
    "id": "ex463",
    "input": "Every clinician can guarantee that the order remains unsigned until the diagnostic
evidence is reviewed.",
    "outputs": [
      {
        "formula": "<<Clinician1>>(order_unsigned_1 U diagnostic_evidence_reviewed_1) &&
<<Clinician2>>(order_unsigned_2 U diagnostic_evidence_reviewed_2) &&
<<Clinician3>>(order_unsigned_3 U diagnostic_evidence_reviewed_3)"
      },
      {
        "formula": "<<Clinician1,Clinician2,Clinician3>>(order_unsigned U
diagnostic_evidence_reviewed)"
      }
    ]
  },
  {
```

```
    "id": "ex464",
    "input": "Every monitor can guarantee that the observation remains active until the deterioration score is confirmed.",
    "outputs": [
      {
        "formula": "<<Monitor1>>(observation_active_1 U deterioration_score_confirmed_1) && <<Monitor2>>(observation_active_2 U deterioration_score_confirmed_2)"
      },
      {
        "formula": "<<Monitor1,Monitor2>>(observation_active U deterioration_score_confirmed)"
      }
    ]
  },
  {
    "id": "ex465",
    "input": "Every pharmacist can guarantee that the prescription remains unverified until the renal adjustment is computed.",
    "outputs": [
      {
        "formula": "<<Pharmacist1>>(prescription_unverified_1 U renal_adjustment_computed_1) && <<Pharmacist2>>(prescription_unverified_2 U renal_adjustment_computed_2) && <<Pharmacist3>>(prescription_unverified_3 U renal_adjustment_computed_3)"
      },
      {
        "formula": "<<Pharmacist1,Pharmacist2,Pharmacist3>>(prescription_unverified U renal_adjustment_computed)"
      }
    ]
  },
  {
    "id": "ex466",
    "input": "Every care coordinator can guarantee that the discharge summary remains incomplete until the follow-up referrals are entered.",
    "outputs": [
      {
        "formula": "<<Coordinator1>>(discharge_summary_incomplete_1 U followup_referrals_entered_1) && <<Coordinator2>>(discharge_summary_incomplete_2 U followup_referrals_entered_2) && <<Coordinator3>>(discharge_summary_incomplete_3 U followup_referrals_entered_3) && <<Coordinator4>>(discharge_summary_incomplete_4 U followup_referrals_entered_4) && <<Coordinator5>>(discharge_summary_incomplete_5 U followup_referrals_entered_5)"
      },
      {
        "formula": "<<Coordinator1,Coordinator2,Coordinator3,Coordinator4,Coordinator5>>(discharge_summary_incomplete U followup_referrals_entered)"
```

```
      }
    ]
  },
  {
    "id": "ex467",
    "input": "They can guarantee that the request will remain under review until the final decision is issued, the applicant and the office.",
    "outputs": [
      {
        "formula": "<<Applicant,Office>>(request_under_review U final_decision_issued)"
      }
    ]
  },
  {
    "id": "ex468",
    "input": "It can guarantee that the discussion will remain open until a formal resolution is adopted, the committee.",
    "outputs": [
      {
        "formula": "<<Committee>>(discussion_open U formal_resolution_adopted)"
      }
    ]
  },
  {
    "id": "ex469",
    "input": "They can guarantee that the negotiation will continue until the written agreement is signed, the buyer and the seller.",
    "outputs": [
      {
        "formula": "<<Buyer,Seller>>(negotiation_active U written_agreement_signed)"
      }
    ]
  },
  {
    "id": "ex470",
    "input": "It can guarantee that the publication process will remain suspended until the revised manuscript is approved, the editorial board.",
    "outputs": [
      {
        "formula": "<<EditorialBoard>>(publication_suspended U revised_manuscript_approved)"
      }
    ]
  },

  {
    "id": "ex471",
```

```
    "input": "The application form, the applicant can guarantee that it will remain incomplete
until the supporting document is uploaded.",
    "outputs": [
      {
        "formula": "<<Applicant>>(application_form_incomplete U
supporting_document_uploaded)"
      }
    ]
  },
  {
    "id": "ex472",
    "input": "The public notice, the authority can guarantee that it will remain provisional until
the consultation period ends.",
    "outputs": [
      {
        "formula": "<<Authority>>(public_notice_provisional U consultation_period_ended)"
      }
    ]
  },
  {
    "id": "ex473",
    "input": "The reservation, the customer service can guarantee that it will remain
unconfirmed until the deposit is received.",
    "outputs": [
      {
        "formula": "<<CustomerService>>(reservation_unconfirmed U deposit_received)"
      }
    ]
  },
  {
    "id": "ex474",
    "input": "The draft report, the analyst can guarantee that it will remain internal until the
external review is completed.",
    "outputs": [
      {
        "formula": "<<Analyst>>(draft_report_internal U external_review_completed)"
      }
    ]
  },

  {
    "id": "ex475",
    "input": "The coordinator can guarantee that the meeting remains tentative until the chair
approves it, and the assistant can too.",
    "outputs": [
      {
        "formula": "<<Coordinator>>(meeting_tentative U chair_approved) &&
<<Assistant>>(meeting_tentative U chair_approved)"
```

```
      }
    ]
  },
  {
    "id": "ex476",
    "input": "The editor can guarantee that the article remains unpublished until the copyright form is returned, and the publisher can too.",
    "outputs": [
      {
        "formula": "<<Editor>>(article_unpublished U copyright_form_returned) && <<Publisher>>(article_unpublished U copyright_form_returned)"
      }
    ]
  },
  {
    "id": "ex477",
    "input": "The organizer can guarantee that the registration remains inactive until the payment is recorded, and the treasurer can too.",
    "outputs": [
      {
        "formula": "<<Organizer>>(registration_inactive U payment_recorded) && <<Treasurer>>(registration_inactive U payment_recorded)"
      }
    ]
  },
  {
    "id": "ex478",
    "input": "The archivist can guarantee that the record remains inaccessible until the release condition is satisfied, and the administrator can too.",
    "outputs": [
      {
        "formula": "<<Archivist>>(record_inaccessible U release_condition_satisfied) && <<Administrator>>(record_inaccessible U release_condition_satisfied)"
      }
    ]
  },
  {
    "id": "ex479",
    "input": "Every reviewer can guarantee that the submission remains undecided until the rebuttal is received.",
    "outputs": [
      {
        "formula": "<<Reviewer1>>(submission_undecided_1 U rebuttal_received_1) && <<Reviewer2>>(submission_undecided_2 U rebuttal_received_2) && <<Reviewer3>>(submission_undecided_3 U rebuttal_received_3)"
      },
      {
```

```
      "formula": "<<Reviewer1,Reviewer2,Reviewer3>>(submission_undecided U rebuttal_received)"
     }
    ]
  },
  {
    "id": "ex480",
    "input": "Every officer can guarantee that the request remains pending until the verification file is complete.",
    "outputs": [
     {
      "formula": "<<Officer1>>(request_pending_1 U verification_file_complete_1) && <<Officer2>>(request_pending_2 U verification_file_complete_2) && <<Officer3>>(request_pending_3 U verification_file_complete_3)"
     },
     {
      "formula": "<<Officer1,Officer2,Officer3>>(request_pending U verification_file_complete)"
     }
    ]
  },
  {
    "id": "ex481",
    "input": "Every participant can guarantee that the agenda remains unsettled until the opening vote is concluded.",
    "outputs": [
     {
      "formula": "<<Participant1>>(agenda_unsettled_1 U opening_vote_concluded_1) && <<Participant2>>(agenda_unsettled_2 U opening_vote_concluded_2) && <<Participant3>>(agenda_unsettled_3 U opening_vote_concluded_3)"
     },
     {
      "formula": "<<Participant1,Participant2,Participant3>>(agenda_unsettled U opening_vote_concluded)"
     }
    ]
  },
  {
    "id": "ex482",
    "input": "Every applicant can guarantee that the file remains incomplete until the missing declaration is submitted.",
    "outputs": [
     {
      "formula": "<<Applicant1>>(file_incomplete_1 U missing_declaration_submitted_1) && <<Applicant2>>(file_incomplete_2 U missing_declaration_submitted_2) && <<Applicant3>>(file_incomplete_3 U missing_declaration_submitted_3)"
     },
     {
```

"formula": "<<Applicant1,Applicant2,Applicant3>>(file_incomplete U
missing_declaration_submitted)"
}
]
},
{
"id": "ex483",
"input": "They can guarantee that at the next step the decision will be issued, the
committee and the chair.",
"outputs": [
{
"formula": "<<Committee,Chair>>X decision_issued"
}
]
},
{
"id": "ex484",
"input": "It can guarantee that at the next step the notice will be published, the authority.",
"outputs": [
{
"formula": "<<Authority>>X notice_published"
}
]
},
{
"id": "ex485",
"input": "They can guarantee that at the next step the contract will be signed, the buyer
and the seller.",
"outputs": [
{
"formula": "<<Buyer,Seller>>X contract_signed"
}
]
},
{
"id": "ex486",
"input": "It can guarantee that at the next step the revised version will be submitted, the
author.",
"outputs": [
{
"formula": "<<Author>>X revised_version_submitted"
}
]
},
{
"id": "ex487",
"input": "The declaration, the applicant can guarantee that at the next step it will be filed.",
"outputs": [

```
      {
        "formula": "<<Applicant>>X declaration_filed"
      }
    ]
  },
  {
    "id": "ex488",
    "input": "The access code, the administrator can guarantee that at the next step it will be delivered.",
    "outputs": [
      {
        "formula": "<<Administrator>>X access_code_delivered"
      }
    ]
  },
  {
    "id": "ex489",
    "input": "The report, the analyst and the reviewer can guarantee that at the next step it will be finalized.",
    "outputs": [
      {
        "formula": "<<Analyst,Reviewer>>X report_finalized"
      }
    ]
  },
  {
    "id": "ex490",
    "input": "The reservation, the booking office can guarantee that at the next step it will be confirmed.",
    "outputs": [
      {
        "formula": "<<BookingOffice>>X reservation_confirmed"
      }
    ]
  },
  {
    "id": "ex491",
    "input": "The applicant can guarantee that at the next step the missing document will be uploaded, and the clerk can too.",
    "outputs": [
      {
        "formula": "<<Applicant>>X missing_document_uploaded && <<Clerk>>X missing_document_uploaded"
      }
    ]
  },
  {
    "id": "ex492",
```

```
    "input": "The chair can guarantee that at the next step the session will be opened, and the coordinator can too.",
    "outputs": [
      {
        "formula": "<<Chair>>X session_opened && <<Coordinator>>X session_opened"
      }
    ]
  },
  {
    "id": "ex493",
    "input": "The editor can guarantee that at the next step the manuscript will be forwarded, and the assistant can too.",
    "outputs": [
      {
        "formula": "<<Editor>>X manuscript_forwarded && <<Assistant>>X manuscript_forwarded"
      }
    ]
  },
  {
    "id": "ex494",
    "input": "The officer can guarantee that at the next step the request will be validated, and the registrar can too.",
    "outputs": [
      {
        "formula": "<<Officer>>X request_validated && <<Registrar>>X request_validated"
      }
    ]
  },
  {
    "id": "ex495",
    "input": "Every reviewer can guarantee that at the next step the evaluation will be submitted.",
    "outputs": [
      {
        "formula": "<<Reviewer1>>X evaluation_submitted_1 && <<Reviewer2>>X evaluation_submitted_2 && <<Reviewer3>>X evaluation_submitted_3"
      },
      {
        "formula": "<<Reviewer1,Reviewer2,Reviewer3>>X evaluation_submitted"
      }
    ]
  },
  {
    "id": "ex496",
    "input": "Every officer can guarantee that at the next step the file will be checked.",
    "outputs": [
      {
```

```
        "formula": "<<Officer1>>X file_checked_1 && <<Officer2>>X file_checked_2 && <<Officer3>>X file_checked_3"
      },
      {
        "formula": "<<Officer1,Officer2,Officer3>>X file_checked"
      }
    ]
  },
  {
    "id": "ex497",
    "input": "Every participant can guarantee that at the next step the motion will be recorded.",
    "outputs": [
      {
        "formula": "<<Participant1>>X motion_recorded_1 && <<Participant2>>X motion_recorded_2 && <<Participant3>>X motion_recorded_3"
      },
      {
        "formula": "<<Participant1,Participant2,Participant3>>X motion_recorded"
      }
    ]
  },
  {
    "id": "ex498",
    "input": "Every applicant can guarantee that at the next step the signature will be provided.",
    "outputs": [
      {
        "formula": "<<Applicant1>>X signature_provided_1 && <<Applicant2>>X signature_provided_2 && <<Applicant3>>X signature_provided_3"
      },
      {
        "formula": "<<Applicant1,Applicant2,Applicant3>>X signature_provided"
      }
    ]
  },
  {
    "id": "ex499",
    "input": "They can guarantee that at the next step the merger will be ratified, the board and the legal office.",
    "outputs": [
      {
        "formula": "<<Board,LegalOffice>>X merger_ratified"
      }
    ]
  },
  {
    "id": "ex500",
```

```
    "input": "It can guarantee that at the next step the maintenance alert will be cleared, the supervisory console.",
    "outputs": [
      {
        "formula": "<<SupervisoryConsole>>X maintenance_alert_cleared"
      }
    ]
  },
  {
    "id": "ex501",
    "input": "They can guarantee that at the next step the briefing note will be distributed, the coordinator and the secretary.",
    "outputs": [
      {
        "formula": "<<Coordinator,Secretary>>X briefing_note_distributed"
      }
    ]
  },
  {
    "id": "ex502",
    "input": "It can guarantee that at the next step the shipment manifest will be released, the customs office.",
    "outputs": [
      {
        "formula": "<<CustomsOffice>>X shipment_manifest_released"
      }
    ]
  },
  {
    "id": "ex503",
    "input": "The compliance certificate, the auditor can guarantee that at the next step it will be issued.",
    "outputs": [
      {
        "formula": "<<Auditor>>X compliance_certificate_issued"
      }
    ]
  },
  {
    "id": "ex504",
    "input": "The hearing transcript, the clerk can guarantee that at the next step it will be archived.",
    "outputs": [
      {
        "formula": "<<Clerk>>X hearing_transcript_archived"
      }
    ]
  },
```

{
"id": "ex505",
"input": "The access badge, the reception desk and the security officer can guarantee that at the next step it will be activated.",
"outputs": [
{
"formula": "<<ReceptionDesk,SecurityOfficer>>X access_badge_activated"
}
]
},
{
"id": "ex506",
"input": "The settlement draft, the mediator can guarantee that at the next step it will be circulated.",
"outputs": [
{
"formula": "<<Mediator>>X settlement_draft_circulated"
}
]
},
{
"id": "ex507",
"input": "The registrar can guarantee that at the next step the enrollment record will be updated, and the admissions office can too.",
"outputs": [
{
"formula": "<<Registrar>>X enrollment_record_updated && <<AdmissionsOffice>>X enrollment_record_updated"
}
]
},
{
"id": "ex508",
"input": "The moderator can guarantee that at the next step the agenda will be published, and the convenor can too.",
"outputs": [
{
"formula": "<<Moderator>>X agenda_published && <<Convenor>>X agenda_published"
}
]
},
{
"id": "ex509",
"input": "The trustee can guarantee that at the next step the payment instruction will be approved, and the controller can too.",
"outputs": [
{

```
        "formula": "<<Trustee>>X payment_instruction_approved && <<Controller>>X
payment_instruction_approved"
      }
    ]
  },
  {
    "id": "ex510",
    "input": "The inspector can guarantee that at the next step the compliance note will be
filed, and the reviewer can too.",
    "outputs": [
      {
        "formula": "<<Inspector>>X compliance_note_filed && <<Reviewer>>X
compliance_note_filed"
      }
    ],
    "difficulty": "hard"
  },
  {
    "id": "ex511",
    "input": "Every assessor can guarantee that at the next step the score sheet will be
submitted.",
    "outputs": [
      {
        "formula": "<<Assessor1>>X score_sheet_submitted_1 && <<Assessor2>>X
score_sheet_submitted_2 && <<Assessor3>>X score_sheet_submitted_3"
      },
      {
        "formula": "<<Assessor1,Assessor2,Assessor3>>X score_sheet_submitted"
      }
    ]
  },
  {
    "id": "ex512",
    "input": "Every delegate can guarantee that at the next step the ballot will be cast.",
    "outputs": [
      {
        "formula": "<<Delegate1>>X ballot_cast_1 && <<Delegate2>>X ballot_cast_2 &&
<<Delegate3>>X ballot_cast_3"
      },
      {
        "formula": "<<Delegate1,Delegate2,Delegate3>>X ballot_cast"
      }
    ]
  },
  {
    "id": "ex513",
    "input": "Every examiner can guarantee that at the next step the report will be signed.",
    "outputs": [
```

```
      {
        "formula": "<<Examiner1>>X report_signed_1 && <<Examiner2>>X report_signed_2 && <<Examiner3>>X report_signed_3 && <<Examiner4>>X report_signed_4"
      },
      {
        "formula": "<<Examiner1,Examiner2,Examiner3,Examiner4>>X report_signed"
      }
    ]
  },
  {
    "id": "ex514",
    "input": "Every subscriber can guarantee that at the next step the confirmation code will be entered.",
    "outputs": [
      {
        "formula": "<<Subscriber1>>X confirmation_code_entered_1 && <<Subscriber2>>X confirmation_code_entered_2"
      },
      {
        "formula": "<<Subscriber1,Subscriber2>>X confirmation_code_entered"
      }
    ]
  },
{
  "id": "ex515",
  "input": "The access controller can guarantee that at the next step the badge will be checked and that the door will remain locked until the identity is verified.",
    "outputs": [
      {
        "formula": "<<AccessController>>(X badge_checked && (door_locked U identity_verified))"
      }
    ]
},
  {
    "id": "ex516",
    "input": "They can guarantee that at the next step the alert will be issued and that the system will remain isolated until the patch is applied, and that sooner or later service will be restored, the operator and the response platform.",
    "outputs": [
      {
        "formula": "<<Operator,ResponsePlatform>>(X alert_issued && (system_isolated U patch_applied) && F service_restored)"
      }
    ]
  },
  {
    "id": "ex517",
```

```
    "input": "It can guarantee that at the next step the backup will start and the volume will remain write-locked until the checksum is verified, and that sooner or later the archive will be sealed, the storage manager.",
    "outputs": [
      {
        "formula": "<<StorageManager>>(X backup_started && (volume_write_locked U checksum_verified) && F archive_sealed)"
      }
    ]
  },
  {
    "id": "ex518",
    "input": "They can guarantee that at the next step the session will be suspended and that it will remain suspended until the token is refreshed, and that sooner or later access will be restored, the client and the gateway.",
    "outputs": [
      {
        "formula": "<<Client,Gateway>>(X session_suspended && (session_suspended U token_refreshed) && F access_restored)"
      }
    ]
  },
  {
    "id": "ex519",
    "input": "It can guarantee that at the next step the valve will close and the pressure will remain stable until maintenance is completed, and that overflow will never occur, the control unit.",
    "outputs": [
      {
        "formula": "<<ControlUnit>>(X valve_closed && (pressure_stable U maintenance_completed) && G !overflow)"
      }
    ]
  },
  {
    "id": "ex520",
    "input": "The incident report, the analyst can guarantee that at the next step it will be filed and it will remain confidential until legal review is completed, and that sooner or later it will be archived.",
    "outputs": [
      {
        "formula": "<<Analyst>>(X incident_report_filed && (incident_report_confidential U legal_review_completed) && F incident_report_archived)"
      }
    ]
  },
  {
    "id": "ex521",
```

"input": "The payment request, the billing service can guarantee that at the next step it will be queued and it will remain pending until approval is granted, and that sooner or later it will be confirmed.",
"outputs": [
{
"formula": "<<BillingService>>(X payment_request_queued && (payment_request_pending U approval_granted) && F payment_request_confirmed)"
}
]
},
{
"id": "ex522",
"input": "The lane assignment, the controller and the dispatcher can guarantee that at the next step it will be updated and that traffic will remain rerouted until the obstruction is removed, and that sooner or later normal flow will resume.",
"outputs": [
{
"formula": "<<Controller,Dispatcher>>(X lane_assignment_updated && (traffic_rerouted U obstruction_removed) && F normal_flow_resumed)"
}
]
},
{
"id": "ex523",
"input": "The draft judgment, the clerk can guarantee that at the next step it will be published and that it will remain provisional until the appeal period expires, and that sooner or later it will become enforceable.",
"outputs": [
{
"formula": "<<Clerk>>(X draft_judgment_published && (draft_judgment_provisional U appeal_period_expired) && F judgment_enforceable)"
}
]
},
{
"id": "ex524",
"input": "The coordinator can guarantee that at the next step the notice will be sent and the issue will remain open until the response arrives, and that sooner or later the meeting will be scheduled, and the secretary can too.",
"outputs": [
{
"formula": "<<Coordinator>>(X notice_sent && (issue_open U response_arrived) && F meeting_scheduled) && <<Secretary>>(X notice_sent && (issue_open U response_arrived) && F meeting_scheduled)"
}
]
},
{

```
    "id": "ex525",
    "input": "The firewall can guarantee that at the next step the connection will be dropped
and that the endpoint will remain quarantined until the scan is completed, and that sooner or
later access will be re-enabled, and the gateway can too.",
    "outputs": [
      {
        "formula": "<<Firewall>>(X connection_dropped && (endpoint_quarantined U
scan_completed) && F access_reenabled) && <<Gateway>>(X connection_dropped &&
(endpoint_quarantined U scan_completed) && F access_reenabled)"
      }
    ]
  },
  {
    "id": "ex526",
    "input": "The editor can guarantee that at the next step the manuscript will be
acknowledged and the review will remain open until all comments are received, and that
sooner or later the decision will be communicated, and the publisher can too.",
    "outputs": [
      {
        "formula": "<<Editor>>(X manuscript_acknowledged && (review_open U
all_comments_received) && F decision_communicated) && <<Publisher>>(X
manuscript_acknowledged && (review_open U all_comments_received) && F
decision_communicated)"
      }
    ]
  },
  {
    "id": "ex527",
    "input": "The supervisor can guarantee that at the next step the alarm will sound and that
the area will remain closed until the inspection is completed, and that sooner or later the
incident will be logged, and the operator can too.",
    "outputs": [
      {
        "formula": "<<Supervisor>>(X alarm_sounded && (area_closed U
inspection_completed) && F incident_logged) && <<Operator>>(X alarm_sounded &&
(area_closed U inspection_completed) && F incident_logged)"
      }
    ]
  },
  {
    "id": "ex528",
    "input": "Every reviewer can guarantee that at the next step the rebuttal will be read and
the submission will remain undecided until the discussion ends, and that sooner or later the
decision will be recorded.",
    "outputs": [
      {
        "formula": "<<Reviewer1>>(X rebuttal_read_1 && (submission_undecided_1 U
discussion_ended_1) && F decision_recorded_1) && <<Reviewer2>>(X rebuttal_read_2 &&
```

(submission_undecided_2 U discussion_ended_2) && F decision_recorded_2) &&
<<Reviewer3>>(X rebuttal_read_3 && (submission_undecided_3 U discussion_ended_3)
&& F decision_recorded_3) && <<Reviewer4>>(X rebuttal_read_4 &&
(submission_undecided_4 U discussion_ended_4) && F decision_recorded_4) &&
<<Reviewer5>>(X rebuttal_read_5 && (submission_undecided_5 U discussion_ended_5)
&& F decision_recorded_5)"
    },
    {
      "formula": "<<Reviewer1,Reviewer2,Reviewer3,Reviewer4,Reviewer5>>(X
rebuttal_read && (submission_undecided U discussion_ended) && F decision_recorded)"
    }
  ]
},
Reviewer 1: I think the decision is unique.
Reviewer 2: I don’t agree since more reviewers express their own decision
Reviewer 1: I agree. Case closed.
{
  "id": "ex529",
  "input": "Every node can guarantee that at the next step the heartbeat will be emitted and the channel will remain active until synchronization is complete, and that sooner or later the snapshot will be recorded.",
  "outputs": [
    {
      "formula": "<<Node1>>(X heartbeat_emitted_1 && (channel_active_1 U
synchronization_complete_1) && F snapshot_recorded_1) && <<Node2>>(X
heartbeat_emitted_2 && (channel_active_2 U synchronization_complete_2) && F
snapshot_recorded_2) && <<Node3>>(X heartbeat_emitted_3 && (channel_active_3 U
synchronization_complete_3) && F snapshot_recorded_3)"
    },
    {
      "formula": "<<Node1,Node2,Node3>>(X heartbeat_emitted && (channel_active U
synchronization_complete) && F snapshot_recorded)"
    }
  ]
},
{
  "id": "ex530",
  "input": "Every clinician can guarantee that at the next step the alert will be acknowledged and the treatment will remain pending until the imaging report is signed, and that sooner or later therapy will start.",
  "outputs": [
    {
      "formula": "<<Clinician1>>(X alert_acknowledged_1 && (treatment_pending_1 U
imaging_report_signed_1) && F therapy_started_1) && <<Clinician2>>(X
alert_acknowledged_2 && (treatment_pending_2 U imaging_report_signed_2) && F
therapy_started_2)"
    },
    {

"formula": "<<Clinician1,Clinician2>>(X alert_acknowledged && (treatment_pending U imaging_report_signed) && F therapy_started)"
}
]
},
{
"id": "ex531",
"input": "Every delegate can guarantee that at the next step the ballot will be recorded and the vote will remain open until the count is completed, and that sooner or later the resolution will be announced.",
"outputs": [
{
"formula": "<<Delegate1>>(X ballot_recorded_1 && (vote_open_1 U count_completed_1) && F resolution_announced_1) && <<Delegate2>>(X ballot_recorded_2 && (vote_open_2 U count_completed_2) && F resolution_announced_2)"
},
{
"formula": "<<Delegate1,Delegate2>>(X ballot_recorded && (vote_open U count_completed) && F resolution_announced)"
}
]
},
{
"id": "ex532",
"input": "They can guarantee that at the next step the audit will begin, and that the evidence will remain sealed until the review is completed, and that no record will ever be altered, the inspector and the compliance system.",
"outputs": [
{
"formula": "<<Inspector,ComplianceSystem>>(X audit_started && (evidence_sealed U review_completed) && G !record_altered)"
}
]
},
Reviewer 1: note, variant of “never” here.
{
"id": "ex533",
"input": "It can guarantee that at the next step the cache will be flushed, and that the service will remain degraded until the warm-up ends, and that sooner or later full capacity will be restored, the deployment manager.",
"outputs": [
{
"formula": "<<DeploymentManager>>(X cache_flushed && (service_degraded U warmup_completed) && F full_capacity_restored)"
}
]
},
{

```
    "id": "ex534",
    "input": "They can guarantee that at the next step the evidence package will be transferred, and that the chain of custody will remain intact until the court receives it, and that tampering will never occur, the forensic officer and the registry.",
    "outputs": [
      {
        "formula": "<<ForensicOfficer,Registry>>(X evidence_package_transferred && (chain_of_custody_intact U court_received_package) && G !tampering)"
      }
    ]
  },
  {
    "id": "ex535",
    "input": "It can guarantee that at the next step the warning banner will appear, and that the account will remain restricted until the owner verifies it, and that sooner or later access will be restored, the identity service.",
    "outputs": [
      {
        "formula": "<<IdentityService>>(X warning_banner_displayed && (account_restricted U owner_verified) && F access_restored)"
      }
    ]
  },
  {
    "id": "ex536",
    "input": "The emergency protocol, the command center can guarantee that at the next step it will be activated, and that the area will remain evacuated until the hazard is cleared, and that sooner or later normal operations will resume.",
    "outputs": [
      {
        "formula": "<<CommandCenter>>(X emergency_protocol_activated && (area_evacuated U hazard_cleared) && F normal_operations_resumed)"
      }
    ]
  },
  {
    "id": "ex537",
    "input": "The consent form, the clinical platform can guarantee that at the next step it will be requested, and that the treatment will remain blocked until the patient signs it, and that unauthorized therapy will never start.",
    "outputs": [
      {
        "formula": "<<ClinicalPlatform>>(X consent_form_requested && (treatment_blocked U patient_signed) && G !unauthorized_therapy_started)"
      }
    ]
  },
  {
```

```
    "id": "ex538",
    "input": "The maintenance ticket, the facility manager can guarantee that at the next step it will be assigned, and that the elevator will remain offline until inspection is completed, and that sooner or later service will be restored.",
    "outputs": [
      {
        "formula": "<<FacilityManager>>(X maintenance_ticket_assigned && (elevator_offline U inspection_completed) && F service_restored)"
      }
    ]
  },
  {
    "id": "ex539",
    "input": "The encryption key, the key manager can guarantee that at the next step it will be rotated, and that the old key will remain disabled until revocation is confirmed, and that plaintext exposure will never occur.",
    "outputs": [
      {
        "formula": "<<KeyManager>>(X encryption_key_rotated && (old_key_disabled U revocation_confirmed) && G !plaintext_exposure)"
      }
    ]
  },
  {
    "id": "ex540",
    "input": "The coordinator can guarantee that at the next step the task will be dispatched, and that the queue will remain frozen until resources are released, and that sooner or later execution will resume, and the scheduler can too.",
    "outputs": [
      {
        "formula": "<<Coordinator>>(X task_dispatched && (queue_frozen U resources_released) && F execution_resumed) && <<Scheduler>>(X task_dispatched && (queue_frozen U resources_released) && F execution_resumed)"
      }
    ]
  },
  {
    "id": "ex541",
    "input": "The doctor can guarantee that at the next step the test will be ordered, and that the diagnosis will remain provisional until results arrive, and that sooner or later treatment will be adjusted, and the decision support system can too.",
    "outputs": [
      {
        "formula": "<<Doctor>>(X test_ordered && (diagnosis_provisional U results_arrived) && F treatment_adjusted) && <<DecisionSupportSystem>>(X test_ordered && (diagnosis_provisional U results_arrived) && F treatment_adjusted)"
      }
    ]
```

```
},
{
  "id": "ex542",
  "input": "The compliance officer can guarantee that at the next step the violation will be
reported, and that the account will remain suspended until remediation is verified, and that
further transactions will never be approved, and the risk engine can too.",
  "outputs": [
    {
      "formula": "<<ComplianceOfficer>>(X violation_reported && (account_suspended U
remediation_verified) && G !further_transactions_approved) && <<RiskEngine>>(X
violation_reported && (account_suspended U remediation_verified) && G
!further_transactions_approved)"
    }
  ]
},
{
  "id": "ex543",
  "input": "The editor can guarantee that at the next step the revision will be requested, and
that the manuscript will remain under review until the authors respond, and that sooner or
later a decision will be issued, and the associate editor can too.",
  "outputs": [
    {
      "formula": "<<Editor>>(X revision_requested && (manuscript_under_review U
authors_responded) && F decision_issued) && <<AssociateEditor>>(X revision_requested
&& (manuscript_under_review U authors_responded) && F decision_issued)"
    }
  ]
},
{
  "id": "ex544",
  "input": "Every controller can guarantee that at the next step the warning will be emitted,
and that the subsystem will remain isolated until diagnostics are completed, and that sooner
or later the subsystem will be reintegrated.",
  "outputs": [
    {
      "formula": "<<Controller1>>(X warning_emitted_1 && (subsystem_isolated_1 U
diagnostics_completed_1) && F subsystem_reintegrated_1) && <<Controller2>>(X
warning_emitted_2 && (subsystem_isolated_2 U diagnostics_completed_2) && F
subsystem_reintegrated_2) && <<Controller3>>(X warning_emitted_3 &&
(subsystem_isolated_3 U diagnostics_completed_3) && F subsystem_reintegrated_3)"
    },
    {
      "formula": "<<Controller1,Controller2,Controller3>>(X warning_emitted &&
(subsystem_isolated U diagnostics_completed) && F subsystem_reintegrated)"
    }
  ]
},
{
```

"id": "ex545",
"input": "Every auditor can guarantee that at the next step the finding will be logged, and that the case will remain open until corrective action is documented, and that no exception will ever be hidden.",
"outputs": [
{
"formula": "<<Auditor1>>(X finding_logged_1 && (case_open_1 U corrective_action_documented_1) && G !exception_hidden_1) && <<Auditor2>>(X finding_logged_2 && (case_open_2 U corrective_action_documented_2) && G !exception_hidden_2) && <<Auditor3>>(X finding_logged_3 && (case_open_3 U corrective_action_documented_3) && G !exception_hidden_3)"
},
{
"formula": "<<Auditor1,Auditor2,Auditor3>>(X finding_logged && (case_open U corrective_action_documented) && G !exception_hidden)"
}
]
},
{
"id": "ex546",
"input": "Every server can guarantee that at the next step the heartbeat will be sent, and that the session will remain alive until the lease expires, and that sooner or later a renewal will be attempted.",
"outputs": [
{
"formula": "<<Server1>>(X heartbeat_sent_1 && (session_alive_1 U lease_expired_1) && F renewal_attempted_1) && <<Server2>>(X heartbeat_sent_2 && (session_alive_2 U lease_expired_2) && F renewal_attempted_2) && <<Server3>>(X heartbeat_sent_3 && (session_alive_3 U lease_expired_3) && F renewal_attempted_3)"
},
{
"formula": "<<Server1,Server2,Server3>>(X heartbeat_sent && (session_alive U lease_expired) && F renewal_attempted)"
}
]
},
{
"id": "ex547",
"input": "Every officer can guarantee that at the next step the restriction will be imposed, and that the file will remain sealed until authorization is granted, and that disclosure will never occur before approval.",
"outputs": [
{
"formula": "<<Officer1>>(X restriction_imposed_1 && (file_sealed_1 U authorization_granted_1) &&G (!disclosure_1 U approval_granted_1)) && <<Officer2>>(X restriction_imposed_2 && (file_sealed_2 U authorization_granted_2) && G (!disclosure_2 U approval_granted_2)) && <<Officer3>>(X restriction_imposed_3 && (file_sealed_3 U authorization_granted_3) && G (!disclosure_3 U approval_granted_3))"

},
{
"formula": "<<Officer1,Officer2,Officer3>>(X restriction_imposed && (file_sealed U authorization_granted) && G (!disclosure U approval_granted))"
}
]
},

Reviewer 1: translation lacks “before approval”, it is an until.
Reviewer 2: If we want to stick with the 1 of 1 translation, I am not able to disagree.

{
"id": "ex548",
"input": "They can guarantee that at the next step the incident channel will be opened and the affected service will remain isolated until the recovery token is released, and that sooner or later the postmortem will be filed, the response team and the service owner.",
"outputs": [
{
"formula": "<<ResponseTeam,ServiceOwner>>(X incident_channel_opened && (affected_service_isolated U recovery_token_released) && F postmortem_filed)"
}
]
},
{
"id": "ex549",
"input": "It can guarantee that at the next step the quarantine flag will be raised and the executable will remain constrained until the sandbox releases it, and that no privileged call will be executed before validation, the endpoint protection module.",
"outputs": [
{
"formula": "<<EndpointProtectionModule>>(X quarantine_flag_raised && (executable_constrained U sandbox_released) && (!privileged_call_executed U validation_completed))"
}
]
},
{
"id": "ex550",
"input": "They can guarantee that at the next step the review board will be notified and the proposal will remain suspended until the legal opinion is released, and that sooner or later the revised protocol will be approved, the principal investigator and the ethics committee.",
"outputs": [
{
"formula": "<<PrincipalInvestigator,EthicsCommittee>>(X review_board_notified && (proposal_suspended U legal_opinion_released) && F revised_protocol_approved)"
}
]
},

{
"id": "ex551",
"input": "It can guarantee that at the next step the fallback route will be installed and the unstable link will remain avoided until the controller releases it, and that packet loss will never exceed the safety threshold, the routing supervisor.",
"outputs": [
{
"formula": "<<RoutingSupervisor>>(X fallback_route_installed && (unstable_link_avoided U controller_released_link) && G !packet_loss_over_threshold)"
}
]
},
{
"id": "ex552",
"input": "The escrow account, the payment service can guarantee that at the next step it will be locked and the funds will remain unavailable until the court releases them, and that eventually the transfer record will be archived.",
"outputs": [
{
"formula": "<<PaymentService>>(X escrow_account_locked && (funds_unavailable U court_released_funds) && F transfer_record_archived)"
}
]
},
{
"id": "ex553",
"input": "The clinical alert, the decision support system can guarantee that at the next step it will be displayed and the medication order will remain blocked until the physician releases it, and that adverse administration will never occur.",
"outputs": [
{
"formula": "<<DecisionSupportSystem>>(X clinical_alert_displayed && (medication_order_blocked U physician_released_order) && G !adverse_administration)"
}
]
},
{
"id": "ex554",
"input": "The deployment artifact, the release manager can guarantee that at the next step it will be signed and the rollout will remain paused until the security scan is completed, and that eventually the package will be released.",
"outputs": [
{
"formula": "<<ReleaseManager>>(X deployment_artifact_signed && (rollout_paused U security_scan_completed) && F package_released)"
}
]
},

{
"id": "ex555",
"input": "The emergency exit, the building controller can guarantee that at the next step it will be unlocked and the alarm will remain active until the fire marshal releases the area, and that evacuation will eventually be completed.",
"outputs": [
{
"formula": "<<BuildingController>>(X emergency_exit_unlocked && (alarm_active U area_released_by_fire_marshal) && F evacuation_completed)"
}
]
},
{
"id": "ex556",
"input": "The auditor can guarantee that at the next step the exception will be logged and the account will remain frozen until remediation is released, and that sooner or later the case will be closed, and the compliance engine can too.",
"outputs": [
{
"formula": "<<Auditor>>(X exception_logged && (account_frozen U remediation_released) && F case_closed) && <<ComplianceEngine>>(X exception_logged && (account_frozen U remediation_released) && F case_closed)"
}
]
},
{
"id": "ex557",
"input": "The dispatcher can guarantee that at the next step the evacuation order will be issued and the zone will remain restricted until the hazard clearance is released, and that no civilian entry will occur before clearance, and the command center can too.",
"outputs": [
{
"formula": "<<Dispatcher>>(X evacuation_order_issued && (zone_restricted U hazard_clearance_released) && (!civilian_entry U clearance_granted)) && <<CommandCenter>>(X evacuation_order_issued && (zone_restricted U hazard_clearance_released) && (!civilian_entry U clearance_granted))"
}
]
},
{
"id": "ex558",
"input": "The database engine can guarantee that at the next step the transaction will be marked pending and the row will remain locked until the commit token is released, and that sooner or later the log will be flushed, and the replication service can too.",
"outputs": [
{

```
      "formula": "<<DatabaseEngine>>(X transaction_marked_pending && (row_locked U commit_token_released) && F log_flushed) && <<ReplicationService>>(X transaction_marked_pending && (row_locked U commit_token_released) && F log_flushed)"
    }
  ]
},
{
  "id": "ex559",
  "input": "The clinician can guarantee that at the next step the imaging request will be submitted and the diagnosis will remain provisional until the radiology report is released, and that sooner or later the treatment plan will be updated, and the hospital system can too.",
  "outputs": [
    {
      "formula": "<<Clinician>>(X imaging_request_submitted && (diagnosis_provisional U radiology_report_released) && F treatment_plan_updated) && <<HospitalSystem>>(X imaging_request_submitted && (diagnosis_provisional U radiology_report_released) && F treatment_plan_updated)"
    }
  ]
},
{
  "id": "ex560",
  "input": "Every reviewer can guarantee that at the next step their individual score will be submitted and their assigned paper will remain under review until their personal report is released, and that the committee decision will remain pending until all reviews are released.",
  "outputs": [
    {
      "formula": "<<Reviewer1>>(X individual_score_submitted_1 && (assigned_paper_under_review_1 U personal_report_released_1)) && <<Reviewer2>>(X individual_score_submitted_2 && (assigned_paper_under_review_2 U personal_report_released_2)) && <<Reviewer3>>(X individual_score_submitted_3 && (assigned_paper_under_review_3 U personal_report_released_3)) && <<Reviewer4>>(X individual_score_submitted_4 && (assigned_paper_under_review_4 U personal_report_released_4)) && <<Reviewer1,Reviewer2,Reviewer3,Reviewer4>>(committee_decision_pending U all_reviews_released)"
    },
    {
      "formula": "<<Reviewer1,Reviewer2,Reviewer3,Reviewer4>>(X individual_scores_submitted && (assigned_papers_under_review U personal_reports_released) && (committee_decision_pending U all_reviews_released))"
    }
  ]
},
{
  "id": "ex561",
```

"input": "Every validator can guarantee that at the next step its local proof will be checked and its assigned block will remain tentative until its certificate is released.",
"outputs": [
{
"formula": "<<Validator1>>(X local_proof_checked_1 && (assigned_block_tentative_1 U certificate_released_1)) && <<Validator2>>(X local_proof_checked_2 && (assigned_block_tentative_2 U certificate_released_2)) && <<Validator3>>(X local_proof_checked_3 && (assigned_block_tentative_3 U certificate_released_3))"
},
{
"formula": "<<Validator1,Validator2,Validator3>>(X local_proof_checked && (assigned_block_tentative U certificate_released))"
}
]
},
{
"id": "ex562",
"input": "Every officer can guarantee that at the next step their individual restriction will be imposed and their assigned file will remain sealed until their authorization is granted.",
"outputs": [
{
"formula": "<<Officer1>>(X individual_restriction_imposed_1 && (assigned_file_sealed_1 U authorization_granted_1)) && <<Officer2>>(X individual_restriction_imposed_2 && (assigned_file_sealed_2 U authorization_granted_2)) && <<Officer3>>(X individual_restriction_imposed_3 && (assigned_file_sealed_3 U authorization_granted_3))"
},
{
"formula": "<<Officer1,Officer2,Officer3>>(X individual_restrictions_imposed && (assigned_files_sealed U authorizations_granted))"
}
]
},
{
"id": "ex563",
"input": "Every clinician can guarantee that at the next step their patient note will be signed and their assigned treatment will remain suspended until their dosage approval is released, and that the shared care plan will remain inactive until the full team releases it.",
"outputs": [
{
"formula": "<<Clinician1>>(X patient_note_signed_1 && (assigned_treatment_suspended_1 U dosage_approval_released_1)) && <<Clinician2>>(X patient_note_signed_2 && (assigned_treatment_suspended_2 U dosage_approval_released_2)) && <<Clinician1,Clinician2>>(shared_care_plan_inactive U full_team_released_plan)"
},
{

"formula": "<<Clinician1,Clinician2>>(X patient_notes_signed && (assigned_treatments_suspended U dosage_approvals_released) && (shared_care_plan_inactive U full_team_released_plan))"
}
]
},
{
"id": "ex564",
"input": "They can guarantee that on the very next transition the evacuation beacon will be armed and the stairwell will stay blocked up to the moment when the smoke sensor clears, and that before long the safety report will be completed, the floor warden and the building console.",
"outputs": [
{
"formula": "<<FloorWarden,BuildingConsole>>(X(evacuation_beacon_armed && (stairwell_blocked U smoke_sensor_cleared)) && F safety_report_completed)"

}
]
},
{
"id": "ex565",
"input": "It can guarantee that in the immediately following state the cooling cycle will be started and the chamber will remain sealed right up until the pressure test succeeds, and that from then on contamination will never be detected, the laboratory controller.",
"outputs": [
{
"formula": "<<LaboratoryController>>(X cooling_cycle_started && (chamber_sealed U pressure_test_succeeded) && G !contamination_detected)"
}
]
},
{
"id": "ex566",
"input": "They can guarantee that on the next transition the emergency channel will be opened and the rescue lane will stay reserved through to the point at which the convoy arrives, and that in due course the incident will be downgraded, the dispatcher and the traffic authority.",
"outputs": [
{
"formula": "<<Dispatcher,TrafficAuthority>>(X emergency_channel_opened && (rescue_lane_reserved U convoy_arrived) && F incident_downgraded)"
}
]
},
{
"id": "ex567",

"input": "It can guarantee that in the immediately succeeding state the inspection drone will be launched and the perimeter will remain restricted pending the clearance signal, and that at all later stages no unauthorized entry will be accepted, the site supervisor.",
"outputs": [
{
"formula": "<<SiteSupervisor>>(X inspection_drone_launched && (perimeter_restricted U clearance_signal_received) && G !unauthorized_entry_accepted)"
}
]
},
{
"id": "ex568",
"input": "The embargoed dataset, the research archive can guarantee that in the immediately following state it will be encrypted and access will remain disabled up to the moment when the disclosure date arrives, and that eventually the public copy will be published.",
"outputs": [
{
"formula": "<<ResearchArchive>>(X embargoed_dataset_encrypted && (access_disabled U disclosure_date_arrived) && F public_copy_published)"
}
]
},
{
"id": "ex569",
"input": "The appeal file, the tribunal clerk can guarantee that on the very next transition it will be registered and the ruling will remain suspended right up until the panel reaches a decision, and that after some finite delay the notification will be sent.",
"outputs": [
{
"formula": "<<TribunalClerk>>(X appeal_file_registered && (ruling_suspended U panel_decision_reached) && F notification_sent)"
}
]
},
{
"id": "ex570",
"input": "The cooling valve, the plant operator can guarantee that in the immediately succeeding state it will be opened and the turbine will remain throttled through to the point at which the vibration level normalizes, and that throughout the remaining execution overload will not occur.",
"outputs": [
{
"formula": "<<PlantOperator>>(X cooling_valve_opened && (turbine_throttled U vibration_normalized) && G !overload)"
}
]
},

{
"id": "ex571",
"input": "The witness statement, the case manager can guarantee that on the next transition it will be sealed and the dossier will remain confidential pending judicial authorization, and that before long the hearing schedule will be updated.",
"outputs": [
{
"formula": "<<CaseManager>>(X witness_statement_sealed && (dossier_confidential U judicial_authorization_granted) && F hearing_schedule_updated)"
}
]
},

{
"id": "ex572",
"input": "The curator can guarantee that in the immediately following state the artifact will be catalogued and the gallery will remain closed up to the moment when humidity stabilizes, and that in due course the exhibition will open, and the conservation officer can too.",
"outputs": [
{
"formula": "<<Curator>>(X artifact_catalogued && (gallery_closed U humidity_stabilized) && F exhibition_opened) && <<ConservationOfficer>>(X artifact_catalogued && (gallery_closed U humidity_stabilized) && F exhibition_opened)"
}
]
},
{
"id": "ex573",
"input": "The port authority can guarantee that on the very next transition the berth will be assigned and the vessel will remain anchored right up until the pilot boards, and that eventually unloading will begin, and the harbor master can too.",
"outputs": [
{
"formula": "<<PortAuthority>>(X berth_assigned && (vessel_anchored U pilot_boarded) && F unloading_started) && <<HarborMaster>>(X berth_assigned && (vessel_anchored U pilot_boarded) && F unloading_started)"
}
]
},
{
"id": "ex574",
"input": "The archivist can guarantee that in the immediately succeeding state the manuscript will be digitized and the original will remain locked through to the point at which preservation is certified, and that from then on physical access will never be granted without clearance, and the records officer can too.",
"outputs": [
{

```
        "formula": "<<Archivist>>(X manuscript_digitized && (original_locked U preservation_certified) && G !physical_access_without_clearance) && <<RecordsOfficer>>(X manuscript_digitized && (original_locked U preservation_certified) && G !physical_access_without_clearance)"
      }
    ]
  },
  {
    "id": "ex575",
    "input": "The procurement lead can guarantee that on the next transition the supplier will be notified and the purchase order will remain provisional pending budget confirmation, and that after some finite delay the contract will be countersigned, and the finance officer can too.",
    "outputs": [
      {
        "formula": "<<ProcurementLead>>(X supplier_notified && (purchase_order_provisional U budget_confirmed) && F contract_countersigned) && <<FinanceOfficer>>(X supplier_notified && (purchase_order_provisional U budget_confirmed) && F contract_countersigned)"
      }
    ]
  },
  {
    "id": "ex576",
    "input": "Every courier can guarantee that on the very next transition their assigned parcel will be scanned and their delivery route will remain active up to the moment when their delivery ends, and that before long their receipt will be issued.",
    "outputs": [
      {
        "formula": "<<Courier1>>(X assigned_parcel_scanned_1 && (route_active_1 U delivery_ended_1) && F receipt_issued_1) && <<Courier2>>(X assigned_parcel_scanned_2 && (route_active_2 U delivery_ended_2) && F receipt_issued_2) && <<Courier3>>(X assigned_parcel_scanned_3 && (route_active_3 U delivery_ended_3) && F receipt_issued_3)"
      },
      {
        "formula": "<<Courier1,Courier2,Courier3>>(X parcel_scanned && (distribution_route_active U delivery_ended) && F receipt_issued)"
      }
    ]
  },
  {
    "id": "ex577",
    "input": "Every lab technician can guarantee that in the immediately following state their sample will be labeled and their assay will remain running right up until their result is validated, and that in due course their report will be filed.",
    "outputs": [
      {
```

"formula": "<<Technician1>>(X sample_labeled_1 && (assay_running_1 U result_validated_1) && F report_filed_1) && <<Technician2>>(X sample_labeled_2 && (assay_running_2 U result_validated_2) && F report_filed_2) && <<Technician3>>(X sample_labeled_3 && (assay_running_3 U result_validated_3) && F report_filed_3)"
},
{
"formula": "<<Technician1,Technician2,Technician3>>(X batch_labeled && (assay_running U result_validated) && F report_filed)"
}
]
},
{
"id": "ex578",
"input": "Every moderator can guarantee that on the next transition their discussion thread will be flagged and their review queue will remain open through to the point at which their decision is logged, and that after some finite delay their notice will be posted.",
"outputs": [
{
"formula": "<<Moderator1>>(X thread_flagged_1 && (review_queue_open_1 U decision_logged_1) && F notice_posted_1) && <<Moderator2>>(X thread_flagged_2 && (review_queue_open_2 U decision_logged_2) && F notice_posted_2) && <<Moderator3>>(X thread_flagged_3 && (review_queue_open_3 U decision_logged_3) && F notice_posted_3)"
},
{
"formula": "<<Moderator1,Moderator2,Moderator3>>(X thread_flagged && (moderation_queue_open U decision_logged) && F notice_posted)"
}
]
},
{
"id": "ex579",
"input": "Every grant officer can guarantee that in the immediately succeeding state their application will be screened and their evaluation track will remain active pending their recommendation, and that eventually their award letter will be generated.",
"outputs": [
{
"formula": "<<GrantOfficer1>>(X application_screened_1 && (evaluation_track_active_1 U recommendation_reached_1) && F award_letter_generated_1) && <<GrantOfficer2>>(X application_screened_2 && (evaluation_track_active_2 U recommendation_reached_2) && F award_letter_generated_2)"
},
{
"formula": "<<GrantOfficer1,GrantOfficer2>>(X application_screened && (evaluation_track_active U panel_recommendation_reached) && F award_letter_generated)"
}
]
},

```
{
  "id": "ex580",
  "input": "They can guarantee that the archive will always remain protected, the librarian and the access system.",
  "outputs": [
    {
      "formula": "<<Librarian,AccessSystem>>G archive_protected"
    }
  ]
},
{
  "id": "ex581",
  "input": "It can guarantee that the control panel will never be exposed, the security module.",
  "outputs": [
    {
      "formula": "<<SecurityModule>>G !control_panel_exposed"
    }
  ]
},
{
  "id": "ex582",
  "input": "They can guarantee that the corridor will stay monitored at all times, the guard and the surveillance system.",
  "outputs": [
    {
      "formula": "<<Guard,SurveillanceSystem>>G corridor_monitored"
    }
  ]
},
{
  "id": "ex583",
  "input": "It can guarantee that invalid requests will be rejected in every state, the validation service.",
  "outputs": [
    {
      "formula": "<<ValidationService>>G invalid_requests_rejected"
    }
  ]
},
{
  "id": "ex584",
  "input": "The vault, the security officer can guarantee that it will always remain closed.",
  "outputs": [
    {
      "formula": "<<SecurityOfficer>>G vault_closed"
    }
  ]
```

},
{
"id": "ex585",
"input": "The patient record, the hospital platform can guarantee that it will never be altered without authorization.",
"outputs": [
{
"formula": "<<HospitalPlatform>>G !record_altered_without_authorization"
}
]
},
{
"id": "ex586",
"input": "The firewall policy, the administrator can guarantee that it will remain enforced throughout the whole execution.",
"outputs": [
{
"formula": "<<Administrator>>G firewall_policy_enforced"
}
]
},
{
"id": "ex587",
"input": "The emergency exit, the building controller can guarantee that it will stay accessible at all times.",
"outputs": [
{
"formula": "<<BuildingController>>G emergency_exit_accessible"
}
]
},
{
"id": "ex588",
"input": "The auditor can guarantee that the report will always remain consistent, and the compliance officer can too.",
"outputs": [
{
"formula": "<<Auditor>>G report_consistent && <<ComplianceOfficer>>G report_consistent"
}
]
},
{
"id": "ex589",
"input": "The gateway can guarantee that malicious traffic will never be forwarded, and the proxy can too.",
"outputs": [
{

"formula": "<<Gateway>>G !malicious_traffic_forwarded && <<Proxy>>G !malicious_traffic_forwarded"
}
]
},
{
"id": "ex590",
"input": "The doctor can guarantee that the prescription will remain valid throughout the whole execution, and the pharmacy system can too.",
"outputs": [
{
"formula": "<<Doctor>>G prescription_valid && <<PharmacySystem>>G prescription_valid"
}
]
},
{
"id": "ex591",
"input": "The moderator can guarantee that harmful content will always be blocked, and the platform can too.",
"outputs": [
{
"formula": "<<Moderator>>G harmful_content_blocked && <<Platform>>G harmful_content_blocked"
}
]
},
{
"id": "ex592",
"input": "Every inspector can guarantee that the facility will always remain compliant.",
"outputs": [
{
"formula": "<<Inspector1>>G facility_compliant_1 && <<Inspector2>>G facility_compliant_2 && <<Inspector3>>G facility_compliant_3"
},
{
"formula": "<<Inspector1,Inspector2,Inspector3>>G facility_compliant"
}
]
},
{
"id": "ex593",
"input": "Every reviewer can guarantee that the submission will never be exposed.",
"outputs": [
{
"formula": "<<Reviewer1>>G !submission_exposed_1 && <<Reviewer2>>G !submission_exposed_2 && <<Reviewer3>>G !submission_exposed_3"
},

```
      {
        "formula": "<<Reviewer1,Reviewer2,Reviewer3>>G !submission_exposed"
      }
    ]
  },
  {
    "id": "ex594",
    "input": "Every controller can guarantee that the system will stay stable at all times.",
    "outputs": [
      {
        "formula": "<<Controller1>>G system_stable_1 && <<Controller2>>G system_stable_2"
      },
      {
        "formula": "<<Controller1,Controller2>>G system_stable"
      }
    ]
  },
  {
    "id": "ex595",
    "input": "Every officer can guarantee that the restricted area will remain secure throughout the whole execution.",
    "outputs": [
      {
        "formula": "<<Officer1>>G restricted_area_secure_1 && <<Officer2>>G restricted_area_secure_2 && <<Officer3>>G restricted_area_secure_3"
      },
      {
        "formula": "<<Officer1,Officer2,Officer3>>G restricted_area_secure"
      }
    ]
  },
  {
    "id": "ex596",
    "input": "They can guarantee that the access log will remain available throughout the whole execution, the administrator and the audit service.",
    "outputs": [
      {
        "formula": "<<Administrator,AuditService>>G access_log_available"
      }
    ]
  },
  {
    "id": "ex597",
    "input": "It can guarantee that the backup channel will always stay encrypted, the communication service.",
    "outputs": [
      {
        "formula": "<<CommunicationService>>G backup_channel_encrypted"
```

```
      }
    ]
  },
  {
    "id": "ex598",
    "input": "They can guarantee that unsafe commands will never be executed, the supervisor and the control interface.",
    "outputs": [
      {
        "formula": "<<Supervisor,ControlInterface>>G !unsafe_commands_executed"
      }
    ]
  },
  {
    "id": "ex599",
    "input": "It can guarantee that the emergency signal will be visible in every state, the warning system.",
    "outputs": [
      {
        "formula": "<<WarningSystem>>G emergency_signal_visible"
      }
    ]
  },
  {
    "id": "ex600",
    "input": "The audit trail, the compliance platform can guarantee that it will always remain complete.",
    "outputs": [
      {
        "formula": "<<CompliancePlatform>>G audit_trail_complete"
      }
    ]
  },
  {
    "id": "ex601",
    "input": "The sterile room, the hospital controller can guarantee that it will never be accessed without clearance.",
    "outputs": [
      {
        "formula": "<<HospitalController>>G !sterile_room_accessed_without_clearance"
      }
    ]
  },
  {
    "id": "ex602",
    "input": "The voting record, the election system can guarantee that it will remain immutable at all times.",
    "outputs": [
```

```
      {
        "formula": "<<ElectionSystem>>G voting_record_immutable"
      }
    ]
  },
  {
    "id": "ex603",
    "input": "The safety barrier, the plant controller can guarantee that it will stay active throughout the whole execution.",
    "outputs": [
      {
        "formula": "<<PlantController>>G safety_barrier_active"
      }
    ]
  },
  {
    "id": "ex604",
    "input": "The registrar can guarantee that the enrollment file will always remain valid, and the academic office can too.",
    "outputs": [
      {
        "formula": "<<Registrar>>G enrollment_file_valid && <<AcademicOffice>>G enrollment_file_valid"
      }
    ]
  },
  {
    "id": "ex605",
    "input": "The security analyst can guarantee that suspicious sessions will never be ignored, and the monitoring console can too.",
    "outputs": [
      {
        "formula": "<<SecurityAnalyst>>G !suspicious_sessions_ignored && <<MonitoringConsole>>G !suspicious_sessions_ignored"
      }
    ]
  },
  {
    "id": "ex606",
    "input": "The nurse can guarantee that the medication label will remain readable at all times, and the pharmacy unit can too.",
    "outputs": [
      {
        "formula": "<<Nurse>>G medication_label_readable && <<PharmacyUnit>>G medication_label_readable"
      }
    ]
  },
```

{
"id": "ex607",
"input": "The archivist can guarantee that restricted documents will always remain sealed, and the records office can too.",
"outputs": [
{
"formula": "<<Archivist>>G restricted_documents_sealed && <<RecordsOffice>>G restricted_documents_sealed"
}
]
},
{
"id": "ex608",
"input": "Every analyst can guarantee that the dashboard will always remain accurate.",
"outputs": [
{
"formula": "<<Analyst1>>G dashboard_accurate_1 && <<Analyst2>>G dashboard_accurate_2 && <<Analyst3>>G dashboard_accurate_3"
},
{
"formula": "<<Analyst1,Analyst2,Analyst3>>G dashboard_accurate"
}
]
},
{
"id": "ex609",
"input": "Every pharmacist can guarantee that the prescription will never be unsafe.",
"outputs": [
{
"formula": "<<Pharmacist1>>G !prescription_unsafe_1 && <<Pharmacist2>>G !prescription_unsafe_2 && <<Pharmacist3>>G !prescription_unsafe_3"
},
{
"formula": "<<Pharmacist1,Pharmacist2,Pharmacist3>>G !prescription_unsafe"
}
]
},
{
"id": "ex610",
"input": "Every delegate can guarantee that the vote will remain confidential throughout the whole execution.",
"outputs": [
{
"formula": "<<Delegate1>>G vote_confidential_1 && <<Delegate2>>G vote_confidential_2 && <<Delegate3>>G vote_confidential_3"
},
{
"formula": "<<Delegate1,Delegate2,Delegate3>>G vote_confidential"

```
    }
   ]
  },
  {
   "id": "ex611",
   "input": "Every technician can guarantee that the calibration state will stay correct in every state.",
   "outputs": [
    {
     "formula": "<<Technician1>>G calibration_state_correct_1 && <<Technician2>>G calibration_state_correct_2 && <<Technician3>>G calibration_state_correct_3"
    },
    {
     "formula": "<<Technician1,Technician2,Technician3>>G calibration_state_correct"
    }
   ]
  },
  {
   "id": "ex612",
   "input": "They can guarantee that the courtroom feed will remain protected at all times, the clerk and the media server.",
   "outputs": [
    {
     "formula": "<<Clerk,MediaServer>>G courtroom_feed_protected"
    }
   ]
  },
  {
   "id": "ex613",
   "input": "It can guarantee that expired credentials will never be accepted, the identity gateway.",
   "outputs": [
    {
     "formula": "<<IdentityGateway>>G !expired_credentials_accepted"
    }
   ]
  },
  {
   "id": "ex614",
   "input": "They can guarantee that the navigation display will stay synchronized throughout the whole execution, the pilot and the avionics system.",
   "outputs": [
    {
     "formula": "<<Pilot,AvionicsSystem>>G navigation_display_synchronized"
    }
   ]
  },
  {
```

```
    "id": "ex615",
    "input": "It can guarantee that the fail-safe mode will remain enabled in every state, the industrial controller.",
    "outputs": [
      {
        "formula": "<<IndustrialController>>G fail_safe_mode_enabled"
      }
    ]
  },
  {
    "id": "ex616",
    "input": "The emergency log, the command platform can guarantee that it will always remain accessible.",
    "outputs": [
      {
        "formula": "<<CommandPlatform>>G emergency_log_accessible"
      }
    ]
  },
  {
    "id": "ex617",
    "input": "The protected folder, the document manager can guarantee that it will never be opened without permission.",
    "outputs": [
      {
        "formula": "<<DocumentManager>>G !protected_folder_opened_without_permission"
      }
    ]
  },
  {
    "id": "ex618",
    "input": "The oxygen supply, the clinical controller can guarantee that it will remain available at all times.",
    "outputs": [
      {
        "formula": "<<ClinicalController>>G oxygen_supply_available"
      }
    ]
  },
  {
    "id": "ex619",
    "input": "The voting terminal, the election supervisor can guarantee that it will stay operational throughout the whole execution.",
    "outputs": [
      {
        "formula": "<<ElectionSupervisor>>G voting_terminal_operational"
      }
    ]
```

```
},
{
  "id": "ex620",
  "input": "The gatekeeper can guarantee that unauthorized guests will never enter, and the reception system can too.",
  "outputs": [
    {
      "formula": "<<Gatekeeper>>G !unauthorized_guests_enter && <<ReceptionSystem>>G !unauthorized_guests_enter"
    }
  ]
},
{
  "id": "ex621",
  "input": "The librarian can guarantee that the catalogue will always remain searchable, and the archive server can too.",
  "outputs": [
    {
      "formula": "<<Librarian>>G catalogue_searchable && <<ArchiveServer>>G catalogue_searchable"
    }
  ]
},
{
  "id": "ex622",
  "input": "The paramedic can guarantee that the stretcher path will remain clear at all times, and the hospital dispatcher can too.",
  "outputs": [
    {
      "formula": "<<Paramedic>>G stretcher_path_clear && <<HospitalDispatcher>>G stretcher_path_clear"
    }
  ]
},
{
  "id": "ex623",
  "input": "The treasurer can guarantee that the balance sheet will never be inconsistent, and the accounting system can too.",
  "outputs": [
    {
      "formula": "<<Treasurer>>G !balance_sheet_inconsistent && <<AccountingSystem>>G !balance_sheet_inconsistent"
    }
  ]
},
{
  "id": "ex624",
```

```
    "input": "Every inspector can guarantee that the checkpoint will remain guarded in every state.",
    "outputs": [
      {
        "formula": "<<Inspector1>>G checkpoint_guarded_1 && <<Inspector2>>G checkpoint_guarded_2"
      },
      {
        "formula": "<<Inspector1,Inspector2>>G checkpoint_guarded"
      }
    ]
  },
  {
    "id": "ex625",
    "input": "Every nurse can guarantee that the monitor will always remain active.",
    "outputs": [
      {
        "formula": "<<Nurse1>>G monitor_active_1 && <<Nurse2>>G monitor_active_2 && <<Nurse3>>G monitor_active_3"
      },
      {
        "formula": "<<Nurse1,Nurse2,Nurse3>>G monitor_active"
      }
    ]
  },
  {
    "id": "ex626",
    "input": "Every administrator can guarantee that the service will never expose private data.",
    "outputs": [
      {
        "formula": "<<Administrator1>>G !private_data_exposed_1 && <<Administrator2>>G !private_data_exposed_2"
      },
      {
        "formula": "<<Administrator1,Administrator2>>G !private_data_exposed"
      }
    ]
  },
  {
    "id": "ex627",
    "input": "Every supervisor can guarantee that the safety protocol will stay enforced throughout the whole execution.",
    "outputs": [
      {
        "formula": "<<Supervisor1>>G safety_protocol_enforced_1 && <<Supervisor2>>G safety_protocol_enforced_2 && <<Supervisor3>>G safety_protocol_enforced_3"
      },
```

```
    {
      "formula": "<<Supervisor1,Supervisor2,Supervisor3>>G safety_protocol_enforced"
    }
   ]
  },
  {
   "id": "ex628",
   "input": "They can guarantee that the evidence room will stay under surveillance
throughout the whole execution, the detective and the custody system.",
   "outputs": [
    {
      "formula": "<<Detective,CustodySystem>>G evidence_room_under_surveillance"
    }
   ]
  },
  {
   "id": "ex629",
   "input": "It can guarantee that revoked badges will never unlock the entrance, the access
controller.",
   "outputs": [
    {
      "formula": "<<AccessController>>G !revoked_badges_unlock_entrance"
    }
   ]
  },
  {
   "id": "ex630",
   "input": "They can guarantee that the sterile corridor will remain uncontaminated at all
times, the surgeon and the ventilation system.",
   "outputs": [
    {
      "formula": "<<Surgeon,VentilationSystem>>G sterile_corridor_uncontaminated"
    }
   ]
  },
  {
   "id": "ex631",
   "input": "It can guarantee that the emergency broadcast will remain audible in every state,
the public warning network.",
   "outputs": [
    {
      "formula": "<<PublicWarningNetwork>>G emergency_broadcast_audible"
    }
   ]
  },

  {
   "id": "ex632",
```

```
   "input": "The sealed ballot box, the election officer can guarantee that it will always remain intact.",
   "outputs": [
    {
     "formula": "<<ElectionOfficer>>G sealed_ballot_box_intact"
    }
   ]
  },
  {
   "id": "ex633",
   "input": "The restricted database, the data steward can guarantee that it will never be queried without authorization.",
   "outputs": [
    {
     "formula": "<<DataSteward>>G !restricted_database_queried_without_authorization"
    }
   ]
  },
  {
   "id": "ex634",
   "input": "The pressure regulator, the plant supervisor can guarantee that it will stay calibrated throughout the whole execution.",
   "outputs": [
    {
     "formula": "<<PlantSupervisor>>G pressure_regulator_calibrated"
    }
   ]
  },
  {
   "id": "ex635",
   "input": "The emergency ward, the hospital coordinator can guarantee that it will remain staffed at all times.",
   "outputs": [
    {
     "formula": "<<HospitalCoordinator>>G emergency_ward_staffed"
    }
   ]
  },

  {
   "id": "ex636",
   "input": "The safety marshal can guarantee that the evacuation route will always remain visible, and the lighting system can too.",
   "outputs": [
    {
     "formula": "<<SafetyMarshal>>G evacuation_route_visible && <<LightingSystem>>G evacuation_route_visible"
    }
```

```
    ]
  },
  {
    "id": "ex637",
    "input": "The database administrator can guarantee that corrupted records will never be committed, and the transaction manager can too.",
    "outputs": [
      {
        "formula": "<<DatabaseAdministrator>>G !corrupted_records_committed && <<TransactionManager>>G !corrupted_records_committed"
      }
    ]
  },
  {
    "id": "ex638",
    "input": "The registrar can guarantee that the student ledger will remain consistent in every state, and the academic platform can too.",
    "outputs": [
      {
        "formula": "<<Registrar>>G student_ledger_consistent && <<AcademicPlatform>>G student_ledger_consistent"
      }
    ]
  },
  {
    "id": "ex639",
    "input": "The harbor officer can guarantee that the navigation channel will stay clear throughout the execution, and the port radar can too.",
    "outputs": [
      {
        "formula": "<<HarborOfficer>>G navigation_channel_clear && <<PortRadar>>G navigation_channel_clear"
      }
    ]
  },
  {
    "id": "ex640",
    "input": "Every archivist can guarantee that the register will always remain complete.",
    "outputs": [
      {
        "formula": "<<Archivist1>>G register_complete_1 && <<Archivist2>>G register_complete_2 && <<Archivist3>>G register_complete_3"
      },
      {
        "formula": "<<Archivist1,Archivist2,Archivist3>>G register_complete"
      }
    ]
  },
```

```
{
  "id": "ex641",
  "input": "Every guard can guarantee that the entrance will never be left unattended.",
  "outputs": [
   {
    "formula": "<<Guard1>>G !entrance_unattended_1 && <<Guard2>>G
!entrance_unattended_2 && <<Guard3>>G !entrance_unattended_3"
   },
   {
    "formula": "<<Guard1,Guard2,Guard3>>G !entrance_unattended"
   }
  ]
 },
 {
  "id": "ex642",
  "input": "Every controller can guarantee that the alarm line will remain operational at all
times.",
  "outputs": [
   {
    "formula": "<<Controller1>>G alarm_line_operational_1 && <<Controller2>>G
alarm_line_operational_2 && <<Controller3>>G alarm_line_operational_3 &&
<<Controller4>>G alarm_line_operational_4"
   },
   {
    "formula": "<<Controller1,Controller2,Controller3,Controller4>>G
alarm_line_operational"
   }
  ]
 },
 {
  "id": "ex643",
  "input": "Every clerk can guarantee that the filing cabinet will stay locked throughout the
whole execution.",
  "outputs": [
   {
    "formula": "<<Clerk1>>G filing_cabinet_locked_1 && <<Clerk2>>G
filing_cabinet_locked_2"
   },
   {
    "formula": "<<Clerk1,Clerk2>>G filing_cabinet_locked"
   }
  ]
 },
 {
  "id": "ex644",
  "input": "They can guarantee that the quarantine ward will remain isolated at all times, the
infection officer and the access system.",
  "outputs": [
```

```
    {
      "formula": "<<InfectionOfficer,AccessSystem>>G quarantine_ward_isolated"
    }
  ]
},
{
  "id": "ex645",
  "input": "It can guarantee that incomplete applications will never be approved, the admissions platform.",
  "outputs": [
    {
      "formula": "<<AdmissionsPlatform>>G !incomplete_applications_approved"
    }
  ]
},
{
  "id": "ex646",
  "input": "They can guarantee that the archive index will stay consistent throughout the whole execution, the records manager and the indexing service.",
  "outputs": [
    {
      "formula": "<<RecordsManager,IndexingService>>G archive_index_consistent"
    }
  ]
},
{
  "id": "ex647",
  "input": "It can guarantee that the warning light will remain visible in every state, the safety console.",
  "outputs": [
    {
      "formula": "<<SafetyConsole>>G warning_light_visible"
    }
  ]
},
{
  "id": "ex648",
  "input": "The restricted corridor, the security desk can guarantee that it will always remain monitored.",
  "outputs": [
    {
      "formula": "<<SecurityDesk>>G restricted_corridor_monitored"
    }
  ]
},
{
  "id": "ex649",
```

```
    "input": "The medical chart, the clinical registry can guarantee that it will never be modified without consent.",
    "outputs": [
      {
        "formula": "<<ClinicalRegistry>>G !medical_chart_modified_without_consent"
      }
    ]
  },
  {
    "id": "ex650",
    "input": "The emergency generator, the facility controller can guarantee that it will stay ready throughout whole the execution.",
    "outputs": [
      {
        "formula": "<<FacilityController>>G emergency_generator_ready"
      }
    ]
  },
  {
    "id": "ex651",
    "input": "The legal archive, the court platform can guarantee that it will remain searchable at all times.",
    "outputs": [
      {
        "formula": "<<CourtPlatform>>G legal_archive_searchable"
      }
    ]
  },
  {
    "id": "ex652",
    "input": "The dispatcher can guarantee that the rescue channel will always remain open, and the radio network can too.",
    "outputs": [
      {
        "formula": "<<Dispatcher>>G rescue_channel_open && <<RadioNetwork>>G rescue_channel_open"
      }
    ]
  },
  {
    "id": "ex653",
    "input": "The examiner can guarantee that invalid answers will never be accepted, and the grading system can too.",
    "outputs": [
      {
        "formula": "<<Examiner>>G !invalid_answers_accepted && <<GradingSystem>>G !invalid_answers_accepted"
      }
```

    ]
  },
  {
    "id": "ex654",
    "input": "The curator can guarantee that the exhibit label will remain readable in every state, and the museum system can too.",
    "outputs": [
      {
        "formula": "<<Curator>>G exhibit_label_readable && <<MuseumSystem>>G exhibit_label_readable"
      }
    ]
  },
  {
    "id": "ex655",
    "input": "The compliance lead can guarantee that audit evidence will stay preserved throughout the whole execution, and the storage service can too.",
    "outputs": [
      {
        "formula": "<<ComplianceLead>>G audit_evidence_preserved && <<StorageService>>G audit_evidence_preserved"
      }
    ]
  },
  {
    "id": "ex656",
    "input": "Every physician can guarantee that the diagnosis will always remain documented.",
    "outputs": [
      {
        "formula": "<<Physician1>>G diagnosis_documented_1 && <<Physician2>>G diagnosis_documented_2"
      },
      {
        "formula": "<<Physician1,Physician2>>G diagnosis_documented"
      }
    ]
  },
  {
    "id": "ex657",
    "input": "Every operator can guarantee that the console will never display unsafe values.",
    "outputs": [
      {
        "formula": "<<Operator1>>G !unsafe_values_displayed_1 && <<Operator2>>G !unsafe_values_displayed_2 && <<Operator3>>G !unsafe_values_displayed_3"
      },
      {
        "formula": "<<Operator1,Operator2,Operator3>>G !unsafe_values_displayed"

```
    }
   ]
  },
  {
   "id": "ex658",
   "input": "Every registrar can guarantee that the enrollment table will remain accurate at all
times.",
   "outputs": [
    {
     "formula": "<<Registrar1>>G enrollment_table_accurate_1 && <<Registrar2>>G
enrollment_table_accurate_2"
    },
    {
     "formula": "<<Registrar1,Registrar2>>G enrollment_table_accurate"
    }
   ]
  },
  {
   "id": "ex659",
   "input": "Every technician can guarantee that the diagnostic panel will stay responsive
throughout the whole execution.",
   "outputs": [
    {
     "formula": "<<Technician1>>G diagnostic_panel_responsive_1 && <<Technician2>>G
diagnostic_panel_responsive_2"
    },
    {
     "formula": "<<Technician1,Technician2>>G diagnostic_panel_responsive"
    }
   ]
  },
  {
   "id": "ex660",
   "input": "They can guarantee that the neonatal unit will remain supervised at all times, the
head nurse and the monitoring station.",
   "outputs": [
    {
     "formula": "<<HeadNurse,MonitoringStation>>G neonatal_unit_supervised"
    }
   ]
  },
  {
   "id": "ex661",
   "input": "It can guarantee that forged signatures will never be accepted, the verification
service.",
   "outputs": [
    {
     "formula": "<<VerificationService>>G !forged_signatures_accepted"
```

```
    }
   ]
  },
  {
   "id": "ex662",
   "input": "They can guarantee that the emergency register will stay synchronized throughout the whole execution, the coordinator and the records server.",
   "outputs": [
    {
     "formula": "<<Coordinator,RecordsServer>>G emergency_register_synchronized"
    }
   ]
  },
  {
   "id": "ex663",
   "input": "It can guarantee that the hazard indicator will remain active in every state, the safety monitor.",
   "outputs": [
    {
     "formula": "<<SafetyMonitor>>G hazard_indicator_active"
    }
   ]
  },

  {
   "id": "ex664",
   "input": "The quarantine log, the hospital platform can guarantee that it will always remain complete.",
   "outputs": [
    {
     "formula": "<<HospitalPlatform>>G quarantine_log_complete"
    }
   ]
  },
  {
   "id": "ex665",
   "input": "The financial ledger, the accounting service can guarantee that it will never contain unbalanced entries.",
   "outputs": [
    {
     "formula": "<<AccountingService>>G !financial_ledger_unbalanced"
    }
   ]
  },
  {
   "id": "ex666",
   "input": "The backup generator, the operations console can guarantee that it will stay available throughout the whole execution.",
```

"outputs": [
{
"formula": "<<OperationsConsole>>G backup_generator_available"
}
]
},
{
"id": "ex667",
"input": "The sealed archive, the records authority can guarantee that it will remain protected at all times.",
"outputs": [
{
"formula": "<<RecordsAuthority>>G sealed_archive_protected"
}
]
},
{
"id": "ex668",
"input": "The reviewer can guarantee that the assessment form will always remain valid, and the quality board can too.",
"outputs": [
{
"formula": "<<Reviewer>>G assessment_form_valid && <<QualityBoard>>G assessment_form_valid"
}
]
},
{
"id": "ex669",
"input": "The security guard can guarantee that the restricted door will never be left open, and the access console can too.",
"outputs": [
{
"formula": "<<SecurityGuard>>G !restricted_door_left_open && <<AccessConsole>>G !restricted_door_left_open"
}
]
},
{
"id": "ex670",
"input": "The pharmacist can guarantee that controlled medication will stay traceable in every state, and the inventory platform can too.",
"outputs": [
{
"formula": "<<Pharmacist>>G controlled_medication_traceable && <<InventoryPlatform>>G controlled_medication_traceable"
}
]

```
},
{
  "id": "ex671",
  "input": "The examiner can guarantee that the answer key will remain confidential
throughout the whole execution, and the exam server can too.",
  "outputs": [
    {
      "formula": "<<Examiner>>G answer_key_confidential && <<ExamServer>>G
answer_key_confidential"
    }
  ]
},
{
  "id": "ex672",
  "input": "Every supervisor can guarantee that the checklist will always remain complete.",
  "outputs": [
    {
      "formula": "<<Supervisor1>>G checklist_complete_1 && <<Supervisor2>>G
checklist_complete_2 && <<Supervisor3>>G checklist_complete_3"
    },
    {
      "formula": "<<Supervisor1,Supervisor2,Supervisor3>>G checklist_complete"
    }
  ]
},
{
  "id": "ex673",
  "input": "Every monitor can guarantee that the alert channel will never become silent.",
  "outputs": [
    {
      "formula": "<<Monitor1>>G !alert_channel_silent_1 && <<Monitor2>>G
!alert_channel_silent_2"
    },
    {
      "formula": "<<Monitor1,Monitor2>>G !alert_channel_silent"
    }
  ]
},
{
  "id": "ex674",
  "input": "Every archivist can guarantee that the catalogue will remain searchable at all
times.",
  "outputs": [
    {
      "formula": "<<Archivist1>>G catalogue_searchable_1 && <<Archivist2>>G
catalogue_searchable_2 && <<Archivist3>>G catalogue_searchable_3"
    },
    {
```

```
        "formula": "<<Archivist1,Archivist2,Archivist3>>G catalogue_searchable"
      }
    ]
  },
  {
    "id": "ex675",
    "input": "Every coordinator can guarantee that the service desk will stay reachable
throughout the whole execution.",
    "outputs": [
      {
        "formula": "<<Coordinator1>>G service_desk_reachable_1 && <<Coordinator2>>G
service_desk_reachable_2 && <<Coordinator3>>G service_desk_reachable_3 &&
<<Coordinator4>>G service_desk_reachable_4"
      },
      {
        "formula": "<<Coordinator1,Coordinator2,Coordinator3,Coordinator4>>G
service_desk_reachable"
      }
    ]
  },
  {
    "id": "ex676",
    "input": "They can guarantee that the isolation room will stay secure at all times, the ward
supervisor and the access terminal.",
    "outputs": [
      {
        "formula": "<<WardSupervisor,AccessTerminal>>G isolation_room_secure"
      }
    ]
  },
  {
    "id": "ex677",
    "input": "It can guarantee that invalid badges will never open the staff entrance, the badge
reader.",
    "outputs": [
      {
        "formula": "<<BadgeReader>>G !invalid_badges_open_staff_entrance"
      }
    ]
  },
  {
    "id": "ex678",
    "input": "They can guarantee that the evacuation map will remain visible throughout the
whole execution, the safety officer and the display system.",
    "outputs": [
      {
        "formula": "<<SafetyOfficer,DisplaySystem>>G evacuation_map_visible"
      }
```

```
    ]
  },
  {
    "id": "ex679",
    "input": "It can guarantee that the sterile cabinet will remain locked in every state, the
laboratory access unit.",
    "outputs": [
      {
        "formula": "<<LaboratoryAccessUnit>>G sterile_cabinet_locked"
      }
    ]
  },
  {
    "id": "ex680",
    "input": "The incident ledger, the response platform can guarantee that it will always
remain consistent.",
    "outputs": [
      {
        "formula": "<<ResponsePlatform>>G incident_ledger_consistent"
      }
    ]
  },
  {
    "id": "ex681",
    "input": "The medication drawer, the pharmacy controller can guarantee that it will never
be opened without clearance.",
    "outputs": [
      {
        "formula": "<<PharmacyController>>G !medication_drawer_opened_without_clearance"
      }
    ]
  },
  {
    "id": "ex682",
    "input": "The court calendar, the scheduling office can guarantee that it will stay accurate
at all times.",
    "outputs": [
      {
        "formula": "<<SchedulingOffice>>G court_calendar_accurate"
      }
    ]
  },
  {
    "id": "ex683",
    "input": "The alarm console, the facility system can guarantee that it will remain
responsive throughout the whole execution.",
    "outputs": [
      {
```

"formula": "<<FacilitySystem>>G alarm_console_responsive"
}
]
},
{
"id": "ex684",
"input": "The data steward can guarantee that private identifiers will never be exposed, and the anonymization service can too.",
"outputs": [
{
"formula": "<<DataSteward>>G !private_identifiers_exposed && <<AnonymizationService>>G !private_identifiers_exposed"
}
]
},
{
"id": "ex685",
"input": "The flight dispatcher can guarantee that the runway status will always remain current, and the tower system can too.",
"outputs": [
{
"formula": "<<FlightDispatcher>>G runway_status_current && <<TowerSystem>>G runway_status_current"
}
]
},
{
"id": "ex686",
"input": "The infection specialist can guarantee that isolation signs will remain posted in every state, and the ward system can too.",
"outputs": [
{
"formula": "<<InfectionSpecialist>>G isolation_signs_posted && <<WardSystem>>G isolation_signs_posted"
}
]
},
{
"id": "ex687",
"input": "The records clerk can guarantee that sealed minutes will stay confidential throughout the whole execution, and the archive platform can too.",
"outputs": [
{
"formula": "<<RecordsClerk>>G sealed_minutes_confidential && <<ArchivePlatform>>G sealed_minutes_confidential"
}
]
},

{
"id": "ex688",
"input": "Every examiner can guarantee that the rubric will always remain visible.",
"outputs": [
{
"formula": "<<Examiner1>>G rubric_visible_1 && <<Examiner2>>G rubric_visible_2 &&
<<Examiner3>>G rubric_visible_3"
},
{
"formula": "<<Examiner1,Examiner2,Examiner3>>G rubric_visible"
}
]
},
{
"id": "ex689",
"input": "Every guard can guarantee that the checkpoint will never be bypassed.",
"outputs": [
{
"formula": "<<Guard1>>G !checkpoint_bypassed_1 && <<Guard2>>G
!checkpoint_bypassed_2"
},
{
"formula": "<<Guard1,Guard2>>G !checkpoint_bypassed"
}
]
},
{
"id": "ex690",
"input": "Every technician can guarantee that the sensor feed will remain calibrated at all
times.",
"outputs": [
{
"formula": "<<Technician1>>G sensor_feed_calibrated_1 && <<Technician2>>G
sensor_feed_calibrated_2 && <<Technician3>>G sensor_feed_calibrated_3"
},
{
"formula": "<<Technician1,Technician2,Technician3>>G sensor_feed_calibrated"
}
]
},
{
"id": "ex691",
"input": "Every coordinator can guarantee that the hotline will stay available throughout the
whole execution.",
"outputs": [
{
"formula": "<<Coordinator1>>G hotline_available_1 && <<Coordinator2>>G
hotline_available_2"

      },
      {
        "formula": "<<Coordinator1,Coordinator2,Coordinator3>>G hotline_available"
      }
    ]
  },
Reviewer 2: The following come from paraphrasis of Declaration of Independence, Gettysburg Address, Churchill's "We shall fight on the beaches", Magna Carta, Federalist No. 51, Sun Tzu's *The Art of War*, e "I Have a Dream".
  {
    "id": "ex692",
    "input": "If the people find that a form of government becomes destructive of their safety and happiness, then they can eventually alter or abolish it and eventually institute a new government that maintains those rights always secure.",
    "outputs": [
      {
        "formula": "government_destructive -> <<People>>F (government_altered || government_abolished) && F new_government_instituted && G rights_secured"
      }
    ]
  },
  {
    "id": "ex693",
    "input": "The living can dedicate themselves to the unfinished work and eventually resolve that the dead shall not have died in vain, and always ensure that government of the people shall not perish from the earth.",
    "outputs": [
      {
        "formula": "<<Living>>dedicated_to_unfinished_work && F !vain_death && G !government_of_people_perishes"
      }
    ]
  },
  {
    "id": "ex694",
    "input": "The defenders can fight on the beaches, on the landing grounds, in the fields, in the streets, and in the hills, and they can guarantee that they will never surrender.",
    "outputs": [
      {
        "formula": "<<Defenders>>(fight_on_beaches && fight_on_landing_grounds && fight_in_fields && fight_in_streets && fight_in_hills) && G !surrender"
      }
    ]
  },
  {
    "id": "ex695",
    "input": "No free man shall be seized, imprisoned, stripped of his rights, exiled, or destroyed except by lawful judgment or by the law of the land.",

"outputs": [
{
"formula": "<<Law>>G ((lawful_judgment || law_of_land) || (!seized && !imprisoned && !rights_stripped && !exiled && !destroyed)) || <<Law_of_the_man>>G ((lawful_judgment || law_of_land) || (!seized && !imprisoned && !rights_stripped && !exiled && !destroyed)) "
}
]
},
Reviewer 2: This example is very hard and useful to analyze as we can assist to a double right dislocation together with a use of or intended as an inclusive or (i.e. not as the humans intend it in reality). Since in natural language humans are prone to treat the “or” as a “xor”, then this translation is extremely loyal to a real world scenario.
Reviewer 1: I agree. A beautiful example.
{
"id": "ex696",
"input": "The citizens can keep watch over at all stages public power and resist abuses when they arise, and ensure that liberty is preserved throughout the whole states.",
"outputs": [
{
"formula": "<<Citizens>>G watch_over_public_power && G (abuse_arises -> abuse_resisted) && G liberty_preserved"
}
]
},
{
"id": "ex697",
"input": "The commander has a strategy to eventually attack when the enemy is unprepared, after some finite time appear when he is not expected, and keep these devices undisclosed until victory is secured.",
"outputs": [
{
"formula": "<<Commander>>(F (appear_unexpectedly && XF attack_unprepared_enemy) && (devices_undisclosed U victory_secured))"
}
]
},
Reviewer 1: XF translates “after some finite time” because it is relativized to a preceding event within the same scope, in order to enforce that the second event occurs strictly after the first one.
Reviewer 2: I agree.
{
"id": "ex698",
"input": "If the enemy takes his ease, then the general can always give him no rest; and if his forces are united, then the general can always separate them; and eventually the path to victory will be opened.",
"outputs": [
{

```
      "formula": "(enemy_takes_ease -><<General>> G !(enemy_rests)) && (forces_united -> <<General>> G forces_separated) && F path_to_victory_opened"
  },
  {
    "id": "ex699",
    "input": "The marchers can refuse to drink from the cup of bitterness, conduct the struggle with dignity and discipline, and continue until justice rolls down like waters.",
    "outputs": [
      {
        "formula": "<<Marchers>>(!bitterness_accepted && dignity_and_discipline && (struggle_continues U justice_arrives))"
      }
    ]
  },
  {
    "id": "ex700",
    "input": "They can guarantee that on the next transition the access token will be refreshed, the client and the identity service.",
    "outputs": [
      {
        "formula": "<<Client,IdentityService>>X access_token_refreshed"
      }
    ]
  },
  {
    "id": "ex701",
    "input": "It can guarantee that in the immediately following state the alert will be acknowledged, the monitoring console.",
    "outputs": [
      {
        "formula": "<<MonitoringConsole>>X alert_acknowledged"
      }
    ]
  },
  {
    "id": "ex702",
    "input": "They can guarantee that in the successor state the file will be transferred, the sender and the receiver.",
    "outputs": [
      {
        "formula": "<<Sender,Receiver>>X file_transferred"
      }
    ]
  },
  {
    "id": "ex703",
    "input": "It can guarantee that after one transition the warning light will be enabled, the safety controller.",
```

```
    "outputs": [
      {
        "formula": "<<SafetyController>>X warning_light_enabled"
      }
    ]
  },
  {
    "id": "ex704",
    "input": "The access token, the identity service can guarantee that it will be refreshed in the immediately following state.",
    "outputs": [
      {
        "formula": "<<IdentityService>>X access_token_refreshed"
      }
    ]
  },
  {
    "id": "ex705",
    "input": "The alarm message, the control center can guarantee that it will be dispatched on the next transition.",
    "outputs": [
      {
        "formula": "<<ControlCenter>>X alarm_message_dispatched"
      }
    ]
  },
  {
    "id": "ex706",
    "input": "The backup channel, the communication system can guarantee that it will be opened in the successor state.",
    "outputs": [
      {
        "formula": "<<CommunicationSystem>>X backup_channel_opened"
      }
    ]
  },
  {
    "id": "ex707",
    "input": "The court notice, the clerk can guarantee that it will be issued after one transition.",
    "outputs": [
      {
        "formula": "<<Clerk>>X court_notice_issued"
      }
    ]
  },
  {
    "id": "ex708",
```

```
    "input": "The registrar can guarantee that in the immediately following state the file will be
checked, and the academic office can too.",
    "outputs": [
      {
        "formula": "<<Registrar>>X file_checked && <<AcademicOffice>>X file_checked"
      }
    ]
  },
  {
    "id": "ex709",
    "input": "The gateway can guarantee that on the next transition the packet will be dropped,
and the firewall can too.",
    "outputs": [
      {
        "formula": "<<Gateway>>X packet_dropped && <<Firewall>>X packet_dropped"
      }
    ]
  },
  {
    "id": "ex710",
    "input": "The physician can guarantee that in the successor state the prescription will be
signed, and the clinical platform can too.",
    "outputs": [
      {
        "formula": "<<Physician>>X prescription_signed && <<ClinicalPlatform>>X
prescription_signed"
      }
    ]
  },
  {
    "id": "ex711",
    "input": "The editor can guarantee that after one transition the manuscript will be
forwarded, and the publisher can too.",
    "outputs": [
      {
        "formula": "<<Editor>>X manuscript_forwarded && <<Publisher>>X
manuscript_forwarded"
      }
    ]
  },
  {
    "id": "ex712",
    "input": "Every reviewer can guarantee that on the next transition the evaluation will be
submitted.",
    "outputs": [
      {
        "formula": "<<Reviewer1>>X evaluation_submitted_1 && <<Reviewer2>>X
evaluation_submitted_2 && <<Reviewer3>>X evaluation_submitted_3"
```

```
      },
      {
        "formula": "<<Reviewer1,Reviewer2,Reviewer3>>X evaluation_submitted"
      }
    ]
  },
  {
    "id": "ex713",
    "input": "Every controller can guarantee that in the immediately following state the signal will be emitted.",
    "outputs": [
      {
        "formula": "<<Controller1>>X signal_emitted_1 && <<Controller2>>X signal_emitted_2 && <<Controller3>>X signal_emitted_3"
      },
      {
        "formula": "<<Controller1,Controller2,Controller3>>X signal_emitted"
      }
    ]
  },
  {
    "id": "ex714",
    "input": "Every officer can guarantee that in the successor state the restriction will be imposed.",
    "outputs": [
      {
        "formula": "<<Officer1>>X restriction_imposed_1 && <<Officer2>>X restriction_imposed_2 && <<Officer3>>X restriction_imposed_3"
      },
      {
        "formula": "<<Officer1,Officer2,Officer3>>X restriction_imposed"
      }
    ]
  },
  {
    "id": "ex715",
    "input": "Every technician can guarantee that after one transition the diagnostic panel will be activated.",
    "outputs": [
      {
        "formula": "<<Technician1>>X diagnostic_panel_activated_1 && <<Technician2>>X diagnostic_panel_activated_2 && <<Technician3>>X diagnostic_panel_activated_3"
      },
      {
        "formula": "<<Technician1,Technician2,Technician3>>X diagnostic_panel_activated"
      }
    ]
  },
```

{
"id": "ex716",
"input": "The scheduler can guarantee that on the next transition the queued task will be dispatched.",
"outputs": [
{
"formula": "<<Scheduler>>X queued_task_dispatched"
}
]
},
{
"id": "ex717",
"input": "The hospital system can guarantee that in the immediately following state the patient record will be updated.",
"outputs": [
{
"formula": "<<HospitalSystem>>X patient_record_updated"
}
]
},
{
"id": "ex718",
"input": "The payment gateway can guarantee that after one transition the transaction will be authorized.",
"outputs": [
{
"formula": "<<PaymentGateway>>X transaction_authorized"
}
]
},
{
"id": "ex719",
"input": "The warehouse system can guarantee that in the successor state the package label will be printed.",
"outputs": [
{
"formula": "<<WarehouseSystem>>X package_label_printed"
}
]
},
{
"id": "ex720",
"input": "The climate controller can guarantee that in the next state the ventilation mode will be enabled.",
"outputs": [
{
"formula": "<<ClimateController>>X ventilation_mode_enabled"
}

```
    ]
  },
  {
    "id": "ex721",
    "input": "They can guarantee that sooner or later the missing invoice will be recovered, the accountant and the archive service.",
    "outputs": [
      {
        "formula": "<<Accountant,ArchiveService>>F missing_invoice_recovered"
      }
    ]
  },
  {
    "id": "ex722",
    "input": "It can guarantee that in due course the access request will be approved, the authorization portal.",
    "outputs": [
      {
        "formula": "<<AuthorizationPortal>>F access_request_approved"
      }
    ]
  },
  {
    "id": "ex723",
    "input": "They can guarantee that before long the emergency report will be delivered, the dispatcher and the response unit.",
    "outputs": [
      {
        "formula": "<<Dispatcher,ResponseUnit>>F emergency_report_delivered"
      }
    ]
  },
  {
    "id": "ex724",
    "input": "It can guarantee that eventually the maintenance ticket will be resolved, the facility platform.",
    "outputs": [
      {
        "formula": "<<FacilityPlatform>>F maintenance_ticket_resolved"
      }
    ]
  },
  {
    "id": "ex725",
    "input": "They can guarantee that sooner or later the appeal will be examined, the clerk and the review panel.",
    "outputs": [
      {
```

"formula": "<<Clerk,ReviewPanel>>F appeal_examined"
}
]
},
{
"id": "ex726",
"input": "It can guarantee that at some future point the shipment will be cleared, the customs service.",
"outputs": [
{
"formula": "<<CustomsService>>F shipment_cleared"
}
]
},
{
"id": "ex727",
"input": "They can guarantee that sooner or later the diagnosis will be confirmed, the physician and the clinical system.",
"outputs": [
{
"formula": "<<Physician,ClinicalSystem>>F diagnosis_confirmed"
}
]
},
{
"id": "ex728",
"input": "It can guarantee that eventually the backup copy will be restored, the storage service.",
"outputs": [
{
"formula": "<<StorageService>>F backup_copy_restored"
}
]
},
{
"id": "ex729",
"input": "They can guarantee that eventually the public notice will be issued, the authority and the registry.",
"outputs": [
{
"formula": "<<Authority,Registry>>F public_notice_issued"
}
]
},
{
"id": "ex730",
"input": "It can guarantee that sooner or later the alarm record will be archived, the monitoring console.",

```
    "outputs": [
      {
        "formula": "<<MonitoringConsole>>F alarm_record_archived"
      }
    ]
  },
  {
    "id": "ex731",
    "input": "The missing invoice, the accountant can guarantee that sooner or later it will be recovered.",
    "outputs": [
      {
        "formula": "<<Accountant>>F missing_invoice_recovered"
      }
    ]
  },
  {
    "id": "ex732",
    "input": "The access request, the authorization portal can guarantee that in due course it will be approved.",
    "outputs": [
      {
        "formula": "<<AuthorizationPortal>>F access_request_approved"
      }
    ]
  },
  {
    "id": "ex733",
    "input": "The emergency report, the dispatcher and the response unit can guarantee that before long it will be delivered.",
    "outputs": [
      {
        "formula": "<<Dispatcher,ResponseUnit>>F emergency_report_delivered"
      }
    ]
  },
  {
    "id": "ex734",
    "input": "The maintenance ticket, the facility platform can guarantee that sooner or later it will be resolved.",
    "outputs": [
      {
        "formula": "<<FacilityPlatform>>F maintenance_ticket_resolved"
      }
    ]
  },
  {
    "id": "ex735",
```

```
    "input": "The appeal, the clerk and the review panel can always guarantee that eventually
it will be examined.",
    "outputs": [
      {
        "formula": "<<Clerk,ReviewPanel>>G (F appeal_examined)"
      }
    ]
  },
  {
    "id": "ex736",
    "input": "The shipment, the customs service can guarantee that at some future point it will
be cleared.",
    "outputs": [
      {
        "formula": "<<CustomsService>>F shipment_cleared"
      }
    ]
  },
  {
    "id": "ex737",
    "input": "The diagnosis, the physician and the clinical system can guarantee that
eventually it will be confirmed.",
    "outputs": [
      {
        "formula": "<<Physician,ClinicalSystem>>F diagnosis_confirmed"
      }
    ]
  },
  {
    "id": "ex738",
    "input": "The backup copy, the storage service can guarantee that eventually it will be
restored.",
    "outputs": [
      {
        "formula": "<<StorageService>>F backup_copy_restored"
      }
    ]
  },
  {
    "id": "ex739",
    "input": "The public notice, the authority and the registry can guarantee that in the end it
will be issued.",
    "outputs": [
      {
        "formula": "<<Authority,Registry>>F public_notice_issued"
      }
    ]
  },
```

```
 {
   "id": "ex740",
   "input": "The alarm record, the monitoring console can guarantee that after finitely many steps it will be archived.",
   "outputs": [
     {
       "formula": "<<MonitoringConsole>>F alarm_record_archived"
     }
   ]
 },

 {
   "id": "ex741",
   "input": "The accountant can guarantee that sooner or later the missing invoice will be recovered, and the archive service can too.",
   "outputs": [
     {
       "formula": "<<Accountant>>F missing_invoice_recovered && <<ArchiveService>>F missing_invoice_recovered"
     }
   ]
 },
 {
   "id": "ex742",
   "input": "The authorization portal can guarantee that in due course the access request will be approved, and the compliance desk can too.",
   "outputs": [
     {
       "formula": "<<AuthorizationPortal>>F access_request_approved && <<ComplianceDesk>>F access_request_approved"
     }
   ]
 },
 {
   "id": "ex743",
   "input": "The dispatcher can guarantee that before long the emergency report will be delivered, and the response unit can too.",
   "outputs": [
     {
       "formula": "<<Dispatcher>>F emergency_report_delivered && <<ResponseUnit>>F emergency_report_delivered"
     }
   ]
 },
 {
   "id": "ex744",
   "input": "The facility platform can guarantee that after some finite delay the maintenance ticket will be resolved, and the service desk can too.",
```

"outputs": [
{
"formula": "<<FacilityPlatform>>F maintenance_ticket_resolved && <<ServiceDesk>>F maintenance_ticket_resolved"
}
]
},
{
"id": "ex745",
"input": "The clerk can guarantee that ultimately the appeal will be examined, and the review panel can too.",
"outputs": [
{
"formula": "<<Clerk>>F appeal_examined && <<ReviewPanel>>F appeal_examined"
}
]
},
{
"id": "ex746",
"input": "The customs service can guarantee that at some future point the shipment will be cleared, and the port authority can too.",
"outputs": [
{
"formula": "<<CustomsService>>F shipment_cleared && <<PortAuthority>>F shipment_cleared"
}
]
},
{
"id": "ex747",
"input": "The physician can guarantee that eventually the diagnosis will be confirmed, and the clinical system can too.",
"outputs": [
{
"formula": "<<Physician>>F diagnosis_confirmed && <<ClinicalSystem>>F diagnosis_confirmed"
}
]
},
{
"id": "ex748",
"input": "The storage service can guarantee that eventually the backup copy will be restored, and the recovery manager can too.",
"outputs": [
{
"formula": "<<StorageService>>F backup_copy_restored && <<RecoveryManager>>F backup_copy_restored"
}

]
},
{
"id": "ex749",
"input": "The authority can guarantee that in the end the public notice will be issued, and the registry can too.",
"outputs": [
{
"formula": "<<Authority>>F public_notice_issued && <<Registry>>F public_notice_issued"
}
]
},
{
"id": "ex750",
"input": "The monitoring console can guarantee that after finitely many steps the alarm record will be archived, and the audit service can too.",
"outputs": [
{
"formula": "<<MonitoringConsole>>F alarm_record_archived && <<AuditService>>F alarm_record_archived"
}
]
},
{
"id": "ex751",
"input": "Every accountant can guarantee that sooner or later the invoice will be recovered.",
"outputs": [
{
"formula": "<<Accountant1>>F invoice_recovered_1 && <<Accountant2>>F invoice_recovered_2 && <<Accountant3>>F invoice_recovered_3"
},
{
"formula": "<<Accountant1,Accountant2,Accountant3>>F invoice_recovered"
}
]
},
{
"id": "ex752",
"input": "Every officer can guarantee that in due course the access request will be approved.",
"outputs": [
{
"formula": "<<Officer1>>F access_request_approved_1 && <<Officer2>>F access_request_approved_2 && <<Officer3>>F access_request_approved_3"
},
{

```
        "formula": "<<Officer1,Officer2,Officer3>>F access_request_approved"
      }
    ]
  },
  {
    "id": "ex753",
    "input": "Every dispatcher can guarantee that before long the emergency report will be
delivered.",
    "outputs": [
      {
        "formula": "<<Dispatcher1>>F emergency_report_delivered_1 && <<Dispatcher2>>F
emergency_report_delivered_2 && <<Dispatcher3>>F emergency_report_delivered_3"
      },
      {
        "formula": "<<Dispatcher1,Dispatcher2,Dispatcher3>>F emergency_report_delivered"
      }
    ]
  },
  {
    "id": "ex754",
    "input": "Every technician can guarantee that after some finite delay the maintenance
ticket will be resolved.",
    "outputs": [
      {
        "formula": "<<Technician1>>F maintenance_ticket_resolved_1 && <<Technician2>>F
maintenance_ticket_resolved_2 && <<Technician3>>F maintenance_ticket_resolved_3"
      },
      {
        "formula": "<<Technician1,Technician2,Technician3>>F maintenance_ticket_resolved"
      }
    ]
  },
  {
    "id": "ex755",
    "input": "Every clerk can guarantee that ultimately the appeal will be examined.",
    "outputs": [
      {
        "formula": "<<Clerk1>>F appeal_examined_1 && <<Clerk2>>F appeal_examined_2 &&
<<Clerk3>>F appeal_examined_3"
      },
      {
        "formula": "<<Clerk1,Clerk2,Clerk3>>F appeal_examined"
      }
    ]
  },
  {
    "id": "ex756",
```

```
    "input": "Every customs agent can guarantee that at some future point the shipment will be
cleared.",
    "outputs": [
      {
        "formula": "<<CustomsAgent1>>F shipment_cleared_1 && <<CustomsAgent2>>F
shipment_cleared_2 && <<CustomsAgent3>>F shipment_cleared_3"
      },
      {
        "formula": "<<CustomsAgent1,CustomsAgent2,CustomsAgent3>>F shipment_cleared"
      }
    ]
  },
  {
    "id": "ex757",
    "input": "Every physician can guarantee that eventually the diagnosis will be confirmed.",
    "outputs": [
      {
        "formula": "<<Physician1>>F diagnosis_confirmed_1 && <<Physician2>>F
diagnosis_confirmed_2 && <<Physician3>>F diagnosis_confirmed_3"
      },
      {
        "formula": "<<Physician1,Physician2,Physician3>>F diagnosis_confirmed"
      }
    ]
  },
  {
    "id": "ex758",
    "input": "Every storage service can guarantee that eventually the backup copy will be
restored.",
    "outputs": [
      {
        "formula": "<<StorageService1>>F backup_copy_restored_1 &&
<<StorageService2>>F backup_copy_restored_2 && <<StorageService3>>F
backup_copy_restored_3"
      },
      {
        "formula": "<<StorageService1,StorageService2,StorageService3>>F
backup_copy_restored"
      }
    ]
  },
  {
    "id": "ex759",
    "input": "Every authority can guarantee that in the end the public notice will be issued.",
    "outputs": [
      {
        "formula": "<<Authority1>>F public_notice_issued_1 && <<Authority2>>F
public_notice_issued_2"
```

    },
    {
      "formula": "<<Authority1,Authority2>>F public_notice_issued"
    }
  ]
},
{
  "id": "ex760",
  "input": "They can guarantee that sooner or later the missing permit will be issued, the applicant and the municipal office.",
  "outputs": [
    {
      "formula": "<<Applicant,MunicipalOffice>>F missing_permit_issued"
    }
  ]
},
{
  "id": "ex761",
  "input": "It can guarantee that in due course the service outage will be resolved, the operations center.",
  "outputs": [
    {
      "formula": "<<OperationsCenter>>F service_outage_resolved"
    }
  ]
},
{
  "id": "ex762",
  "input": "They can guarantee that before long the damaged bridge will be inspected, the engineer and the safety authority.",
  "outputs": [
    {
      "formula": "<<Engineer,SafetyAuthority>>F damaged_bridge_inspected"
    }
  ]
},
{
  "id": "ex763",
  "input": "It can guarantee that after finitely many steps the certificate will be renewed, the licensing platform.",
  "outputs": [
    {
      "formula": "<<LicensingPlatform>>F certificate_renewed"
    }
  ]
},
{
  "id": "ex764",

```
    "input": "They can guarantee that ultimately the grant proposal will be evaluated, the coordinator and the funding board.",
    "outputs": [
      {
        "formula": "<<Coordinator,FundingBoard>>F grant_proposal_evaluated"
      }
    ]
  },
  {
    "id": "ex765",
    "input": "It can guarantee that at some future point the quarantine order will be lifted, the health authority.",
    "outputs": [
      {
        "formula": "<<HealthAuthority>>F quarantine_order_lifted"
      }
    ]
  },
  {
    "id": "ex766",
    "input": "They can guarantee that eventually the training record will be validated, the instructor and the certification system.",
    "outputs": [
      {
        "formula": "<<Instructor,CertificationSystem>>F training_record_validated"
      }
    ]
  },
  {
    "id": "ex767",
    "input": "It can guarantee that eventually the rejected transaction will be reviewed, the banking platform.",
    "outputs": [
      {
        "formula": "<<BankingPlatform>>F rejected_transaction_reviewed"
      }
    ]
  },
  {
    "id": "ex768",
    "input": "They can guarantee that in the end the evacuation plan will be approved, the fire officer and the building manager.",
    "outputs": [
      {
        "formula": "<<FireOfficer,BuildingManager>>F evacuation_plan_approved"
      }
    ]
  },
```

{
"id": "ex769",
"input": "It can guarantee that after some finite delay the corrupted archive will be restored, the recovery service.",
"outputs": [
{
"formula": "<<RecoveryService>>F corrupted_archive_restored"
}
]
},
{
"id": "ex770",
"input": "The missing permit, the applicant can guarantee that sooner or later it will be issued.",
"outputs": [
{
"formula": "<<Applicant>>F missing_permit_issued"
}
]
},
{
"id": "ex771",
"input": "The service outage, the operations center can guarantee that in due course it will be resolved.",
"outputs": [
{
"formula": "<<OperationsCenter>>F service_outage_resolved"
}
]
},
{
"id": "ex772",
"input": "The damaged bridge, the engineer and the safety authority can guarantee that before long it will be inspected.",
"outputs": [
{
"formula": "<<Engineer,SafetyAuthority>>F damaged_bridge_inspected"
}
]
},
{
"id": "ex773",
"input": "The certificate, the licensing platform can guarantee that after finitely many steps it will be renewed.",
"outputs": [
{
"formula": "<<LicensingPlatform>>F certificate_renewed"
}

]
},
{
"id": "ex774",
"input": "The grant proposal, the coordinator and the funding board can guarantee that ultimately it will be evaluated.",
"outputs": [
{
"formula": "<<Coordinator,FundingBoard>>F grant_proposal_evaluated"
}
]
},
{
"id": "ex775",
"input": "The quarantine order, the health authority can guarantee that at some future point it will be lifted.",
"outputs": [
{
"formula": "<<HealthAuthority>>F quarantine_order_lifted"
}
]
},
{
"id": "ex776",
"input": "The training record, the instructor and the certification system can guarantee that in time it will be validated.",
"outputs": [
{
"formula": "<<Instructor,CertificationSystem>>F training_record_validated"
}
]
},
{
"id": "ex777",
"input": "The rejected transaction, the banking platform can guarantee that eventually it will be reviewed.",
"outputs": [
{
"formula": "<<BankingPlatform>>F rejected_transaction_reviewed"
}
]
},
{
"id": "ex778",
"input": "The evacuation plan, the fire officer and the building manager can guarantee that in the end it will be approved.",
"outputs": [
{

"formula": "<<FireOfficer,BuildingManager>>F evacuation_plan_approved"
}
]
},
{
"id": "ex779",
"input": "The corrupted archive, the recovery service can guarantee that after some finite delay it will be restored.",
"outputs": [
{
"formula": "<<RecoveryService>>F corrupted_archive_restored"
}
]
},
{
"id": "ex780",
"input": "The applicant can guarantee that sooner or later the missing permit will be issued, and the municipal office can too.",
"outputs": [
{
"formula": "<<Applicant>>F missing_permit_issued && <<MunicipalOffice>>F missing_permit_issued"
}
]
},
{
"id": "ex781",
"input": "The operations center can guarantee that in due course the service outage will be resolved, and the field team can too.",
"outputs": [
{
"formula": "<<OperationsCenter>>F service_outage_resolved && <<FieldTeam>>F service_outage_resolved"
}
]
},
{
"id": "ex782",
"input": "The engineer can guarantee that before long the damaged bridge will be inspected, and the safety authority can too.",
"outputs": [
{
"formula": "<<Engineer>>F damaged_bridge_inspected && <<SafetyAuthority>>F damaged_bridge_inspected"
}
]
},
{

"id": "ex783",
"input": "The licensing platform can guarantee that after finitely many steps the certificate will be renewed, and the registrar can too.",
"outputs": [
{
"formula": "<<LicensingPlatform>>F certificate_renewed && <<Registrar>>F certificate_renewed"
}
]
},
{
"id": "ex784",
"input": "The coordinator can guarantee that ultimately the grant proposal will be evaluated, and the funding board can too.",
"outputs": [
{
"formula": "<<Coordinator>>F grant_proposal_evaluated && <<FundingBoard>>F grant_proposal_evaluated"
}
]
},
{
"id": "ex785",
"input": "The health authority can guarantee that at some future point the quarantine order will be lifted, and the clinic network can too.",
"outputs": [
{
"formula": "<<HealthAuthority>>F quarantine_order_lifted && <<ClinicNetwork>>F quarantine_order_lifted"
}
]
},
{
"id": "ex786",
"input": "The instructor can guarantee that eventually the training record will be validated, and the certification system can too.",
"outputs": [
{
"formula": "<<Instructor>>F training_record_validated && <<CertificationSystem>>F training_record_validated"
}
]
},
{
"id": "ex787",
"input": "The banking platform can guarantee that eventually the rejected transaction will be reviewed, and the compliance unit can too.",
"outputs": [

{
"formula": "<<BankingPlatform>>F rejected_transaction_reviewed &&
<<ComplianceUnit>>F rejected_transaction_reviewed"
}
]
},
{
"id": "ex788",
"input": "The fire officer can guarantee that in the end the evacuation plan will be
approved, and the building manager can too.",
"outputs": [
{
"formula": "<<FireOfficer>>F evacuation_plan_approved && <<BuildingManager>>F
evacuation_plan_approved"
}
]
},
{
"id": "ex789",
"input": "The recovery service can guarantee that after some finite delay the corrupted
archive will be restored, and the backup manager can too.",
"outputs": [
{
"formula": "<<RecoveryService>>F corrupted_archive_restored &&
<<BackupManager>>F corrupted_archive_restored"
}
]
},
{
"id": "ex790",
"input": "Every applicant can guarantee that sooner or later the permit will be issued.",
"outputs": [
{
"formula": "<<Applicant1>>F permit_issued_1 && <<Applicant2>>F permit_issued_2"
},
{
"formula": "<<Applicant1,Applicant2>>F permit_issued"
}
]
},
{
"id": "ex791",
"input": "Every operator can guarantee that in due course the outage will be resolved.",
"outputs": [
{
"formula": "<<Operator1>>F outage_resolved_1 && <<Operator2>>F
outage_resolved_2 && <<Operator3>>F outage_resolved_3 && <<Operator4>>F
outage_resolved_4 && <<Operator5>>F outage_resolved_5"

},
{
"formula": "<<Operator1,Operator2,Operator3,Operator4,Operator5>>F
outage_resolved"
}
]
},
{
"id": "ex792",
"input": "Every engineer can guarantee that before long the bridge will be inspected.",
"outputs": [
{
"formula": "<<Engineer1>>F bridge_inspected_1 && <<Engineer2>>F
bridge_inspected_2 && <<Engineer3>>F bridge_inspected_3"
},
{
"formula": "<<Engineer1,Engineer2,Engineer3>>F bridge_inspected"
}
]
},
{
"id": "ex793",
"input": "Every registrar can guarantee that after finitely many steps the certificate will be
renewed.",
"outputs": [
{
"formula": "<<Registrar1>>F certificate_renewed_1 && <<Registrar2>>F
certificate_renewed_2 && <<Registrar3>>F certificate_renewed_3"
},
{
"formula": "<<Registrar1,Registrar2,Registrar3>>F certificate_renewed"
}
]
},
{
"id": "ex794",
"input": "Every coordinator can guarantee that ultimately the proposal will be evaluated.",
"outputs": [
{
"formula": "<<Coordinator1>>F proposal_evaluated_1 && <<Coordinator2>>F
proposal_evaluated_2 && <<Coordinator3>>F proposal_evaluated_3"
},
{
"formula": "<<Coordinator1,Coordinator2,Coordinator3>>F proposal_evaluated"
}
]
},
{

```
    "id": "ex795",
    "input": "Every health officer can guarantee that at some future point the order will be lifted.",
    "outputs": [
      {
        "formula": "<<HealthOfficer1>>F order_lifted_1 && <<HealthOfficer2>>F order_lifted_2 && <<HealthOfficer3>>F order_lifted_3"
      },
      {
        "formula": "<<HealthOfficer1,HealthOfficer2,HealthOfficer3>>F order_lifted"
      }
    ]
  },
  {
    "id": "ex796",
    "input": "Every instructor can guarantee that eventually the record will be validated.",
    "outputs": [
      {
        "formula": "<<Instructor1>>F record_validated_1 && <<Instructor2>>F record_validated_2 && <<Instructor3>>F record_validated_3"
      },
      {
        "formula": "<<Instructor1,Instructor2,Instructor3>>F record_validated"
      }
    ]
  },
  {
    "id": "ex797",
    "input": "Every analyst can guarantee that eventually the transaction will be reviewed.",
    "outputs": [
      {
        "formula": "<<Analyst1>>F transaction_reviewed_1 && <<Analyst2>>F transaction_reviewed_2 && <<Analyst3>>F transaction_reviewed_3"
      },
      {
        "formula": "<<Analyst1,Analyst2,Analyst3>>F transaction_reviewed"
      }
    ]
  },
  {
    "id": "ex798",
    "input": "Every fire officer can guarantee that in the end the plan will be approved.",
    "outputs": [
      {
        "formula": "<<FireOfficer1>>F plan_approved_1 && <<FireOfficer2>>F plan_approved_2 && <<FireOfficer3>>F plan_approved_3"
      },
      {
```

```
        "formula": "<<FireOfficer1,FireOfficer2,FireOfficer3>>F plan_approved"
      }
    ]
  },
  {
    "id": "ex799",
    "input": "Every recovery service can guarantee that after some finite delay the archive will
be restored.",
    "outputs": [
      {
        "formula": "<<RecoveryService1>>F archive_restored_1 && <<RecoveryService2>>F
archive_restored_2 && <<RecoveryService3>>F archive_restored_3"
      },
      {
        "formula": "<<RecoveryService1,RecoveryService2,RecoveryService3>>F
archive_restored"
      }
    ]
  },
  {
    "id": "ex800",
    "input": "They can guarantee that sooner or later the missing license will be found, the
registrar and the records office.",
    "outputs": [
      {
        "formula": "<<Registrar,RecordsOffice>>F missing_license_found"
      }
    ]
  },
  {
    "id": "ex801",
    "input": "It can guarantee that in due course the delayed shipment will be located, the
logistics dashboard.",
    "outputs": [
      {
        "formula": "<<LogisticsDashboard>>F delayed_shipment_located"
      }
    ]
  },
  {
    "id": "ex802",
    "input": "They can guarantee that before long the damaged equipment will be replaced,
the technician and the procurement unit.",
    "outputs": [
      {
        "formula": "<<Technician,ProcurementUnit>>F damaged_equipment_replaced"
      }
    ]
```

```
},
{
  "id": "ex803",
  "input": "It can guarantee that after finitely many steps the rejected form will be corrected, the submission portal.",
  "outputs": [
    {
      "formula": "<<SubmissionPortal>>F rejected_form_corrected"
    }
  ]
},
{
  "id": "ex804",
  "input": "They can guarantee that ultimately the safety complaint will be investigated, the inspector and the oversight board.",
  "outputs": [
    {
      "formula": "<<Inspector,OversightBoard>>F safety_complaint_investigated"
    }
  ]
},
{
  "id": "ex805",
  "input": "It can guarantee that at some future point the locked account will be recovered, the identity service.",
  "outputs": [
    {
      "formula": "<<IdentityService>>F locked_account_recovered"
    }
  ]
},
{
  "id": "ex806",
  "input": "They can guarantee that eventually the medical referral will be processed, the doctor and the scheduling desk.",
  "outputs": [
    {
      "formula": "<<Doctor,SchedulingDesk>>F medical_referral_processed"
    }
  ]
},
{
  "id": "ex807",
  "input": "It can guarantee that eventually the corrupted profile will be rebuilt, the user management system.",
  "outputs": [
    {
      "formula": "<<UserManagementSystem>>F corrupted_profile_rebuilt"
```

```
    }
   ]
  },
  {
   "id": "ex808",
   "input": "They can guarantee that in the end the research dataset will be published, the data curator and the repository.",
   "outputs": [
    {
     "formula": "<<DataCurator,Repository>>F research_dataset_published"
    }
   ]
  },
  {
   "id": "ex809",
   "input": "It can guarantee that after some finite delay the pending refund will be credited, the payment processor.",
   "outputs": [
    {
     "formula": "<<PaymentProcessor>>F pending_refund_credited"
    }
   ]
  },
  {
   "id": "ex810",
   "input": "The missing license, the registrar can guarantee that sooner or later it will be found.",
   "outputs": [
    {
     "formula": "<<Registrar>>F missing_license_found"
    }
   ]
  },
  {
   "id": "ex811",
   "input": "The delayed shipment, the logistics dashboard can guarantee that in due course it will be located.",
   "outputs": [
    {
     "formula": "<<LogisticsDashboard>>F delayed_shipment_located"
    }
   ]
  },
  {
   "id": "ex812",
   "input": "The damaged equipment, the technician and the procurement unit can guarantee that before long it will be replaced.",
   "outputs": [
```

{
"formula": "<<Technician,ProcurementUnit>>F damaged_equipment_replaced"
}
]
},
{
"id": "ex813",
"input": "The rejected form, the submission portal can guarantee that after finitely many steps it will be corrected.",
"outputs": [
{
"formula": "<<SubmissionPortal>>F rejected_form_corrected"
}
]
},
{
"id": "ex814",
"input": "The safety complaint, the inspector and the oversight board can guarantee that ultimately it will be investigated.",
"outputs": [
{
"formula": "<<Inspector,OversightBoard>>F safety_complaint_investigated"
}
]
},
{
"id": "ex815",
"input": "The locked account, the identity service can guarantee that at some future point it will be recovered.",
"outputs": [
{
"formula": "<<IdentityService>>F locked_account_recovered"
}
]
},
{
"id": "ex816",
"input": "The medical referral, the doctor and the scheduling desk can guarantee that eventually it will be processed.",
"outputs": [
{
"formula": "<<Doctor,SchedulingDesk>>F medical_referral_processed"
}
]
},
{
"id": "ex817",

```
    "input": "The corrupted profile, the user management system can guarantee that eventually it will be rebuilt.",
    "outputs": [
      {
        "formula": "<<UserManagementSystem>>F corrupted_profile_rebuilt"
      }
    ]
  },
  {
    "id": "ex818",
    "input": "The research dataset, the data curator and the repository can guarantee that in the end it will be published.",
    "outputs": [
      {
        "formula": "<<DataCurator,Repository>>F research_dataset_published"
      }
    ]
  },
  {
    "id": "ex819",
    "input": "The pending refund, the payment processor can guarantee that after some finite delay it will be credited.",
    "outputs": [
      {
        "formula": "<<PaymentProcessor>>F pending_refund_credited"
      }
    ]
  },
  {
    "id": "ex820",
    "input": "The registrar can guarantee that sooner or later the missing license will be found, and the records office can too.",
    "outputs": [
      {
        "formula": "<<Registrar>>F missing_license_found && <<RecordsOffice>>F missing_license_found"
      }
    ]
  },
  {
    "id": "ex821",
    "input": "The logistics dashboard can guarantee that in due course the delayed shipment will be located, and the warehouse system can too.",
    "outputs": [
      {
        "formula": "<<LogisticsDashboard>>F delayed_shipment_located && <<WarehouseSystem>>F delayed_shipment_located"
      }
```

]
},
{
"id": "ex822",
"input": "The technician can guarantee that before long the damaged equipment will be replaced, and the procurement unit can too.",
"outputs": [
{
"formula": "<<Technician>>F damaged_equipment_replaced && <<ProcurementUnit>>F damaged_equipment_replaced"
}
]
},
{
"id": "ex823",
"input": "The submission portal can guarantee that after finitely many steps the rejected form will be corrected, and the help desk can too.",
"outputs": [
{
"formula": "<<SubmissionPortal>>F rejected_form_corrected && <<HelpDesk>>F rejected_form_corrected"
}
]
},
{
"id": "ex824",
"input": "The inspector can guarantee that ultimately the safety complaint will be investigated, and the oversight board can too.",
"outputs": [
{
"formula": "<<Inspector>>F safety_complaint_investigated && <<OversightBoard>>F safety_complaint_investigated"
}
]
},
{
"id": "ex825",
"input": "The identity service can guarantee that at some future point the locked account will be recovered, and the support team can too.",
"outputs": [
{
"formula": "<<IdentityService>>F locked_account_recovered && <<SupportTeam>>F locked_account_recovered"
}
]
},
{
"id": "ex826",

"input": "The doctor can guarantee that eventually the medical referral will be processed, and the scheduling desk can too.",
"outputs": [
{
"formula": "<<Doctor>>F medical_referral_processed && <<SchedulingDesk>>F medical_referral_processed"
}
]
},
{
"id": "ex827",
"input": "The user management system can guarantee that eventually the corrupted profile will be rebuilt, and the recovery service can too.",
"outputs": [
{
"formula": "<<UserManagementSystem>>F corrupted_profile_rebuilt && <<RecoveryService>>F corrupted_profile_rebuilt"
}
]
},
{
"id": "ex828",
"input": "The data curator can guarantee that in the end the research dataset will be published, and the repository can too.",
"outputs": [
{
"formula": "<<DataCurator>>F research_dataset_published && <<Repository>>F research_dataset_published"
}
]
},
{
"id": "ex829",
"input": "The payment processor can guarantee that after some finite delay the pending refund will be credited, and the billing service can too.",
"outputs": [
{
"formula": "<<PaymentProcessor>>F pending_refund_credited && <<BillingService>>F pending_refund_credited"
}
]
},
{
"id": "ex830",
"input": "Every registrar can guarantee that sooner or later the license will be found.",
"outputs": [
{

```
      "formula": "<<Registrar1>>F license_found_1 && <<Registrar2>>F license_found_2 &&
<<Registrar3>>F license_found_3"
    },
    {
      "formula": "<<Registrar1,Registrar2,Registrar3>>F license_found"
    }
  ]
 },
 {
  "id": "ex831",
  "input": "Every operator can guarantee that in due course the shipment will be located.",
  "outputs": [
    {
      "formula": "<<Operator1>>F shipment_located_1 && <<Operator2>>F
shipment_located_2 && <<Operator3>>F shipment_located_3"
    },
    {
      "formula": "<<Operator1,Operator2,Operator3>>F shipment_located"
    }
  ]
 },
 {
  "id": "ex832",
  "input": "Every technician can guarantee that before long the equipment will be replaced.",
  "outputs": [
    {
      "formula": "<<Technician1>>F equipment_replaced_1 && <<Technician2>>F
equipment_replaced_2 && <<Technician3>>F equipment_replaced_3"
    },
    {
      "formula": "<<Technician1,Technician2,Technician3>>F equipment_replaced"
    }
  ]
 },
 {
  "id": "ex833",
  "input": "Every clerk can guarantee that after finitely many steps the form will be
corrected.",
  "outputs": [
    {
      "formula": "<<Clerk1>>F form_corrected_1 && <<Clerk2>>F form_corrected_2 &&
<<Clerk3>>F form_corrected_3"
    },
    {
      "formula": "<<Clerk1,Clerk2,Clerk3>>F form_corrected"
    }
  ]
 },
```

```
 {
   "id": "ex834",
   "input": "Every inspector can guarantee that ultimately the complaint will be investigated.",
   "outputs": [
     {
       "formula": "<<Inspector1>>F complaint_investigated_1 && <<Inspector2>>F complaint_investigated_2 && <<Inspector3>>F complaint_investigated_3"
     },
     {
       "formula": "<<Inspector1,Inspector2,Inspector3>>F complaint_investigated"
     }
   ]
 },
 {
   "id": "ex835",
   "input": "Every support agent can guarantee that at some future point the account will be recovered.",
   "outputs": [
     {
       "formula": "<<SupportAgent1>>F account_recovered_1 && <<SupportAgent2>>F account_recovered_2 && <<SupportAgent3>>F account_recovered_3"
     },
     {
       "formula": "<<SupportAgent1,SupportAgent2,SupportAgent3>>F account_recovered"
     }
   ]
 },
 {
   "id": "ex836",
   "input": "Every physician can guarantee that eventually the referral will be processed.",
   "outputs": [
     {
       "formula": "<<Physician1>>F referral_processed_1 && <<Physician2>>F referral_processed_2 && <<Physician3>>F referral_processed_3"
     },
     {
       "formula": "<<Physician1,Physician2,Physician3>>F referral_processed"
     }
   ]
 },
 {
   "id": "ex837",
   "input": "Every administrator can guarantee that eventually the profile will be rebuilt.",
   "outputs": [
     {
       "formula": "<<Administrator1>>F profile_rebuilt_1 && <<Administrator2>>F profile_rebuilt_2"
     },
```

```
      {
        "formula": "<<Administrator1,Administrator2>>F profile_rebuilt"
      }
    ]
  },
  {
    "id": "ex838",
    "input": "Every curator can guarantee that in the end the dataset will be published.",
    "outputs": [
      {
        "formula": "<<Curator1>>F dataset_published_1 && <<Curator2>>F
dataset_published_2 && <<Curator3>>F dataset_published_3"
      },
      {
        "formula": "<<Curator1,Curator2,Curator3>>F dataset_published"
      }
    ]
  },
  {
    "id": "ex839",
    "input": "Every processor can guarantee that after some finite delay the refund will be
credited.",
    "outputs": [
      {
        "formula": "<<Processor1>>F refund_credited_1 && <<Processor2>>F
refund_credited_2"
      },
      {
        "formula": "<<Processor1,Processor2>>F refund_credited"
      }
    ]
  },
  {
    "id": "ex840",
    "input": "They can guarantee that sooner or later the missing passport will be returned, the
consular officer and the document office.",
    "outputs": [
      {
        "formula": "<<ConsularOfficer,DocumentOffice>>F missing_passport_returned"
      }
    ]
  },
  {
    "id": "ex841",
    "input": "It can guarantee that in due course the suspended subscription will be restored,
the account service.",
    "outputs": [
      {
```

"formula": "<<AccountService>>F suspended_subscription_restored"
}
]
},
{
"id": "ex842",
"input": "They can guarantee that before long the damaged pipeline will be repaired, the field engineer and the utility company.",
"outputs": [
{
"formula": "<<FieldEngineer,UtilityCompany>>F damaged_pipeline_repaired"
}
]
},
{
"id": "ex843",
"input": "It can guarantee that after finitely many steps the delayed application will be processed, the intake portal.",
"outputs": [
{
"formula": "<<IntakePortal>>F delayed_application_processed"
}
]
},
{
"id": "ex844",
"input": "They can guarantee that ultimately the missing evidence will be catalogued, the investigator and the evidence registry.",
"outputs": [
{
"formula": "<<Investigator,EvidenceRegistry>>F missing_evidence_catalogued"
}
]
},
{
"id": "ex845",
"input": "It can guarantee that at some future point the frozen profile will be reactivated, the profile manager.",
"outputs": [
{
"formula": "<<ProfileManager>>F frozen_profile_reactivated"
}
]
},
{
"id": "ex846",
"input": "They can guarantee that eventually the final transcript will be released, the examiner and the academic office.",

"outputs": [
{
"formula": "<<Examiner,AcademicOffice>>F final_transcript_released"
}
]
},
{
"id": "ex847",
"input": "It can guarantee that eventually the broken sensor will be recalibrated, the maintenance console.",
"outputs": [
{
"formula": "<<MaintenanceConsole>>F broken_sensor_recalibrated"
}
]
},
{
"id": "ex848",
"input": "The missing passport, the officer can guarantee that sooner or later it will be returned.",
"outputs": [
{
"formula": "<<Officer>>F missing_passport_returned"
}
]
},
{
"id": "ex849",
"input": "The suspended subscription, the account service can guarantee that in due course it will be restored.",
"outputs": [
{
"formula": "<<AccountService>>F suspended_subscription_restored"
}
]
},
{
"id": "ex850",
"input": "The damaged pipeline, the field engineer and the utility company can guarantee that before long it will be repaired.",
"outputs": [
{
"formula": "<<FieldEngineer,UtilityCompany>>F damaged_pipeline_repaired"
}
]
},
{
"id": "ex851",

"input": "The delayed application, the intake portal can guarantee that after finitely many steps it will be processed.",
"outputs": [
{
"formula": "<<IntakePortal>>F delayed_application_processed"
}
]
},
{
"id": "ex852",
"input": "The missing evidence, the investigator and the evidence registry can guarantee that ultimately it will be catalogued.",
"outputs": [
{
"formula": "<<Investigator,EvidenceRegistry>>F missing_evidence_catalogued"
}
]
},
{
"id": "ex853",
"input": "The frozen profile, the profile manager can guarantee that at some future point it will be reactivated.",
"outputs": [
{
"formula": "<<ProfileManager>>F frozen_profile_reactivated"
}
]
},
{
"id": "ex854",
"input": "The final transcript, the examiner and the academic office can guarantee that eventually it will be released.",
"outputs": [
{
"formula": "<<Examiner,AcademicOffice>>F final_transcript_released"
}
]
},
{
"id": "ex855",
"input": "The broken sensor, the maintenance console can guarantee that eventually it will be recalibrated.",
"outputs": [
{
"formula": "<<MaintenanceConsole>>F broken_sensor_recalibrated"
}
]
},

{
"id": "ex856",
"input": "The consular officer can guarantee that sooner or later the missing passport will be returned, and the document office can too.",
"outputs": [
{
"formula": "<<ConsularOfficer>>F missing_passport_returned && <<DocumentOffice>>F missing_passport_returned"
}
]
},
{
"id": "ex857",
"input": "The account service can guarantee that in due course the suspended subscription will be restored, and the billing desk can too.",
"outputs": [
{
"formula": "<<AccountService>>F suspended_subscription_restored && <<BillingDesk>>F suspended_subscription_restored"
}
]
},
{
"id": "ex858",
"input": "The field engineer can guarantee that before long the damaged pipeline will be repaired, and the utility company can too.",
"outputs": [
{
"formula": "<<FieldEngineer>>F damaged_pipeline_repaired && <<UtilityCompany>>F damaged_pipeline_repaired"
}
]
},
{
"id": "ex859",
"input": "The intake portal can guarantee that after finitely many steps the delayed application will be processed, and the review desk can too.",
"outputs": [
{
"formula": "<<IntakePortal>>F delayed_application_processed && <<ReviewDesk>>F delayed_application_processed"
}
]
},
{
"id": "ex860",
"input": "The investigator can guarantee that ultimately the missing evidence will be catalogued, and the evidence registry can too.",

"outputs": [
{
"formula": "<<Investigator>>F missing_evidence_catalogued &&
<<EvidenceRegistry>>F missing_evidence_catalogued"
}
]
},
{
"id": "ex861",
"input": "The profile manager can guarantee that at some future point the frozen profile will be reactivated, and the support center can too.",
"outputs": [
{
"formula": "<<ProfileManager>>F frozen_profile_reactivated && <<SupportCenter>>F frozen_profile_reactivated"
}
]
},
{
"id": "ex862",
"input": "The examiner can guarantee that eventually the final transcript will be released, and the academic office can too.",
"outputs": [
{
"formula": "<<Examiner>>F final_transcript_released && <<AcademicOffice>>F final_transcript_released"
}
]
},
{
"id": "ex863",
"input": "The maintenance console can guarantee that eventually the broken sensor will be recalibrated, and the calibration service can too.",
"outputs": [
{
"formula": "<<MaintenanceConsole>>F broken_sensor_recalibrated &&
<<CalibrationService>>F broken_sensor_recalibrated"
}
]
},
{
"id": "ex864",
"input": "Every police officer can guarantee that sooner or later the wallet will be returned.",
"outputs": [
{
"formula": "<<PoliceOfficer1>>F wallet_returned_1 && <<PoliceOfficer2>>F wallet_returned_2"

```
      },
      {
        "formula": "<<PoliceOfficer1,PoliceOfficer2>>F wallet_returned"
      }
    ]
  },
  {
    "id": "ex865",
    "input": "Every account manager can guarantee that in due course the subscription will be
restored.",
    "outputs": [
      {
        "formula": "<<AccountManager1>>F subscription_restored_1 &&
<<AccountManager2>>F subscription_restored_2 && <<AccountManager3>>F
subscription_restored_3"
      },
      {
        "formula": "<<AccountManager1,AccountManager2,AccountManager3>>F
subscription_restored"
      }
    ]
  },
  {
    "id": "ex866",
    "input": "Every engineer can guarantee that before long the pipeline will be repaired.",
    "outputs": [
      {
        "formula": "<<Engineer1>>F pipeline_repaired_1 && <<Engineer2>>F
pipeline_repaired_2 && <<Engineer3>>F pipeline_repaired_3"
      },
      {
        "formula": "<<Engineer1,Engineer2,Engineer3>>F pipeline_repaired"
      }
    ]
  },
  {
    "id": "ex867",
    "input": "Every clerk can guarantee that after finitely many steps the application will be
processed.",
    "outputs": [
      {
        "formula": "<<Clerk1>>F application_processed_1 && <<Clerk2>>F
application_processed_2 && <<Clerk3>>F application_processed_3"
      },
      {
        "formula": "<<Clerk1,Clerk2,Clerk3>>F application_processed"
      }
    ]
```

```
},
{
  "id": "ex868",
  "input": "Every investigator can guarantee that eventually the evidence will be
catalogued.",
  "outputs": [
    {
      "formula": "<<Investigator1>>F evidence_catalogued_1 && <<Investigator2>>F
evidence_catalogued_2 && <<Investigator3>>F evidence_catalogued_3 &&
<<Investigator4>>F evidence_catalogued_4"
    },
    {
      "formula": "<<Investigator1,Investigator2,Investigator3,Investigator4>>F
evidence_catalogued"
    }
  ]
},
{
  "id": "ex869",
  "input": "Every profile manager can guarantee that at some future point the profile will be
reactivated.",
  "outputs": [
    {
      "formula": "<<ProfileManager1>>F profile_reactivated_1 && <<ProfileManager2>>F
profile_reactivated_2 && <<ProfileManager3>>F profile_reactivated_3"
    },
    {
      "formula": "<<ProfileManager1,ProfileManager2,ProfileManager3>>F
profile_reactivated"
    }
  ]
},
{
  "id": "ex870",
  "input": "Every examiner can guarantee that eventually the transcript will be released.",
  "outputs": [
    {
      "formula": "<<Examiner1>>F transcript_released_1 && <<Examiner2>>F
transcript_released_2 && <<Examiner3>>F transcript_released_3"
    },
    {
      "formula": "<<Examiner1,Examiner2,Examiner3>>F transcript_released"
    }
  ]
},
{
  "id": "ex871",
  "input": "Every technician can guarantee that eventually the sensor will be recalibrated.",
```

"outputs": [
{
"formula": "<<Technician1>>F sensor_recalibrated_1 && <<Technician2>>F sensor_recalibrated_2"
},
{
"formula": "<<Technician1,Technician2>>F sensor_recalibrated"
}
]
},

{
"id": "ex872",
"input": "The recovery coordinator can guarantee that before long the unresolved case will be closed.",
"outputs": [
{
"formula": "<<RecoveryCoordinator>>F unresolved_case_closed"
}
]
},

{
"id": "ex873",
"input": "They can guarantee that on the next transition the permit will be signed, the clerk and the director.",
"outputs": [
{
"formula": "<<Clerk,Director>>X permit_signed"
}
]
},
{
"id": "ex874",
"input": "It can guarantee that in the immediately following state the ticket will be validated, the booking terminal.",
"outputs": [
{
"formula": "<<BookingTerminal>>X ticket_validated"
}
]
},
{
"id": "ex875",
"input": "They can guarantee that at the next step the invoice will be approved, the accountant and the supervisor.",
"outputs": [
{

"formula": "<<Accountant,Supervisor>>X invoice_approved"
}
]
},
{
"id": "ex876",
"input": "It can guarantee that in the immediate successor state the alarm will be acknowledged, the monitoring panel.",
"outputs": [
{
"formula": "<<MonitoringPanel>>X alarm_acknowledged"
}
]
},
{
"id": "ex877",
"input": "They can guarantee that on the very next transition the report will be submitted, the analyst and the reviewer.",
"outputs": [
{
"formula": "<<Analyst,Reviewer>>X report_submitted"
}
]
},
{
"id": "ex878",
"input": "It can guarantee that one step later the gate will be unlocked, the access terminal.",
"outputs": [
{
"formula": "<<AccessTerminal>>X gate_unlocked"
}
]
},
{
"id": "ex879",
"input": "They can guarantee that upon the next transition the package will be scanned, the courier and the depot system.",
"outputs": [
{
"formula": "<<Courier,DepotSystem>>X package_scanned"
}
]
},
{
"id": "ex880",
"input": "It can guarantee that in the next stage the backup will be triggered, the recovery daemon.",

```
    "outputs": [
      {
        "formula": "<<RecoveryDaemon>>X backup_triggered"
      }
    ]
  },
  {
    "id": "ex881",
    "input": "They can guarantee that in the next configuration the valve will be closed, the technician and the control console.",
    "outputs": [
      {
        "formula": "<<Technician,ControlConsole>>X valve_closed"
      }
    ]
  },
  {
    "id": "ex882",
    "input": "It can guarantee that immediately afterward the record will be indexed, the archive service.",
    "outputs": [
      {
        "formula": "<<ArchiveService>>X record_indexed"
      }
    ]
  },

  {
    "id": "ex883",
    "input": "The permit, the clerk can guarantee that on the next transition it will be signed.",
    "outputs": [
      {
        "formula": "<<Clerk>>X permit_signed"
      }
    ]
  },
  {
    "id": "ex884",
    "input": "The ticket, the booking terminal can guarantee that in the immediately following state it will be validated.",
    "outputs": [
      {
        "formula": "<<BookingTerminal>>X ticket_validated"
      }
    ]
  },
  {
    "id": "ex885",
```

```
  "input": "The invoice, the accountant and the supervisor can guarantee that at the next step it will be approved.",
  "outputs": [
   {
    "formula": "<<Accountant,Supervisor>>X invoice_approved"
   }
  ]
 },
 {
  "id": "ex886",
  "input": "The alarm, the monitoring panel can guarantee that in the immediate successor state it will be acknowledged.",
  "outputs": [
   {
    "formula": "<<MonitoringPanel>>X alarm_acknowledged"
   }
  ]
 },
 {
  "id": "ex887",
  "input": "The report, the analyst and the reviewer can guarantee that on the very next transition it will be submitted.",
  "outputs": [
   {
    "formula": "<<Analyst,Reviewer>>X report_submitted"
   }
  ]
 },
 {
  "id": "ex888",
  "input": "The gate, the access terminal can guarantee that one step later it will be unlocked.",
  "outputs": [
   {
    "formula": "<<AccessTerminal>>X gate_unlocked"
   }
  ]
 },
 {
  "id": "ex889",
  "input": "The package, the courier and the depot system can guarantee that upon the next transition it will be scanned.",
  "outputs": [
   {
    "formula": "<<Courier,DepotSystem>>X package_scanned"
   }
  ]
 },
```

{
"id": "ex890",
"input": "The backup, the recovery daemon can guarantee that at the next tick it will be triggered.",
"outputs": [
{
"formula": "<<RecoveryDaemon>>X backup_triggered"
}
]
},
{
"id": "ex891",
"input": "The valve, the technician and the control console can guarantee that in the next configuration it will be closed.",
"outputs": [
{
"formula": "<<Technician,ControlConsole>>X valve_closed"
}
]
},
{
"id": "ex892",
"input": "The record, the archive service can guarantee that immediately afterward it will be indexed.",
"outputs": [
{
"formula": "<<ArchiveService>>X record_indexed"
}
]
},
{
"id": "ex893",
"input": "The clerk can guarantee that on the next transition the permit will be signed, and the director can too.",
"outputs": [
{
"formula": "<<Clerk>>X permit_signed && <<Director>>X permit_signed"
}
]
},
{
"id": "ex894",
"input": "The booking terminal can guarantee that in the immediately following state the ticket will be validated, and the agent console can too.",
"outputs": [
{
"formula": "<<BookingTerminal>>X ticket_validated && <<AgentConsole>>X ticket_validated"

}
]
},
{
"id": "ex895",
"input": "The accountant can guarantee that at the next step the invoice will be approved, and the supervisor can too.",
"outputs": [
{
"formula": "<<Accountant>>X invoice_approved && <<Supervisor>>X invoice_approved"
}
]
},
{
"id": "ex896",
"input": "The monitoring panel can guarantee that in the immediate successor state the alarm will be acknowledged, and the duty officer can too.",
"outputs": [
{
"formula": "<<MonitoringPanel>>X alarm_acknowledged && <<DutyOfficer>>X alarm_acknowledged"
}
]
},
{
"id": "ex897",
"input": "The analyst can guarantee that on the very next transition the report will be submitted, and the reviewer can too.",
"outputs": [
{
"formula": "<<Analyst>>X report_submitted && <<Reviewer>>X report_submitted"
}
]
},
{
"id": "ex898",
"input": "The access terminal can guarantee that one step later the gate will be unlocked, and the security desk can too.",
"outputs": [
{
"formula": "<<AccessTerminal>>X gate_unlocked && <<SecurityDesk>>X gate_unlocked"
}
]
},
{
"id": "ex899",

```
    "input": "The courier can guarantee that upon the next transition the package will be scanned, and the depot system can too.",
    "outputs": [
      {
        "formula": "<<Courier>>X package_scanned && <<DepotSystem>>X package_scanned"
      }
    ]
  },
  {
    "id": "ex900",
    "input": "The recovery system can guarantee that in the next state the backup will be triggered, and the storage manager can too.",
    "outputs": [
      {
        "formula": "<<RecoverySystem>>X backup_triggered && <<StorageManager>>X backup_triggered"
      }
    ]
  },
  {
    "id": "ex901",
    "input": "The technician can guarantee that in the next configuration the valve will be closed, and the control console can too.",
    "outputs": [
      {
        "formula": "<<Technician>>X valve_closed && <<ControlConsole>>X valve_closed"
      }
    ]
  },
  {
    "id": "ex902",
    "input": "The archive service can guarantee that immediately afterward the record will be indexed, and the catalogue manager can too.",
    "outputs": [
      {
        "formula": "<<ArchiveService>>X record_indexed && <<CatalogueManager>>X record_indexed"
      }
    ]
  },
  {
    "id": "ex903",
    "input": "Every clerk can guarantee that on the next transition the permit will be signed.",
    "outputs": [
      {
        "formula": "<<Clerk1>>X permit_signed_1 && <<Clerk2>>X permit_signed_2"
      },
```

```
      {
        "formula": "<<Clerk1,Clerk2>>X permit_signed"
      }
    ]
  },
  {
    "id": "ex904",
    "input": "Every terminal can guarantee that in the immediately following state the ticket will
be validated.",
    "outputs": [
      {
        "formula": "<<Terminal1>>X ticket_validated_1 && <<Terminal2>>X ticket_validated_2
&& <<Terminal3>>X ticket_validated_3"
      },
      {
        "formula": "<<Terminal1, Terminal2, Terminal3>>X ticket_validated"
      }
    ]
  },
  {
    "id": "ex905",
    "input": "Every accountant can guarantee that at the next step the invoice will be
approved.",
    "outputs": [
      {
        "formula": "<<Accountant1>>X invoice_approved_1 && <<Accountant2>>X
invoice_approved_2 && <<Accountant3>>X invoice_approved_3"
      },
      {
        "formula": "<<Accountant1, Accountant2, Accountant3>>X invoice_approved"
      }
    ]
  },
  {
    "id": "ex906",
    "input": "Every monitor can guarantee that in the immediate successor state the alarm will
be acknowledged.",
    "outputs": [
      {
        "formula": "<<Monitor1>>X alarm_acknowledged_1 && <<Monitor2>>X
alarm_acknowledged_2 && <<Monitor3>>X alarm_acknowledged_3 && <<Monitor4>>X
alarm_acknowledged_4 && <<Monitor5>>X alarm_acknowledged_5"
      },
      {
        "formula": "<<Monitor1, Monitor2, Monitor3, Monitor4, Monitor5>>X
alarm_acknowledged"
      }
    ]
```

},
{
"id": "ex907",
"input": "Every analyst can guarantee that on the very next transition the report will be submitted.",
"outputs": [
{
"formula": "<<Analyst1>>X report_submitted_1 && <<Analyst2>>X report_submitted_2 && <<Analyst3>>X report_submitted_3"
},
{
"formula": "<<Analyst1,Analyst2,Analyst3>>X report_submitted"
}
]
},
{
"id": "ex908",
"input": "Every access terminal can guarantee that one step later the gate will be unlocked.",
"outputs": [
{
"formula": "<<AccessTerminal1>>X gate_unlocked_1 && <<AccessTerminal2>>X gate_unlocked_2 && <<AccessTerminal3>>X gate_unlocked_3"
},
{
"formula": "<<AccessTerminal1,AccessTerminal2,AccessTerminal3>>X gate_unlocked"
}
]
},
{
"id": "ex909",
"input": "Every courier can guarantee that upon the next transition the package will be scanned.",
"outputs": [
{
"formula": "<<Courier1>>X package_scanned_1 && <<Courier2>>X package_scanned_2"
},
{
"formula": "<<Courier1,Courier2>>X package_scanned"
}
]
},
{
"id": "ex910",
"input": "Every recovery system can guarantee that in the next step the backup will be triggered.",
"outputs": [

```
      {
        "formula": "<<RecoverySystem1>>X backup_triggered_1 && <<RecoverySystem2>>X
backup_triggered_2 && <<RecoverySystem3>>X backup_triggered_3"
      },
      {
        "formula": "<<RecoverySystem1,RecoverySystem2,RecoverySystem3>>X
backup_triggered"
      }
    ]
  },
  {
    "id": "ex911",
    "input": "Every technician can guarantee that in the next configuration the valve will be
closed.",
    "outputs": [
      {
        "formula": "<<Technician1>>X valve_closed_1 && <<Technician2>>X valve_closed_2
&& <<Technician3>>X valve_closed_3 && <<Technician4>>X valve_closed_4 &&
<<Technician5>>X valve_closed_5 && <<Technician6>>X valve_closed_6"
      },
      {
        "formula":
"<<Technician1,Technician2,Technician3,Technician4,Technician5,Technician6>>X
valve_closed"
      }
    ]
  },
  {
    "id": "ex912",
    "input": "Every archive service can guarantee that immediately afterward the record will be
indexed.",
    "outputs": [
      {
        "formula": "<<ArchiveService1>>X record_indexed_1 && <<ArchiveService2>>X
record_indexed_2 && <<ArchiveService3>>X record_indexed_3"
      },
      {
        "formula": "<<ArchiveService1,ArchiveService2,ArchiveService3>>X record_indexed"
      }
    ]
  },
  {
    "id": "ex913",
    "input": "They can guarantee that in the immediate successor state the access badge will
be printed, the receptionist and the badge system.",
    "outputs": [
      {
        "formula": "<<Receptionist,BadgeSystem>>X access_badge_printed"
```

```
      }
    ]
  },
  {
    "id": "ex914",
    "input": "It can guarantee that on the next transition the error flag will be cleared, the diagnostic console.",
    "outputs": [
      {
        "formula": "<<DiagnosticConsole>>X error_flag_cleared"
      }
    ]
  },
  {
    "id": "ex915",
    "input": "They can guarantee that at the next step the delivery label will be generated, the warehouse bot and the shipping service.",
    "outputs": [
      {
        "formula": "<<WarehouseBot,ShippingService>>X delivery_label_generated"
      }
    ]
  },
  {
    "id": "ex916",
    "input": "It can guarantee that after the patient wristband will be scanned, the bedside terminal.",
    "outputs": [
      {
        "formula": "<<BedsideTerminal>>X patient_wristband_scanned"
      }
    ]
  },
  {
    "id": "ex917",
    "input": "They can guarantee that upon entering the next state the evidence tag will be attached, the investigator and the registry system.",
    "outputs": [
      {
        "formula": "<<Investigator,RegistrySystem>>X evidence_tag_attached"
      }
    ]
  },
  {
    "id": "ex918",
    "input": "It can guarantee that right after this step the consent prompt will be displayed, the clinical kiosk.",
    "outputs": [
```

```
    {
      "formula": "<<ClinicalKiosk>>X consent_prompt_displayed"
    }
  ]
},
{
  "id": "ex919",
  "input": "They can guarantee that at the next tick the meeting link will be created, the
organizer and the calendar service.",
  "outputs": [
    {
      "formula": "<<Organizer,CalendarService>>X meeting_link_created"
    }
  ]
},
{
  "id": "ex920",
  "input": "It can guarantee that in the following configuration the safety latch will be
engaged, the machine controller.",
  "outputs": [
    {
      "formula": "<<MachineController>>X safety_latch_engaged"
    }
  ]
},
{
  "id": "ex921",
  "input": "They can guarantee that immediately afterward the inspection note will be
uploaded, the field agent and the reporting app.",
  "outputs": [
    {
      "formula": "<<FieldAgent,ReportingApp>>X inspection_note_uploaded"
    }
  ]
},
{
  "id": "ex922",
  "input": "It can guarantee that in the immediately succeeding state the recovery token will
be issued, the authentication server.",
  "outputs": [
    {
      "formula": "<<AuthenticationServer>>X recovery_token_issued"
    }
  ]
},

{
  "id": "ex923",
```

```
   "input": "The access badge, the receptionist can guarantee that in the immediate successor state it will be printed.",
   "outputs": [
    {
     "formula": "<<Receptionist>>X access_badge_printed"
    }
   ]
  },
  {
   "id": "ex924",
   "input": "The error flag, the diagnostic console can guarantee that on the next transition it will be cleared.",
   "outputs": [
    {
     "formula": "<<DiagnosticConsole>>X error_flag_cleared"
    }
   ]
  },
  {
   "id": "ex925",
   "input": "The delivery label, the warehouse clerk and the shipping service can guarantee that at the next step it will be generated.",
   "outputs": [
    {
     "formula": "<<WarehouseClerk,ShippingService>>X delivery_label_generated"
    }
   ]
  },
  {
   "id": "ex926",
   "input": "The patient wristband, the bedside terminal can guarantee that one transition later it will be scanned.",
   "outputs": [
    {
     "formula": "<<BedsideTerminal>>X patient_wristband_scanned"
    }
   ]
  },
  {
   "id": "ex927",
   "input": "The evidence tag, the investigator and the registry system can guarantee that in the next state it will be attached.",
   "outputs": [
    {
     "formula": "<<Investigator,RegistrySystem>>X evidence_tag_attached"
    }
   ]
  },
```

{
"id": "ex928",
"input": "The consent prompt, the clinical kiosk can guarantee that right after this step it will be displayed.",
"outputs": [
{
"formula": "<<ClinicalKiosk>>X consent_prompt_displayed"
}
]
},
{
"id": "ex929",
"input": "The meeting link, the organizer and the calendar service can guarantee that in the next stage it will be created.",
"outputs": [
{
"formula": "<<Organizer,CalendarService>>X meeting_link_created"
}
]
},
{
"id": "ex930",
"input": "The safety latch, the machine controller can guarantee that in the following configuration it will be engaged.",
"outputs": [
{
"formula": "<<MachineController>>X safety_latch_engaged"
}
]
},
{
"id": "ex931",
"input": "The inspection note, the field agent and the reporting app can guarantee that immediately afterward it will be uploaded.",
"outputs": [
{
"formula": "<<FieldAgent,ReportingApp>>X inspection_note_uploaded"
}
]
},
{
"id": "ex932",
"input": "The recovery token, the authentication server can guarantee that in the immediately succeeding state it will be issued.",
"outputs": [
{
"formula": "<<AuthenticationServer>>X recovery_token_issued"
}

]
},
{
"id": "ex933",
"input": "The receptionist can guarantee that in the immediate successor state the access badge will be printed, and the badge system can too.",
"outputs": [
{
"formula": "<<Receptionist>>X access_badge_printed && <<BadgeSystem>>X access_badge_printed"
}
]
},
{
"id": "ex934",
"input": "The diagnostic console can guarantee that on the next transition the error flag will be cleared, and the repair module can too.",
"outputs": [
{
"formula": "<<DiagnosticConsole>>X error_flag_cleared && <<RepairModule>>X error_flag_cleared"
}
]
},
{
"id": "ex935",
"input": "The warehouse clerk can guarantee that at the next step the delivery label will be generated, and the shipping service can too.",
"outputs": [
{
"formula": "<<WarehouseClerk>>X delivery_label_generated && <<ShippingService>>X delivery_label_generated"
}
]
},
{
"id": "ex936",
"input": "The bedside terminal can guarantee that one transition later the patient wristband will be scanned, and the nurse station can too.",
"outputs": [
{
"formula": "<<BedsideTerminal>>X patient_wristband_scanned && <<NurseStation>>X patient_wristband_scanned"
}
]
},
{
"id": "ex937",

"input": "The investigator can guarantee that upon entering the next state the evidence tag will be attached, and the registry system can too.",
"outputs": [
{
"formula": "<<Investigator>>X evidence_tag_attached && <<RegistrySystem>>X evidence_tag_attached"
}
]
},
{
"id": "ex938",
"input": "The clinical kiosk can guarantee that right after this step the consent prompt will be displayed, and the patient portal can too.",
"outputs": [
{
"formula": "<<ClinicalKiosk>>X consent_prompt_displayed && <<PatientPortal>>X consent_prompt_displayed"
}
]
},
{
"id": "ex939",
"input": "The organizer can guarantee that in the next state the meeting link will be created, and the calendar service can too.",
"outputs": [
{
"formula": "<<Organizer>>X meeting_link_created && <<CalendarService>>X meeting_link_created"
}
]
},
{
"id": "ex940",
"input": "The machine controller can guarantee that in the following configuration the safety latch will be engaged, and the safety relay can too.",
"outputs": [
{
"formula": "<<MachineController>>X safety_latch_engaged && <<SafetyRelay>>X safety_latch_engaged"
}
]
},
{
"id": "ex941",
"input": "The field agent can guarantee that immediately afterward the inspection note will be uploaded, and the reporting app can too.",
"outputs": [
{

```
        "formula": "<<FieldAgent>>X inspection_note_uploaded && <<ReportingApp>>X inspection_note_uploaded"
      }
    ]
  },
  {
    "id": "ex942",
    "input": "The authentication server can guarantee that in the immediately succeeding state the recovery token will be issued, and the account portal can too.",
    "outputs": [
      {
        "formula": "<<AuthenticationServer>>X recovery_token_issued && <<AccountPortal>>X recovery_token_issued"
      }
    ]
  },

  {
    "id": "ex943",
    "input": "Every receptionist can guarantee that in the immediate successor state the access badge will be printed.",
    "outputs": [
      {
        "formula": "<<Receptionist1>>X access_badge_printed_1 && <<Receptionist2>>X access_badge_printed_2 && <<Receptionist3>>X access_badge_printed_3"
      },
      {
        "formula": "<<Receptionist1,Receptionist2,Receptionist3>>X access_badge_printed"
      }
    ]
  },
  {
    "id": "ex944",
    "input": "Every diagnostic console can guarantee that on the next transition the error flag will be cleared.",
    "outputs": [
      {
        "formula": "<<DiagnosticConsole1>>X error_flag_cleared_1 && <<DiagnosticConsole2>>X error_flag_cleared_2 && <<DiagnosticConsole3>>X error_flag_cleared_3"
      },
      {
        "formula": "<<DiagnosticConsole1,DiagnosticConsole2,DiagnosticConsole3>>X error_flag_cleared"
      }
    ]
  },
  {
```

```
    "id": "ex945",
    "input": "Every warehouse clerk can guarantee that at the next step the delivery label will
be generated.",
    "outputs": [
      {
        "formula": "<<WarehouseClerk1>>X delivery_label_generated_1 &&
<<WarehouseClerk2>>X delivery_label_generated_2"
      },
      {
        "formula": "<<WarehouseClerk1,WarehouseClerk2>>X delivery_label_generated"
      }
    ]
  },
  {
    "id": "ex946",
    "input": "Every bedside terminal can guarantee that one transition later the patient
wristband will be scanned.",
    "outputs": [
      {
        "formula": "<<BedsideTerminal1>>X patient_wristband_scanned_1 &&
<<BedsideTerminal2>>X patient_wristband_scanned_2 && <<BedsideTerminal3>>X
patient_wristband_scanned_3"
      },
      {
        "formula": "<<BedsideTerminal1,BedsideTerminal2,BedsideTerminal3>>X
patient_wristband_scanned"
      }
    ]
  },
  {
    "id": "ex947",
    "input": "Every investigator can guarantee that upon entering the next state the evidence
tag will be attached.",
    "outputs": [
      {
        "formula": "<<Investigator1>>X evidence_tag_attached_1 && <<Investigator2>>X
evidence_tag_attached_2 && <<Investigator3>>X evidence_tag_attached_3 &&
<<Investigator4>>X evidence_tag_attached_4"
      },
      {
        "formula": "<<Investigator1,Investigator2,Investigator3,Investigator4>>X
evidence_tag_attached"
      }
    ]
  },
  {
    "id": "ex948",
```

```
    "input": "Every clinical kiosk can guarantee that right after this step the consent prompt will be displayed.",
    "outputs": [
      {
        "formula": "<<ClinicalKiosk1>>X consent_prompt_displayed_1 && <<ClinicalKiosk2>>X consent_prompt_displayed_2 && <<ClinicalKiosk3>>X consent_prompt_displayed_3"
      },
      {
        "formula": "<<ClinicalKiosk1,ClinicalKiosk2,ClinicalKiosk3>>X consent_prompt_displayed"
      }
    ]
  },
  {
    "id": "ex949",
    "input": "Every organizer can guarantee that in the next state the meeting link will be created.",
    "outputs": [
      {
        "formula": "<<Organizer1>>X meeting_link_created_1 && <<Organizer2>>X meeting_link_created_2 && <<Organizer3>>X meeting_link_created_3"
      },
      {
        "formula": "<<Organizer1,Organizer2,Organizer3>>X meeting_link_created"
      }
    ]
  },
  {
    "id": "ex950",
    "input": "Every machine controller can guarantee that in the following configuration the safety latch will be engaged.",
    "outputs": [
      {
        "formula": "<<MachineController1>>X safety_latch_engaged_1 && <<MachineController2>>X safety_latch_engaged_2 && <<MachineController3>>X safety_latch_engaged_3"
      },
      {
        "formula": "<<MachineController1,MachineController2,MachineController3>>X safety_latch_engaged"
      }
    ]
  },
  {
    "id": "ex951",
    "input": "Every field agent can guarantee that immediately afterward the inspection note will be uploaded.",
    "outputs": [
```

```
      {
        "formula": "<<FieldAgent1>>X inspection_note_uploaded_1 && <<FieldAgent2>>X
inspection_note_uploaded_2 && <<FieldAgent3>>X inspection_note_uploaded_3"
      },
      {
        "formula": "<<FieldAgent1,FieldAgent2,FieldAgent3>>X inspection_note_uploaded"
      }
    ]
  },
  {
    "id": "ex952",
    "input": "Every authentication server can guarantee that in the immediately succeeding
state the recovery token will be issued.",
    "outputs": [
      {
        "formula": "<<AuthenticationServer1>>X recovery_token_issued_1 &&
<<AuthenticationServer2>>X recovery_token_issued_2"
      },
      {
        "formula": "<<AuthenticationServer1,AuthenticationServer2>>X recovery_token_issued"
      }
    ]
  },
  {
    "id": "ex953",
    "input": "The dispatch hub can guarantee that on the next transition the route code will be
assigned.",
    "outputs": [
      {
        "formula": "<<DispatchHub>>X route_code_assigned"
      }
    ]
  },
  {
    "id": "ex954",
    "input": "The evidence scanner can guarantee that in the immediate successor state the
barcode will be captured.",
    "outputs": [
      {
        "formula": "<<EvidenceScanner>>X barcode_captured"
      }
    ]
  },
  {
    "id": "ex955",
    "input": "The intake desk can guarantee that the request will remain pending until the
missing attachment is received.",
    "outputs": [
```

    {
      "formula": "<<IntakeDesk>>(request_pending U missing_attachment_received)"
    }
  ]
},
{
  "id": "ex956",
  "input": "The service portal can guarantee that the account will remain limited until identity verification is completed.",
  "outputs": [
    {
      "formula": "<<ServicePortal>>(account_limited U identity_verification_completed)"
    }
  ]
},
{
  "id": "ex957",
  "input": "The archive system can guarantee that the record will remain locked until the retention check is completed.",
  "outputs": [
    {
      "formula": "<<ArchiveSystem>>(record_locked U retention_check_completed)"
    }
  ],
  "difficulty": "hard"
},
{
  "id": "ex958",
  "input": "The coordinator can guarantee that the agenda will remain provisional until the chair approves it.",
  "outputs": [
    {
      "formula": "<<Coordinator>>(agenda_provisional U chair_approved)"
    }
  ]
},
{
  "id": "ex959",
  "input": "The monitoring service can guarantee that the alert will remain active until the operator acknowledges it.",
  "outputs": [
    {
      "formula": "<<MonitoringService>>(alert_active U operator_acknowledged)"
    }
  ]
},
{
  "id": "ex960",

"input": "The dispatch office can guarantee that the route will remain suspended until the hazard is cleared.",
"outputs": [
{
"formula": "<<DispatchOffice>>(route_suspended U hazard_cleared)"
}
]
},
{
"id": "ex961",
"input": "The payment gateway can guarantee that the transfer will remain blocked until the fraud review is completed.",
"outputs": [
{
"formula": "<<PaymentGateway>>(transfer_blocked U fraud_review_completed)"
}
]
},
{
"id": "ex962",
"input": "The laboratory console can guarantee that the sample will remain unprocessed until the consent flag is enabled.",
"outputs": [
{
"formula": "<<LaboratoryConsole>>(sample_unprocessed U consent_flag_enabled)"
}
]
},
{
"id": "ex963",
"input": "The maintenance unit can guarantee that the machine will remain offline until the inspection certificate is issued.",
"outputs": [
{
"formula": "<<MaintenanceUnit>>(machine_offline U inspection_certificate_issued)"
}
]
},
{
"id": "ex964",
"input": "The review board can guarantee that the case will remain open until the final recommendation is recorded.",
"outputs": [
{
"formula": "<<ReviewBoard>>(case_open U final_recommendation_recorded)"
}
]
},

```
 {
   "id": "ex965",
   "input": "The registrar can guarantee that the enrollment will remain incomplete until the
fee is paid.",
   "outputs": [
     {
       "formula": "<<Registrar>>(enrollment_incomplete U fee_paid)"
     }
   ]
 },
 {
   "id": "ex966",
   "input": "The security console can guarantee that the badge will remain inactive until
clearance is granted.",
   "outputs": [
     {
       "formula": "<<SecurityConsole>>(badge_inactive U clearance_granted)"
     }
   ]
 },
 {
   "id": "ex967",
   "input": "The clinical platform can guarantee that the prescription will remain unsigned until
the physician confirms it.",
   "outputs": [
     {
       "formula": "<<ClinicalPlatform>>(prescription_unsigned U physician_confirmed)"
     }
   ]
 },
 {
   "id": "ex968",
   "input": "The collector can guarantee that the Pokemon box will remain sealed until the
restock is granted.",
   "outputs": [
     {
       "formula": "<<collector>>(box_sealed U restock_granted)"
     }
   ]
 },
 {
   "id": "ex969",
   "input": "The recovery service can guarantee that at the next step the restore job will start
and that sooner or later the corrupted profile will be rebuilt.",
   "outputs": [
     {
       "formula": "<<RecoveryService>>(X restore_job_started && F
corrupted_profile_rebuilt)"
```

}
]
},
{
"id": "ex970",
"input": "The operations center can guarantee that at the next step both the incident queue will be opened and the supervisor will be notified, that the response channel will always remain available, and that unsafe escalation will never occur, that the audit trail will always be preserved, that the command log will remain visible in every state, that the fallback rule will stay enabled throughout the execution, and that the ticket will remain pending until validation is completed, the service will remain degraded until recovery is confirmed, the alert will remain active until the operator acknowledges it, and the case will remain open until the report is filed.",
"outputs": [
{
"formula": "<<OperationsCenter>>(X (incident_queue_opened && supervisor_notified) && G response_channel_available && G !unsafe_escalation && G audit_trail_preserved && G command_log_visible && G fallback_rule_enabled && (ticket_pending U validation_completed) && (service_degraded U recovery_confirmed) && (alert_active U operator_acknowledged) && (case_open U report_filed))"
}
]
},

Reviewer 2: I think that “both” is perfect to encapsulate the conjunction inside the next operator.

Reviewer 1: I agree.

{
"id": "ex971",
"input": "The hospital platform can guarantee that at the next step the triage form will be created and the nurse will be alerted, that the patient record will always remain protected, that unauthorized treatment will never start, that the medication log will always be retained, that the monitoring screen will remain readable in every state, that the safety protocol will stay enforced throughout the execution, and that the order will remain suspended until allergy screening is completed, the bed request will remain pending until admission is approved, the alert will remain active until the clinician acknowledges it, and the discharge plan will remain incomplete until referrals are entered.",
"outputs": [
{
"formula": "<<HospitalPlatform>>(X triage_form_created && X nurse_alerted && G patient_record_protected && G !unauthorized_treatment_started && G medication_log_retained && G monitoring_screen_readable && G safety_protocol_enforced && (order_suspended U allergy_screening_completed) && (bed_request_pending U admission_approved) && (alert_active U clinician_acknowledged) && (discharge_plan_incomplete U referrals_entered))"
}
]
},
{

"id": "ex972",
"input": "The security gateway can guarantee that at the next step the session will be challenged and the token will be checked, that the firewall policy will always remain enforced, that malicious payloads will never be forwarded, that the access log will always be preserved, that the quarantine rule will remain active in every state, that the denial rule will stay enabled throughout the execution, and that the request will remain blocked until identity is verified, the credential will remain untrusted until attestation is validated, the device will remain quarantined until scanning is completed, and the flow will remain inspected until the signature is matched.",
"outputs": [
{
"formula": "<<SecurityGateway>>(X session_challenged && X token_checked && G firewall_policy_enforced && G !malicious_payloads_forwarded && G access_log_preserved && G quarantine_rule_active && G denial_rule_enabled && (request_blocked U identity_verified) && (credential_untrusted U attestation_validated) && (device_quarantined U scan_completed) && (flow_inspected U signature_matched))"
}
]
},
{
"id": "ex973",
"input": "The archive service can guarantee that at the next step the document will be indexed and the checksum will be computed, that the catalogue will always remain searchable, that protected files will never be exposed, that the retention policy will always be enforced, that the access trail will remain complete in every state, that the storage mirror will stay synchronized throughout the execution, and that the manuscript will remain sealed until review is completed, the folder will remain locked until authorization is granted, the export will remain disabled until clearance is issued, and the record will remain pending until metadata is approved.",
"outputs": [
{
"formula": "<<ArchiveService>>(X document_indexed && X checksum_computed && G catalogue_searchable && G !protected_files_exposed && G retention_policy_enforced && G access_trail_complete && G storage_mirror_synchronized && (manuscript_sealed U review_completed) && (folder_locked U authorization_granted) && (export_disabled U clearance_issued) && (record_pending U metadata_approved))"
}
]
},
{
"id": "ex974",
"input": "The court system can guarantee that at the next step the docket will be updated and the clerk will be notified, that the hearing record will always remain complete, that sealed evidence will never be disclosed, that the filing rule will always be respected, that the transcript will remain accessible in every state, that the custody chain will stay intact throughout the execution, and that the motion will remain pending until the judge rules, the file will remain sealed until release is authorized, the hearing will remain suspended until the panel arrives, and the appeal will remain open until the final order is issued.",

"outputs": [
{
"formula": "<<CourtSystem>>(X docket_updated && X clerk_notified && G hearing_record_complete && G !sealed_evidence_disclosed && G filing_rule_respected && G transcript_accessible && G custody_chain_intact && (motion_pending U judge_ruled) && (file_sealed U release_authorized) && (hearing_suspended U panel_arrived) && (appeal_open U final_order_issued))"
}
]
},
{
"id": "ex975",
"input": "The logistics hub can guarantee that at the next step the parcel will be scanned and the route will be assigned, that the tracking page will always remain available, that lost-package status will never be hidden, that the delivery log will always be preserved, that the route board will remain current in every state, that the depot signal will stay active throughout the execution, and that the shipment will remain on hold until customs clears it, the van will remain parked until loading is completed, the route will remain blocked until the hazard is removed, and the receipt will remain pending until delivery is confirmed.",
"outputs": [
{
"formula": "<<LogisticsHub>>(X parcel_scanned && X route_assigned && G tracking_page_available && G !lost_package_status_hidden && G delivery_log_preserved && G route_board_current && G depot_signal_active && (shipment_on_hold U customs_cleared) && (van_parked U loading_completed) && (route_blocked U hazard_removed) && (receipt_pending U delivery_confirmed))"
}
]
},
{
"id": "ex976",
"input": "The payment platform can guarantee that at the next step the transaction will be flagged and the payer will be notified, that the ledger will always remain consistent, that unauthorized transfers will never be executed, that the audit record will always be preserved, that the risk rule will remain active in every state, that the fraud screen will stay enabled throughout the execution, and that the payment will remain blocked until review is completed, the refund will remain pending until approval is granted, the account will remain restricted until identity is verified, and the receipt will remain withheld until settlement is confirmed.",
"outputs": [
{
"formula": "<<PaymentPlatform>>(X (transaction_flagged && payer_notified) && G ledger_consistent && G !unauthorized_transfers_executed && G audit_record_preserved && G risk_rule_active && G fraud_screen_enabled && (payment_blocked U review_completed) && (refund_pending U approval_granted) && (account_restricted U identity_verified) && (receipt_withheld U settlement_confirmed))"

}

]
},
{
"id": "ex977",
"input": "The education portal can guarantee that at the next step the exam page will be opened and the timer will be started, that the gradebook will always remain consistent, that invalid certificates will never be issued, that the submission log will always be retained, that the rubric will remain visible in every state, that the plagiarism rule will stay enforced throughout the execution, and that the exam will remain locked until enrollment is confirmed, the answer sheet will remain hidden until grading is completed, the appeal will remain pending until review ends, and the certificate will remain withheld until payment is recorded.",
"outputs": [
{
"formula": "<<EducationPortal>>(X exam_page_opened && X timer_started && G gradebook_consistent && G !invalid_certificates_issued && G submission_log_retained && G rubric_visible && G plagiarism_rule_enforced && (exam_locked U enrollment_confirmed) && (answer_sheet_hidden U grading_completed) && (appeal_pending U review_ended) && (certificate_withheld U payment_recorded))"
}
]
},
{
"id": "ex978",
"input": "The building controller can guarantee that at the next step the alarm will sound and at the step after the next one the exit sign will illuminate, that the evacuation route will always remain visible, that emergency doors will never be blocked, that the safety log will always be preserved, that the hazard map will remain current in every state, that the warning signal will stay active throughout the execution, and that the stairwell will remain closed until smoke is cleared, the elevator will remain disabled until inspection is completed, the area will remain evacuated until clearance is granted, and the incident will remain open until the report is submitted.",
"outputs": [
{
"formula": "<<BuildingController>>(X alarm_sounded && XX exit_sign_illuminated && G evacuation_route_visible && G !emergency_doors_blocked && G safety_log_preserved && G hazard_map_current && G warning_signal_active && (stairwell_closed U smoke_cleared) && (elevator_disabled U inspection_completed) && (area_evacuated U clearance_granted) && (incident_open U report_submitted))"
}
]
},
{
"id": "ex979",
"input": "The research repository can guarantee that at the next step the dataset will be registered and the license will be attached, that the metadata will always remain complete, that embargoed material will never be exposed, that the citation record will always be preserved, that the access rule will remain active in every state, that the integrity check will stay enabled throughout the execution, and that the dataset will remain private until the

embargo ends, the review will remain open until approval is granted, the version will remain provisional until validation is completed, and the release will remain pending until the curator confirms it.",
"outputs": [
{
"formula": "<<ResearchRepository>>(X dataset_registered && X license_attached && G metadata_complete && G !embargoed_material_exposed && G citation_record_preserved && G access_rule_active && G integrity_check_enabled && (dataset_private U embargo_ended) && (review_open U approval_granted) && (version_provisional U validation_completed) && (release_pending U curator_confirmed))"
}
]
},
{
"id": "ex980",
"input": "The traffic authority can guarantee that at the next step the signal plan will be loaded and the detour notice will be posted, that the emergency lane will always remain clear, that invalid route signs will never be displayed, that the incident board will always be updated, that the congestion rule will remain active in every state, that the pedestrian warning will stay enabled throughout the execution, and that the lane will remain closed until debris is removed, the detour will remain active until traffic normalizes, the crossing will remain blocked until the convoy passes, and the alert will remain open until the operator closes it.",
"outputs": [
{
"formula": "<<TrafficAuthority>>(X signal_plan_loaded && X detour_notice_posted && G emergency_lane_clear && G !invalid_route_signs_displayed && G incident_board_updated && G congestion_rule_active && G pedestrian_warning_enabled && (lane_closed U debris_removed) && (detour_active U traffic_normalized) && (crossing_blocked U convoy_passed) && (alert_open U operator_closed))"
}
]
},
{
"id": "ex981",
"input": "The cloud orchestrator can guarantee that at the next step the container will be started and the health probe will be scheduled, that the service mesh will always remain configured, that unsafe deployments will never be promoted, that the event log will always be retained, that the scaling rule will remain active in every state, that the rollback option will stay enabled throughout the execution, and that the release will remain paused until testing is completed, the node will remain drained until migration ends, the workload will remain queued until capacity is available, and the incident will remain open until service is restored.",
"outputs": [
{
"formula": "<<CloudOrchestrator>>(X container_started && X health_probe_scheduled && G service_mesh_configured && G !unsafe_deployments_promoted && G event_log_retained && G scaling_rule_active && G rollback_option_enabled &&

(release_paused U testing_completed) && (node_drained U migration_ended) && (workload_queued U capacity_available) && (incident_open U service_restored))"
      }
    ]
  },
  {
    "id": "ex982",
    "input": "The laboratory system can guarantee that at the next step the sample will be labeled and the assay will be started, that the chain of custody will always remain intact, that contaminated results will never be released, that the instrument log will always be preserved, that the calibration rule will remain active in every state, that the safety interlock will stay enabled throughout the execution, and that the sample will remain stored until analysis begins, the report will remain draft until validation ends, the container will remain sealed until inspection is completed, and the result will remain withheld until authorization is granted.",
    "outputs": [
      {
        "formula": "<<LaboratorySystem>>(X sample_labeled && X assay_started && G chain_of_custody_intact && G !contaminated_results_released && G instrument_log_preserved && G calibration_rule_active && G safety_interlock_enabled && (sample_stored U analysis_started) && (report_draft U validation_ended) && (container_sealed U inspection_completed) && (result_withheld U authorization_granted))"
      }
    ]
  },
  {
    "id": "ex983",
    "input": "The museum system can guarantee that at the next step the exhibit will be catalogued and the label will be printed, that the climate record will always remain complete, that restricted artifacts will never be handled without clearance, that the loan log will always be preserved, that the lighting rule will remain active in every state, that the alarm sensor will stay enabled throughout the execution, and that the gallery will remain closed until humidity stabilizes, the artifact will remain crated until transport begins, the case will remain locked until the curator approves it, and the display will remain hidden until installation is completed.",
    "outputs": [
      {
        "formula": "<<MuseumSystem>>(X exhibit_catalogued && X label_printed && G climate_record_complete && G !restricted_artifacts_handled_without_clearance && G loan_log_preserved && G lighting_rule_active && G alarm_sensor_enabled && (gallery_closed U humidity_stabilized) && (artifact_crated U transport_started) && (case_locked U curator_approved) && (display_hidden U installation_completed))"
      }
    ]
  },
  {
    "id": "ex984",
    "input": "The election platform can guarantee that at the next step the ballot will be registered and the voter receipt will be generated, that the tally log will always remain

consistent, that duplicate votes will never be accepted, that the audit rule will always be enforced, that the precinct status will remain visible in every state, that the sealing rule will stay active throughout the execution, and that the ballot will remain encrypted until counting begins, the challenge will remain pending until review ends, the box will remain sealed until observers arrive, and the result will remain provisional until certification is completed.",
"outputs": [
{
"formula": "<<ElectionPlatform>>(X ballot_registered && X voter_receipt_generated && G tally_log_consistent && G !duplicate_votes_accepted && G audit_rule_enforced && G precinct_status_visible && G sealing_rule_active && (ballot_encrypted U counting_started) && (challenge_pending U review_ended) && (box_sealed U observers_arrived) && (result_provisional U certification_completed))"
}
]
},
{
"id": "ex985",
"input": "The railway controller can guarantee that at the next step the signal will change and the platform notice will be updated, that the track circuit will always remain monitored, that unsafe departures will never be authorized, that the timetable log will always be retained, that the braking rule will remain active in every state, that the crossing alarm will stay enabled throughout the execution, and that the train will remain halted until clearance is received, the platform will remain closed until boarding begins, the route will remain locked until the switch is confirmed, and the delay will remain open until the train departs.",
"outputs": [
{
"formula": "<<RailwayController>>(X signal_changed && X platform_notice_updated && G track_circuit_monitored && G !unsafe_departures_authorized && G timetable_log_retained && G braking_rule_active && G crossing_alarm_enabled && (train_halted U clearance_received) && (platform_closed U boarding_started) && (route_locked U switch_confirmed) && (delay_open U train_departed))"
}
]
},
{
"id": "ex986",
"input": "The library platform can guarantee that at the next step the request will be logged and the patron will be notified, that the catalogue will always remain searchable, that restricted books will never be loaned without permission, that the borrowing record will always be preserved, that the privacy rule will remain active in every state, that the reservation queue will stay enabled throughout the execution, and that the book will remain reserved until pickup expires, the account will remain limited until the fine is paid, the archive will remain closed until clearance is granted, and the loan will remain pending until the librarian confirms it.",
"outputs": [
{
"formula": "<<LibraryPlatform>>(X request_logged && X patron_notified && G catalogue_searchable && G !restricted_books_loaned_without_permission && G

borrowing_record_preserved && G privacy_rule_active && G reservation_queue_enabled && (book_reserved U pickup_expired) && (account_limited U fine_paid) && (archive_closed U clearance_granted) && (loan_pending U librarian_confirmed))"
      }
    ]
  },
  {
    "id": "ex987",
    "input": "The insurance system can guarantee that at the next step the claim will be opened and the assessor will be assigned, that the claim log will always remain complete, that invalid payouts will never be issued, that the document trail will always be preserved, that the fraud rule will remain active in every state, that the appeal option will stay available throughout the execution, and that the claim will remain pending until evidence is reviewed, the payment will remain blocked until approval is granted, the policy will remain active until cancellation is confirmed, and the case will remain open until settlement is recorded.",
    "outputs": [
      {
        "formula": "<<InsuranceSystem>>(X claim_opened && X assessor_assigned && G claim_log_complete && G !invalid_payouts_issued && G document_trail_preserved && G fraud_rule_active && G appeal_option_available && (claim_pending U evidence_reviewed) && (payment_blocked U approval_granted) && (policy_active U cancellation_confirmed) && (case_open U settlement_recorded))"
      }
    ]
  },
  {
    "id": "ex988",
    "input": "The harbor authority can guarantee that at the next step the berth will be assigned and the pilot will be notified, that the navigation channel will always remain clear, that unsafe docking will never be authorized, that the vessel log will always be retained, that the weather rule will remain active in every state, that the port signal will stay enabled throughout the execution, and that the ship will remain anchored until clearance is granted, the cargo will remain sealed until inspection is completed, the crane will remain idle until loading begins, and the manifest will remain pending until customs approves it.",
    "outputs": [
      {
        "formula": "<<HarborAuthority>>(X (berth_assigned && pilot_notified) && G navigation_channel_clear && G !unsafe_docking_authorized && G vessel_log_retained && G weather_rule_active && G port_signal_enabled && (ship_anchored U clearance_granted) && (cargo_sealed U inspection_completed) && (crane_idle U loading_started) && (manifest_pending U customs_approved))"
      }
    ]
  },
  {
    "id": "ex989",
    "input": "The civic portal can guarantee that at the next step the petition will be filed and the tracking number will be created, that the case history will always remain available, that

anonymous signatures will never be counted, that the submission log will always be preserved, that the eligibility rule will remain active in every state, that the notification service will stay enabled throughout the execution, and that the petition will remain pending until signatures are verified, the hearing will remain unscheduled until quorum is met, the draft will remain hidden until publication is approved, and the case will remain open until the council records its decision.",
"outputs": [
{
"formula": "<<CivicPortal>>(X petition_filed && X tracking_number_created && G case_history_available && G !anonymous_signatures_counted && G submission_log_preserved && G eligibility_rule_active && G notification_service_enabled && (petition_pending U signatures_verified) && (hearing_unscheduled U quorum_met) && (draft_hidden U publication_approved) && (case_open U council_decision_recorded))"
}
]
},
{
"id": "ex990",
"input": "The broadcast system can guarantee that at the next step the bulletin will be queued and the producer will be alerted, that the emergency feed will always remain available, that false warnings will never be aired, that the transmission log will always be retained, that the caption rule will remain active in every state, that the backup transmitter will stay enabled throughout the execution, and that the bulletin will remain pending until verification is completed, the channel will remain reserved until the alert ends, the message will remain draft until the editor approves it, and the story will remain open until the update is published.",
"outputs": [
{
"formula": "<<BroadcastSystem>>(X bulletin_queued && X producer_alerted && G emergency_feed_available && G !false_warnings_aired && G transmission_log_retained && G caption_rule_active && G backup_transmitter_enabled && (bulletin_pending U verification_completed) && (channel_reserved U alert_ended) && (message_draft U editor_approved) && (story_open U update_published))"
}
]
},
{
"id": "ex991",
"input": "The inspection service can guarantee that at the next step the site will be registered and the checklist will be opened, that the compliance record will always remain accurate, that unsafe certificates will never be issued, that the evidence log will always be preserved, that the sampling rule will remain active in every state, that the retest option will stay available throughout the execution, and that the site will remain restricted until hazards are removed, the certificate will remain pending until measurements are reviewed, the report will remain draft until the inspector signs it, and the case will remain open until corrective action is recorded.",
"outputs": [
{

"formula": "<<InspectionService>>(X site_registered && X checklist_opened && G compliance_record_accurate && G !unsafe_certificates_issued && G evidence_log_preserved && G sampling_rule_active && G retest_option_available && (site_restricted U hazards_removed) && (certificate_pending U measurements_reviewed) && (report_draft U inspector_signed) && (case_open U corrective_action_recorded))"
}
]
},
{
"id": "ex992",
"input": "The help desk can guarantee that at the next step the user will be notified, that the ticket will always remain traceable, that invalid replies will never be accepted, that the conversation log will always be retained, and that the request will remain pending until the missing detail is supplied, the account will remain limited until identity is checked, and the case will remain open until the resolution is confirmed.",
"outputs": [
{
"formula": "<<HelpDesk>>(X user_notified && G ticket_traceable && G !invalid_replies_accepted && G conversation_log_retained && G request_pending && (missing_detail_supplied U request_resolved) && (identity_checked U account_limited) && (resolution_confirmed U case_open))"
}
]
},
{
"id": "ex993",
"input": "The clinic desk can guarantee that at the next step the appointment will be created, that the schedule will always remain visible, that duplicate bookings will never be accepted, that the patient note will always be preserved, and that the referral will remain pending until approval is granted, the slot will remain reserved until confirmation arrives, and the case will remain open until the visit is completed.",
"outputs": [
{
"formula": "<<ClinicDesk>>(X appointment_created && G schedule_visible && G !duplicate_bookings_accepted && G patient_note_preserved && G referral_pending && (approval_granted U slot_reserved) && (confirmation_arrived U case_open) && (visit_completed U case_closed))"
}
]
},
{
"id": "ex994",
"input": "The customs desk can guarantee that at the next step the declaration will be registered, that the manifest will always remain available, that prohibited cargo will never be released, that the inspection log will always be retained, and that the shipment will remain blocked until scanning ends, the container will remain sealed until clearance is issued, and the case will remain open until duties are paid.",
"outputs": [

{
"formula": "<<CustomsDesk>>(X declaration_registered && G manifest_available && G !prohibited_cargo_released && G inspection_log_retained && G shipment_blocked && (shipment_blocked U scanning_ended) && (container_sealed U clearance_issued) && (case_open U duties_paid))"
}
]
},
{
"id": "ex995",
"input": "The data platform can guarantee that at the next step the dataset will be copied, that the backup will always remain encrypted, that private fields will never be exposed, that the lineage log will always be preserved, and that the export will remain disabled until masking is completed, the job will remain queued until capacity is available, and the version will remain provisional until validation ends.",
"outputs": [
{
"formula": "<<DataPlatform>>(X dataset_copied && G backup_encrypted && G !private_fields_exposed && G lineage_log_preserved && G export_disabled && (export_disabled U masking_completed) && (job_queued U capacity_available) && (version_provisional U validation_ended))"
}
]
},
{
"id": "ex996",
"input": "The maintenance desk can guarantee that at the next step the work order will be opened, that the machine will always remain traceable, that unsafe restarts will never be allowed, that the service log will always be retained, and that the machine will remain offline until inspection is completed, the part will remain reserved until repair begins, and the ticket will remain open until the technician closes it.",
"outputs": [
{
"formula": "<<MaintenanceDesk>>(X work_order_opened && G machine_traceable && G !unsafe_restarts_allowed && G service_log_retained && G machine_offline && (machine_offline U inspection_completed) && (part_reserved U repair_started) && (ticket_open U technician_closed))"
}
]
},
{
"id": "ex997",
"input": "The records desk can guarantee that at the next step the file will be stamped, that the registry will always remain searchable, that sealed pages will never be copied, that the access log will always be preserved, and that the file will remain locked until release is authorized, the request will remain pending until identity is verified, and the case will remain open until the archive confirms receipt.",
"outputs": [

{
"formula": "<<RecordsDesk>>(X file_stamped && G registry_searchable && G !sealed_pages_copied && G access_log_preserved && G file_locked && (file_locked U release_authorized) && (request_pending U identity_verified) && (case_open U archive_confirmed_receipt))"
}
]
},
{
"id": "ex998",
"input": "The billing service can guarantee that at the next step the invoice will be drafted, that the account will always remain balanced, that invalid charges will never be posted, that the payment log will always be retained, and that the invoice will remain pending until review is completed, the refund will remain blocked until approval is granted, and the case will remain open until settlement is recorded.",
"outputs": [
{
"formula": "<<BillingService>>(X invoice_drafted && G account_balanced && G !invalid_charges_posted && G payment_log_retained && G invoice_pending && (invoice_pending U review_completed) && (refund_blocked U approval_granted) && (case_open U settlement_recorded))"
}
]
},
{
"id": "ex999",
"input": "The access office can guarantee that at the next step the badge will be printed, that the entry log will always remain complete, that revoked credentials will never be accepted, that the visitor record will always be retained, and that the badge will remain inactive until clearance is granted, the door will remain locked until supervision arrives, and the request will remain pending until the host approves it.",
"outputs": [
{
"formula": "<<AccessOffice>>(X badge_printed && G entry_log_complete && G !revoked_credentials_accepted && G visitor_record_retained && G badge_inactive && (badge_inactive U clearance_granted) && (door_locked U supervision_arrived) && (request_pending U host_approved))"
}
]
},
{
"id": "ex1000",
"input": "The coordination unit can guarantee that at the next step the master ticket will be opened, that the control dashboard will always remain available, that unsafe closure will never occur, that the history log will always be retained, that the escalation path will remain visible in every state, that the request will remain pending until validation ends, the assignment will remain provisional until ownership is accepted, the communication channel will remain active until the incident is closed, the evidence bundle will remain sealed until

```
approval is granted, the recovery plan will remain draft until review is completed, and the status page will remain locked until the coordinator releases it.",
    "outputs": [
      {
        "formula": "<<CoordinationUnit>>(X master_ticket_opened && G control_dashboard_available && G !unsafe_closure && G history_log_retained && G escalation_path_visible && (request_pending U validation_ended) && (assignment_provisional U ownership_accepted) && (communication_channel_active U incident_closed) && (evidence_bundle_sealed U approval_granted) && (recovery_plan_draft U review_completed) && (status_page_locked U coordinator_released))"
      }
    ]
  },
  {
    "id": "ex1001",
    "input": "The telescope controller can, but the target selector cannot cannot, guarantee that on the next transition the exposure sequence will start.",
    "outputs": [
      {
        "formula": "<<TelescopeController>>X exposure_sequence_started && !<<TargetSelector>>X exposure_sequence_started"
      }
    ]
  },
  {
    "id": "ex1002",
    "input": "The submarine navigator can, but the navigation planner cannot, guarantee that in the immediately following state the ballast valve will be adjusted.",
    "outputs": [
      {
        "formula": "<<SubmarineNavigator>>X ballast_valve_adjusted && !<<NavigationPlanner>>X ballast_valve_adjusted"
      }
    ]
  },
  {
    "id": "ex1003",
    "input": "The translation engine can, but the style checker cannot, guarantee that at the next step the ambiguous term will be resolved.",
    "outputs": [
      {
        "formula": "<<TranslationEngine>>X ambiguous_term_resolved && !<<StyleChecker>>X ambiguous_term_resolved"
      }
    ]
  },
  {
    "id": "ex1004",
```

```
   "input": "The voting clerk can, but the poll observer cannot, guarantee that in the immediate successor state the spoiled ballot will be marked.",
   "outputs": [
    {
     "formula": "<<VotingClerk>>X spoiled_ballot_marked && !<<PollObserver>>X spoiled_ballot_marked"
    }
   ]
  },
  {
   "id": "ex1005",
   "input": "The seismic monitor can, but the network manager cannot, guarantee that on the very next transition the tremor warning will be raised.",
   "outputs": [
    {
     "formula": "<<SeismicMonitor>>X tremor_warning_raised && !<<NetworkManager>>X tremor_warning_raised"
    }
   ]
  },
  {
   "id": "ex1006",
   "input": "The aquarium controller can, but the feeding scheduler cannot, guarantee that one step later the oxygen pump will be activated.",
   "outputs": [
    {
     "formula": "<<AquariumController>>X oxygen_pump_activated && !<<FeedingScheduler>>X oxygen_pump_activated"
    }
   ]
  },
  {
   "id": "ex1007",
   "input": "The automated theater lighting console can, but the automated audio console cannot, guarantee that at the next tick the stage lights will be dimmed.",
   "outputs": [
    {
     "formula": "<<LightingConsole>>X stage_lights_dimmed && !<<AudioConsole>>X stage_lights_dimmed"
    }
   ]
  },
  {
   "id": "ex1008",
   "input": "The satellite scheduler can, but the payload manager cannot, guarantee that upon the next transition the imaging task will be queued.",
   "outputs": [
    {
```

```
      "formula": "<<SatelliteScheduler>>X imaging_task_queued && !<<PayloadManager>>X
imaging_task_queued"
    }
  ]
 },
 {
  "id": "ex1009",
  "input": "The compiler optimizer can, but the syntax highlighter cannot, guarantee that in
the following configuration the redundant branch will be removed.",
  "outputs": [
    {
      "formula": "<<CompilerOptimizer>>X redundant_branch_removed &&
!<<SyntaxHighlighter>>X redundant_branch_removed"
    }
  ]
 },
 {
  "id": "ex1010",
  "input": "The library sorter can, but the barcode scanner cannot, guarantee that
immediately afterward the returned volume will be shelved.",
  "outputs": [
    {
      "formula": "<<LibrarySorter>>X returned_volume_shelved && !<<BarcodeScanner>>X
returned_volume_shelved"
    }
  ]
 },
 {
  "id": "ex1011",
  "input": "The customs scanner can, but the cargo handler cannot, guarantee that in the
next state the suspicious crate will be flagged.",
  "outputs": [
    {
      "formula": "<<CustomsScanner>>X suspicious_crate_flagged && !<<CargoHandler>>X
suspicious_crate_flagged"
    }
  ]
 },
 {
  "id": "ex1012",
  "input": "The observatory dome controller can, but the climate predictor cannot, guarantee
that on the next transition the shutter will be closed.",
  "outputs": [
    {
      "formula": "<<DomeController>>X shutter_closed && !<<ClimatePredictor>>X
shutter_closed"
    }
  ]
```

},
{
"id": "ex1013",
"input": "The railway dispatcher can, but the gate coordinator cannot, guarantee that at the next step the platform change will be announced.",
"outputs": [
{
"formula": "<<RailwayDispatcher>>X platform_change_announced && !<<GateCoordinator>>X platform_change_announced"
}
]
},
{
"id": "ex1014",
"input": "The kiln controller can, but the material feed cannot, guarantee that in the immediate successor state the cooling phase will begin.",
"outputs": [
{
"formula": "<<KilnController>>X cooling_phase_started && !<<MaterialFeed>>X cooling_phase_started"
}
]
},
{
"id": "ex1015",
"input": "The food safety inspector can, but the batch packager cannot, guarantee that on the very next transition the contaminated batch will be isolated.",
"outputs": [
{
"formula": "<<FoodSafetyInspector>>X contaminated_batch_isolated && !<<BatchPackager>>X contaminated_batch_isolated"
}
]
},
{
"id": "ex1016",
"input": "The auction platform can, but the fraud detector cannot, guarantee that one transition later the reserve price will be checked.",
"outputs": [
{
"formula": "<<AuctionPlatform>>X reserve_price_checked && !<<FraudDetector>>X reserve_price_checked"
}
]
},
{
"id": "ex1017",

"input": "The greenhouse scheduler can, but the ventilation fan cannot, guarantee that at the next tick the misting cycle will start.",
"outputs": [
{
"formula": "<<GreenhouseScheduler>>X misting_cycle_started && !<<VentilationFan>>X misting_cycle_started"
}
]
},
{
"id": "ex1018",
"input": "The laboratory freezer controller can, but the cryo pump cannot, guarantee that in the next configuration the alarm threshold will be lowered.",
"outputs": [
{
"formula": "<<FreezerController>>X alarm_threshold_lowered && !<<CryoPump>>X alarm_threshold_lowered"
}
]
},
{
"id": "ex1019",
"input": "The harbor crane controller can, but the dock camera cannot, guarantee that upon entering the next state the container lock will be engaged.",
"outputs": [
{
"formula": "<<CraneController>>X container_lock_engaged && !<<DockCamera>>X container_lock_engaged"
}
]
},
{
"id": "ex1020",
"input": "The patent office system can, but the payment gateway cannot, guarantee that immediately afterward the filing timestamp will be assigned.",
"outputs": [
{
"formula": "<<PatentOfficeSystem>>X filing_timestamp_assigned && !<<PaymentGateway>>X filing_timestamp_assigned"
}
]
},
{
"id": "ex1021",
"input": "The stormwater controller can, but the sewer inflow sensor cannot, guarantee that at the next step the overflow gate will be opened.",
"outputs": [
{

```
      "formula": "<<StormwaterController>>X overflow_gate_opened &&
!<<SewerInflowSensor>>X overflow_gate_opened"
     }
   ]
  },
  {
   "id": "ex1022",
   "input": "The legal filing clerk can, but the archival bot cannot, guarantee that on the next
transition the appeal notice will be recorded.",
   "outputs": [
     {
      "formula": "<<FilingClerk>>X appeal_notice_recorded && !<<ArchivalBot>>X
appeal_notice_recorded"
     }
   ]
  },
  {
   "id": "ex1023",
   "input": "The museum registrar can, but the security patrol cannot, guarantee that in the
immediately succeeding state the loan receipt will be generated.",
   "outputs": [
     {
      "formula": "<<MuseumRegistrar>>X loan_receipt_generated && !<<SecurityPatrol>>X
loan_receipt_generated"
     }
   ]
  },
  {
   "id": "ex1024",
   "input": "The air-quality controller can, but the filter monitor cannot, guarantee that right
after this step the filtration mode will be enabled.",
   "outputs": [
     {
      "formula": "<<AirQualityController>>X filtration_mode_enabled && !<<FilterMonitor>>X
filtration_mode_enabled"
     }
   ]
  },

  {
   "id": "ex1025",
   "input": "The lighthouse controller can, but the fog sensor cannot, guarantee that the
beacon will always remain visible.",
   "outputs": [
     {
      "formula": "<<LighthouseController>>G beacon_visible && !<<FogSensor>>G
beacon_visible"
     }
```

```
    ]
  },
  {
    "id": "ex1026",
    "input": "The treaty archive can, but the encryption module cannot, guarantee that signed
records will never be overwritten.",
    "outputs": [
      {
        "formula": "<<TreatyArchive>>G !signed_records_overwritten &&
!<<EncryptionModule>>G !signed_records_overwritten"
      }
    ]
  },
  {
    "id": "ex1027",
    "input": "The avalanche warning service can, but the snow controller cannot, guarantee
that the hazard level will remain published at all times.",
    "outputs": [
      {
        "formula": "<<AvalancheWarningService>>G hazard_level_published &&
!<<SnowController>>G hazard_level_published"
      }
    ]
  },
  {
    "id": "ex1028",
    "input": "The automated maritime registry can, but the the operator cannot, guarantee that
vessel ownership will remain traceable in every state.",
    "outputs": [
      {
        "formula": "<<MaritimeRegistry>>G vessel_ownership_traceable && !<<Operator>>G
vessel_ownership_traceable"
      }
    ]
  },
  {
    "id": "ex1029",
    "input": "The manuscript repository can, but the access auditor cannot, guarantee that
embargoed drafts will never be exposed.",
    "outputs": [
      {
        "formula": "<<ManuscriptRepository>>G !embargoed_drafts_exposed &&
!<<AccessAuditor>>G !embargoed_drafts_exposed"
      }
    ]
  },
  {
    "id": "ex1030",
```

"input": "The pharmacy vault can, but the inventory tracker cannot, guarantee that controlled substances will remain logged throughout the whole execution.",
"outputs": [
{
"formula": "<<PharmacyVault>>G controlled_substances_logged && !<<InventoryTracker>>G controlled_substances_logged"
}
]
},
{
"id": "ex1031",
"input": "The border checkpoint system can, but the biometric validator cannot, guarantee that forged passes will never be accepted.",
"outputs": [
{
"formula": "<<BorderCheckpointSystem>>G !forged_passes_accepted && !<<BiometricValidator>>G !forged_passes_accepted"
}
]
},
{
"id": "ex1032",
"input": "The telescope archive can, but the the backup manager cannot, guarantee that raw observation files will remain preserved at all times.",
"outputs": [
{
"formula": "<<TelescopeArchive>>G raw_observation_files_preserved && !<<BackupManager>>G raw_observation_files_preserved"
}
]
},
{
"id": "ex1033",
"input": "The accreditation board can, but the registry synchronizer cannot, guarantee that revoked credentials will never appear valid.",
"outputs": [
{
"formula": "<<AccreditationBoard>>G !revoked_credentials_valid && !<<RegistrySynchronizer>>G !revoked_credentials_valid"
}
]
},
{
"id": "ex1034",
"input": "The reservoir controller can, but the telemetry logger cannot, guarantee that the spillway status will remain monitored in every state.",
"outputs": [
{

```
        "formula": "<<ReservoirController>>G spillway_status_monitored &&
!<<TelemetryLogger>>G spillway_status_monitored"
      }
    ]
  },
  {
    "id": "ex1035",
    "input": "The digital notary can, but the cryptographic validator cannot, guarantee that
certified signatures will remain verifiable throughout the execution.",
    "outputs": [
      {
        "formula": "<<DigitalNotary>>G certified_signatures_verifiable &&
!<<CryptoValidator>>G certified_signatures_verifiable"
      }
    ]
  },
  {
    "id": "ex1036",
    "input": "The prison access system can, but the guard alone cannot, guarantee that
restricted cells will never unlock without authorization.",
    "outputs": [
      {
        "formula": "<<PrisonAccessSystem>>G !restricted_cells_unlock_without_authorization
&& !<<Guard>>G !restricted_cells_unlock_without_authorization"
      }
    ]
  },
  {
    "id": "ex1037",
    "input": "The cathedral fire monitor can, but the zone tester cannot, guarantee that smoke
alarms will remain armed at all times.",
    "outputs": [
      {
        "formula": "<<CathedralFireMonitor>>G smoke_alarms_armed && !<<ZoneTester>>G
smoke_alarms_armed"
      }
    ]
  },
  {
    "id": "ex1038",
    "input": "The transcript authority can, but the grade submitter cannot, guarantee that exam
results will never be altered after certification.",
    "outputs": [
      {
        "formula": "<<TranscriptAuthority>>(certification -> XG !exam_results_altered) &&
!<<GradeSubmitter>>(certification -> XG !exam_results_altered)"
      }
    ]
```

},
Original formula discussed: "<<TranscriptAuthority>>G !exam_results_altered_after_certification && !<<TimetablePage>>G !exam_results_altered_after_certification"

Reviewer 1: We should translate “after”.
Reviewer 2: I agree with Reviewer 1. In fact, I think we should translate it as “<<TranscriptAuthority>>certification XG !exam_results_altered && !<<GradeSubmitter>>XG !exam_results_altered”.
Reviewer 1: I agree.

{
"id": "ex1039",
"input": "The fuel depot controller can, but the pressure sensor cannot, guarantee that leak alarms will remain enabled throughout the execution.",
"outputs": [
{
"formula": "<<FuelDepotController>>G leak_alarms_enabled && !<<PressureSensor>>G leak_alarms_enabled"
}
]
},
{
"id": "ex1040",
"input": "The archive custodian can, but the tracker tag cannot, guarantee that rare manuscripts will never be removed without clearance.",
"outputs": [
{
"formula": "<<ArchiveCustodian>>G !rare_manuscripts_removed_without_clearance && !<<TrackerTag>>G !rare_manuscripts_removed_without_clearance"
}
]
},
{
"id": "ex1041",
"input": "The water-quality lab can, but the publishing router cannot, guarantee that contamination reports will remain accessible in every state.",
"outputs": [
{
"formula": "<<WaterQualityLab>>G contamination_reports_accessible && !<<PublishingRouter>>G contamination_reports_accessible"
}
]
},
{
"id": "ex1042",
"input": "The diplomatic cable system can, but the encryption gateway cannot, guarantee that classified cables will never be routed publicly.",

```
    "outputs": [
      {
        "formula": "<<DiplomaticCableSystem>>G !classified_cables_routed_publicly &&
!<<EncryptionGateway>>G !classified_cables_routed_publicly"
      }
    ]
  },
  {
    "id": "ex1043",
    "input": "The spacecraft life-support module can, but the ventilation regulator cannot,
guarantee that oxygen circulation will remain active at all times.",
    "outputs": [
      {
        "formula": "<<LifeSupportModule>>G oxygen_circulation_active &&
!<<VentilationRegulator>>G oxygen_circulation_active"
      }
    ]
  },
  {
    "id": "ex1044",
    "input": "The conservation lab can, but the blinds actuator cannot, guarantee that fragile
pigments will never be exposed to unsafe light.",
    "outputs": [
      {
        "formula": "<<ConservationLab>>G !fragile_pigments_exposed_to_unsafe_light &&
!<<BlindsActuator>>G !fragile_pigments_exposed_to_unsafe_light"
      }
    ]
  },
  {
    "id": "ex1045",
    "input": "The stockroom controller can, but the cooling unit cannot, guarantee that
cold-chain items will remain refrigerated throughout the execution.",
    "outputs": [
      {
        "formula": "<<StockroomController>>G cold_chain_items_refrigerated &&
!<<CoolingUnit>>G cold_chain_items_refrigerated"
      }
    ]
  },
  {
    "id": "ex1046",
    "input": "The identity authority can, but the login handler cannot, guarantee that expired
identities will never authenticate.",
    "outputs": [
      {
        "formula": "<<IdentityAuthority>>G !expired_identities_authenticate &&
!<<LoginHandler>>G !expired_identities_authenticate"
```

}
]
},
{
"id": "ex1047",
"input": "The runway safety system can, but the radar tracker cannot, guarantee that active runways will remain conflict-free in every state.",
"outputs": [
{
"formula": "<<RunwaySafetySystem>>G active_runways_conflict_free && !<<RadarTracker>>G active_runways_conflict_free"
}
]
},
{
"id": "ex1048",
"input": "The heritage database can, but the replication daemon cannot, guarantee that provenance records will never disappear.",
"outputs": [
{
"formula": "<<HeritageDatabase>>G !provenance_records_disappear && !<<ReplicationDaemon>>G !provenance_records_disappear"
}
]
},
{
"id": "ex1049",
"input": "The grant portal can, but the review assignment engine cannot, guarantee that in due course the eligibility decision will be issued.",
"outputs": [
{
"formula": "<<GrantPortal>>F eligibility_decision_issued && !<<ReviewAssignmentEngine>>F eligibility_decision_issued"
}
]
},
{
"id": "ex1050",
"input": "The medical referral desk can, but the triage automaton cannot, guarantee that sooner or later the specialist appointment will be assigned.",
"outputs": [
{
"formula": "<<ReferralDesk>>F specialist_appointment_assigned && !<<TriageAutomaton>>F specialist_appointment_assigned"
}
]
},
{

```
    "id": "ex1051",
    "input": "The ocean buoy network can, but the wave analyzer cannot, guarantee that
eventually the wave anomaly will be reported.",
    "outputs": [
      {
        "formula": "<<BuoyNetwork>>F wave_anomaly_reported && !<<WaveAnalyzer>>F
wave_anomaly_reported"
      }
    ]
  },
  {
    "id": "ex1052",
    "input": "The manuscript review system can, but the notification scheduler cannot,
guarantee that before long the editorial decision will be recorded.",
    "outputs": [
      {
        "formula": "<<ReviewSystem>>F editorial_decision_recorded &&
!<<NotificationScheduler>>F editorial_decision_recorded"
      }
    ]
  },
  {
    "id": "ex1053",
    "input": "The archaeological survey unit can, but the signal processor cannot, guarantee
that at some future point the buried marker will be located.",
    "outputs": [
      {
        "formula": "<<SurveyUnit>>F buried_marker_located && !<<SignalProcessor>>F
buried_marker_located"
      }
    ]
  },
  {
    "id": "ex1054",
    "input": "The immigration case system can, but the document ingestor cannot, guarantee
that after finitely many steps the residency file will be reviewed.",
    "outputs": [
      {
        "formula": "<<CaseSystem>>F residency_file_reviewed && !<<DocumentIngestor>>F
residency_file_reviewed"
      }
    ]
  },
  {
    "id": "ex1055",
    "input": "The insurance triage engine can, but the photo analyzer cannot, guarantee that
eventually the damage assessment will be completed.",
    "outputs": [
```

```
      {
        "formula": "<<InsuranceEngine>>F damage_assessment_completed &&
!<<PhotoAnalyzer>>F damage_assessment_completed"
      }
    ]
  },
  {
    "id": "ex1056",
    "input": "The airport baggage router can, but the tag reader cannot, guarantee that
eventually the misplaced suitcase will be redirected.",
    "outputs": [
      {
        "formula": "<<BaggageRouter>>F misplaced_suitcase_redirected &&
!<<TagReader>>F misplaced_suitcase_redirected"
      }
    ]
  },
  {
    "id": "ex1057",
    "input": "The telescope allocation committee can, but the proposal ranker cannot,
guarantee that sooner or later the observation slot will be granted.",
    "outputs": [
      {
        "formula": "<<AllocationCommittee>>F observation_slot_granted &&
!<<ProposalRanker>>F observation_slot_granted"
      }
    ]
  },
  {
    "id": "ex1058",
    "input": "The procurement office can, but the budget umpire cannot, guarantee that in due
course the emergency order will be approved.",
    "outputs": [
      {
        "formula": "<<ProcurementOffice>>F emergency_order_approved &&
!<<BudgetUmpire>>F emergency_order_approved"
      }
    ]
  },
  {
    "id": "ex1059",
    "input": "The port inspection desk can, but the manifest parser cannot, guarantee that
eventually the cargo manifest will be cleared.",
    "outputs": [
      {
        "formula": "<<PortInspectionDesk>>F cargo_manifest_cleared &&
!<<ManifestParser>>F cargo_manifest_cleared"
      }
```

```
    ]
  },
  {
    "id": "ex1060",
    "input": "The scholarship committee can, but the media coordinator cannot, guarantee that
before long the award notice will be published.",
    "outputs": [
      {
        "formula": "<<ScholarshipCommittee>>F award_notice_published &&
!<<MediaCoordinator>>F award_notice_published"
      }
    ]
  },
  {
    "id": "ex1061",
    "input": "The disaster relief coordinator can, but the fleet dispatcher cannot, guarantee that
at some later stage the supply convoy will arrive.",
    "outputs": [
      {
        "formula": "<<ReliefCoordinator>>F supply_convoy_arrived && !<<FleetDispatcher>>F
supply_convoy_arrived"
      }
    ]
  },
  {
    "id": "ex1062",
    "input": "The genome annotation tool can, but the alignment engine cannot, guarantee that
at some later stage the variant will be classified.",
    "outputs": [
      {
        "formula": "<<AnnotationTool>>F variant_classified && !<<AlignmentEngine>>F
variant_classified"
      }
    ]
  },
  {
    "id": "ex1063",
    "input": "The city permits office can, but the zoning checker cannot, guarantee that after
some finite delay the excavation permit will be issued.",
    "outputs": [
      {
        "formula": "<<PermitsOffice>>F excavation_permit_issued && !<<ZoningChecker>>F
excavation_permit_issued"
      }
    ]
  },
  {
    "id": "ex1064",
```

"input": "The meteorological center can, but the alert broadcaster cannot, guarantee that sooner or later the storm bulletin will be updated.",
"outputs": [
{
"formula": "<<MeteorologicalCenter>>F storm_bulletin_updated && !<<AlertBroadcaster>>F storm_bulletin_updated"
}
]
},
{
"id": "ex1065",
"input": "The railway claims office can, but the ticket validator cannot, guarantee that in due course the delay compensation will be processed.",
"outputs": [
{
"formula": "<<ClaimsOffice>>F delay_compensation_processed && !<<TicketValidator>>F delay_compensation_processed"
}
]
},
{
"id": "ex1066",
"input": "The vineyard irrigation planner can, but the moisture sensor grid cannot, guarantee that eventually the dry parcel will be watered.",
"outputs": [
{
"formula": "<<IrrigationPlanner>>F dry_parcel_watered && !<<MoistureSensorGrid>>F dry_parcel_watered"
}
]
},
{
"id": "ex1067",
"input": "The naval rescue center can, but the sonar array cannot, guarantee that before long the distress vessel will be located.",
"outputs": [
{
"formula": "<<RescueCenter>>F distress_vessel_located && !<<SonarArray>>F distress_vessel_located"
}
]
},
{
"id": "ex1068",
"input": "The accreditation portal can, but the audit uploader cannot, guarantee that eventually the compliance certificate will be renewed.",
"outputs": [
{

"formula": "<<AccreditationPortal>>F compliance_certificate_renewed && !<<AuditUploader>>F compliance_certificate_renewed"
}
]
},
{
"id": "ex1069",
"input": "The wildfire command center can, but the siren network cannot, guarantee that at some future point the evacuation notice will be delivered.",
"outputs": [
{
"formula": "<<WildfireCommandCenter>>F evacuation_notice_delivered && !<<SirenNetwork>>F evacuation_notice_delivered"
}
]
},
{
"id": "ex1070",
"input": "The fishery monitoring service can, but the AIS correlator cannot, guarantee that eventually the illegal trawler will be identified.",
"outputs": [
{
"formula": "<<FisheryMonitoringService>>F illegal_trawler_identified && !<<AISCorrelator>>F illegal_trawler_identified"
}
]
},
{
"id": "ex1071",
"input": "The campus housing office can, but the preference matcher cannot, guarantee that sooner or later the room assignment will be confirmed.",
"outputs": [
{
"formula": "<<HousingOffice>>F room_assignment_confirmed && !<<PreferenceMatcher>>F room_assignment_confirmed"
}
]
},
{
"id": "ex1072",
"input": "The translation review board can, but the context extractor cannot, guarantee that in due course the disputed phrase will be approved.",
"outputs": [
{
"formula": "<<TranslationReviewBoard>>F disputed_phrase_approved && !<<ContextExtractor>>F disputed_phrase_approved"
}
]

```
},
{
  "id": "ex1073",
  "input": "The public-health system can, but the symptom screener cannot, guarantee that
eventually the exposure report will be filed.",
  "outputs": [
    {
      "formula": "<<PublicHealthSystem>>F exposure_report_filed &&
!<<SymptomScreener>>F exposure_report_filed"
    }
  ]
},
{
  "id": "ex1074",
  "input": "The conservation permit board can, but the ecology surveyor cannot, guarantee
that before long the habitat assessment will be completed.",
  "outputs": [
    {
      "formula": "<<PermitBoard>>F habitat_assessment_completed &&
!<<EcologySurveyor>>F habitat_assessment_completed"
    }
  ]
},
{
  "id": "ex1075",
  "input": "The telescope mirror actuator can, but the wavefront sensor cannot, guarantee
that after finitely many steps the mirror alignment will be corrected.",
  "outputs": [
    {
      "formula": "<<MirrorActuator>>F mirror_alignment_corrected &&
!<<WavefrontSensor>>F mirror_alignment_corrected"
    }
  ]
},
{
  "id": "ex1076",
  "input": "The civic ombudsman can, but the case router cannot, guarantee that eventually
the citizen complaint will be answered.",
  "outputs": [
    {
      "formula": "<<Ombudsman>>F citizen_complaint_answered && !<<CaseRouter>>F
citizen_complaint_answered"
    }
  ]
},
{
  "id": "ex1077",
```

```
    "input": "The ferry dispatcher can, but the gangway controller cannot, guarantee that
boarding will remain suspended until the safety inspection is completed.",
    "outputs": [
      {
        "formula": "<<FerryDispatcher>>(boarding_suspended U safety_inspection_completed)
&& !<<GangwayController>>(boarding_suspended U safety_inspection_completed)"
      }
    ]
  },
  {
    "id": "ex1078",
    "input": "The manuscript curator can, but the climate regulator cannot, guarantee that the
folio will remain sealed until restoration is finished.",
    "outputs": [
      {
        "formula": "<<ManuscriptCurator>>(folio_sealed U restoration_finished) &&
!<<ClimateRegulator>>(folio_sealed U restoration_finished)"
      }
    ]
  },
  {
    "id": "ex1079",
    "input": "The vaccine cold-room controller can, but the thermal regulator cannot,
guarantee that the shipment will remain chilled until pickup begins.",
    "outputs": [
      {
        "formula": "<<ColdRoomController>>(shipment_chilled U pickup_started) &&
!<<ThermalRegulator>>(shipment_chilled U pickup_started)"
      }
    ]
  },
  {
    "id": "ex1080",
    "input": "The court archivist can, but the redaction engine cannot, guarantee that the
transcript will remain confidential until the appeal is closed.",
    "outputs": [
      {
        "formula": "<<CourtArchivist>>(transcript_confidential U appeal_closed) &&
!<<RedactionEngine>>(transcript_confidential U appeal_closed)"
      }
    ]
  },
  {
    "id": "ex1081",
    "input": "The polar station controller can, but the fuel injector cannot, guarantee that the
generator will remain active until the battery bank is recharged.",
    "outputs": [
      {
```

```
        "formula": "<<PolarStationController>>(generator_active U battery_bank_recharged) && !<<FuelInjector>>(generator_active U battery_bank_recharged)"
      }
    ]
  },
  {
    "id": "ex1082",
    "input": "The harbor quarantine officer can, but the patrol cutter cannot, guarantee that the vessel will remain isolated until inspection clearance is granted.",
    "outputs": [
      {
        "formula": "<<QuarantineOfficer>>(vessel_isolated U inspection_clearance_granted) && !<<PatrolCutter>>(vessel_isolated U inspection_clearance_granted)"
      }
    ]
  },
  {
    "id": "ex1083",
    "input": "The telescope cooling system can, but the cryo heater cannot, guarantee that the detector will remain idle until the thermal level stabilizes.",
    "outputs": [
      {
        "formula": "<<CoolingSystem>>(detector_idle U thermal_level_stabilized) && !<<CryoHeater>>(detector_idle U thermal_level_stabilized)"
      }
    ]
  },
  {
    "id": "ex1084",
    "input": "The passport office can, but the fraud scanner cannot, guarantee that the application will remain pending until identity evidence is verified.",
    "outputs": [
      {
        "formula": "<<PassportOffice>>(application_pending U identity_evidence_verified) && !<<FraudScanner>>(application_pending U identity_evidence_verified)"
      }
    ]
  },
  {
    "id": "ex1085",
    "input": "The canyon rescue unit can, but the barrier actuator cannot, guarantee that the route will remain closed until the injured hiker is evacuated.",
    "outputs": [
      {
        "formula": "<<RescueUnit>>(route_closed U injured_hiker_evacuated) && !<<BarrierActuator>>(route_closed U injured_hiker_evacuated)"
      }
    ]
```

},
{
"id": "ex1086",
"input": "The art transport coordinator can, but the GPS beacon cannot, guarantee that the crate will remain locked until the courier arrives.",
"outputs": [
{
"formula": "<<TransportCoordinator>>(crate_locked U courier_arrived) && !<<GPSBeacon>>(crate_locked U courier_arrived)"
}
]
},
{
"id": "ex1087",
"input": "The floodgate supervisor can, but the debris skimmer cannot, guarantee that the spillway will remain open until pressure returns to normal.",
"outputs": [
{
"formula": "<<FloodgateSupervisor>>(spillway_open U pressure_normalized) && !<<DebrisSkimmer>>(spillway_open U pressure_normalized)"
}
]
},
{
"id": "ex1088",
"input": "The cryptographic key manager can, but the entropy generator cannot, guarantee that the old key will remain disabled until rotation is complete.",
"outputs": [
{
"formula": "<<KeyManager>>(old_key_disabled U rotation_complete) && !<<EntropyGenerator>>(old_key_disabled U rotation_complete)"
}
]
},
{
"id": "ex1089",
"input": "The mining safety officer can, but the air scrubber cannot, guarantee that the shaft will remain evacuated until methane readings are safe.",
"outputs": [
{
"formula": "<<SafetyOfficer>>(shaft_evacuated U methane_readings_safe) && !<<AirScrubber>>(shaft_evacuated U methane_readings_safe)"
}
]
},
{
"id": "ex1090",

```
    "input": "The archival digitization unit can, but the laser measurer cannot, guarantee that
the scan job will remain paused until the fragile page is stabilized.",
    "outputs": [
      {
        "formula": "<<DigitizationUnit>>(scan_job_paused U fragile_page_stabilized) &&
!<<LaserMeasurer>>(scan_job_paused U fragile_page_stabilized)"
      }
    ]
  },
  {
    "id": "ex1091",
    "input": "The clinical ethics board can, but the lexicon parser cannot, guarantee that the
protocol will remain under review until consent language is revised.",
    "outputs": [
      {
        "formula": "<<EthicsBoard>>(protocol_under_review U consent_language_revised) &&
!<<LexiconParser>>(protocol_under_review U consent_language_revised)"
      }
    ]
  },
  {
    "id": "ex1092",
    "input": "The orbital docking controller can, but the RCS thruster cannot, guarantee that
approach mode will remain limited until alignment is confirmed.",
    "outputs": [
      {
        "formula": "<<DockingController>>(approach_mode_limited U alignment_confirmed) &&
!<<RCSThruster>>(approach_mode_limited U alignment_confirmed)"
      }
    ]
  },
  {
    "id": "ex1093",
    "input": "The food recall coordinator can, but the distribution tracker cannot, guarantee that
the product will remain blocked until the supplier response is received.",
    "outputs": [
      {
        "formula": "<<RecallCoordinator>>(product_blocked U supplier_response_received) &&
!<<DistributionTracker>>(product_blocked U supplier_response_received)"
      }
    ]
  },
  {
    "id": "ex1094",
    "input": "The glacier monitoring station can, but the laser altimeter cannot, guarantee that
the trail will remain restricted until the icefall risk is reduced.",
    "outputs": [
      {
```

```
      "formula": "<<GlacierMonitoringStation>>(trail_restricted U icefall_risk_reduced) &&
!<<LaserAltimeter>>(trail_restricted U icefall_risk_reduced)"
     }
    ]
  },
  {
    "id": "ex1095",
    "input": "The procurement compliance desk can, but the entity matcher cannot, guarantee
that the contract will remain unsigned until conflict screening is completed.",
    "outputs": [
     {
      "formula": "<<ComplianceDesk>>(contract_unsigned U conflict_screening_completed)
&& !<<EntityMatcher>>(contract_unsigned U conflict_screening_completed)"
     }
    ]
  },
  {
    "id": "ex1096",
    "input": "The tunnel ventilation controller can, but the exhaust fan cannot, guarantee that
the tunnel will remain closed until smoke extraction is completed.",
    "outputs": [
     {
      "formula": "<<VentilationController>>(tunnel_closed U smoke_extraction_completed)
&& !<<ExhaustFan>>(tunnel_closed U smoke_extraction_completed)"
     }
    ]
  },
  {
    "id": "ex1097",
    "input": "The diplomatic registry can, but the tamper detector cannot, guarantee that the
note will remain sealed until the envoy signs it.",
    "outputs": [
     {
      "formula": "<<DiplomaticRegistry>>(note_sealed U envoy_signed) &&
!<<TamperDetector>>(note_sealed U envoy_signed)"
     }
    ],
    "difficulty": "hard"
  },
  {
    "id": "ex1098",
    "input": "The emergency shelter coordinator can, but the logistics stager cannot,
guarantee that intake will remain open until the last evacuee is registered.",
    "outputs": [
     {
      "formula": "<<ShelterCoordinator>>(intake_open U last_evacuee_registered) &&
!<<LogisticsStager>>(intake_open U last_evacuee_registered)"
     }
```

```
    ],
    "difficulty": "hard"
  },
  {
    "id": "ex1099",
    "input": "The bridge inspection unit can, but the strain gauge cannot, guarantee that the
crossing will remain closed until the load test is passed.",
    "outputs": [
      {
        "formula": "<<BridgeInspectionUnit>>(crossing_closed U load_test_passed) &&
!<<StrainGauge>>(crossing_closed U load_test_passed)"
      }
    ]
  },
  {
    "id": "ex1100",
    "input": "The observatory scheduler can, but the ephemeris calculator cannot, guarantee
that the observation request will remain queued until the target window opens.",
    "outputs": [
      {
        "formula": "<<ObservatoryScheduler>>(observation_request_queued U
target_window_opened) && !<<EphemerisCalculator>>(observation_request_queued U
target_window_opened)"
      }
    ]
  }
]
```